

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\401-E-30001B- Top- RP.pvdb

Date of Analysis : Aug 12,2026 9:43am

PV Elite 28, January 2026



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FileName : 401-E-30001B- Top- RP

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Vessel Design Summary:

Step: 31 9:43am Aug 12,2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 700.000 mm ID
Vessel Design Length, Tangent to Tangent 494.45 cm
Specified Datum Line Distance 0.00 cm
Shell Side Design Temperature 155 °C
Channel Side Design Temperature 155 °C
Shell Side Design Pressure 10.197 kgf/cm²
Channel Side Design Pressure 46.908 kgf/cm²
Shop Shell Side Test Pressure 17.377 kgf/cm²
Field Shell Side Test Pressure 13.257 kgf/cm²
Shop Channel Side Test Pressure 62.772 kgf/cm²
Field Channel Side Test Pressure 60.980 kgf/cm²

Wind Design Code IS 875 (PART 3) Amd. 1&2 (2020)
Earthquake Design Code IS 1893 RSM Amd. 1&2 (2018)

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Shell	SA-240 304L	...	...	S30403	
Head	SA-240 304L	...	...	S30403	
Flange	SA-266 4	...	...	K03017	
Flange	SA-965 F304L	...	...	S30403	
Nozzle	SA-266 4	...	...	K03017	
Nozzle	SA-312 TP304L	...	...	S30403	
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-105	...	...	K03504	
Nozzle Flg	SA-182 F304L	...	> 130	S30403	
Tubes	SA-213 TP304L	...	...	S30403	
Tubesheet	SA-965 F304L	...	...	S30403	
Flg Bolting	SA-193 B7	...	<= 64	G41400	
Hrz Bolting	SA-193 B7	...	<= 64	G41400	
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
+Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (kgf/cm² & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Chnl rear cover	46.908	0.00	48.000	1.5000	N/A	No
Chnl rear flang	46.908	0.00	48.000	1.5000	N/A	No
Chnl. rear	46.909	0.00	64.000	1.5000	N/A	No
Shell flange	10.197	0.00	13.000	0.0000	N/A	No
Shell	10.267	0.00	33.000	0.0000	N/A	No
shell head	10.267	0.00	17.000	0.0000	No	No

Liquid Level: 70.00 cm Dens.: 999.542 kg/m³ Sp. Gr.: 1.000



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Vessel Design Summary:

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Element Types and Properties:

Element Type	"To" Elev cm	Element Length cm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Body Flg	9.9	9.5	73.0	95.0	90.5	43.4	1.00	1.00
Body Flg	23.5	13.1	90.0	108.0	101.9	79.5	1.00	1.00
Cylinder	123.5	100.0	18.0	18.0	13.5	...	1.00	1.00
Body Flg	142.6	8.1	6.0	72.0	67.6	67.6	1.00	1.00
Cylinder	489.4	346.9	10.0	10.0	3.1	...	1.00	1.00
Ellipse	494.4	5.0	6.0	5.1	3.1	...	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N1- Channel in	WRC-297	0.945	Shell	Passed
N2B- channel o	WRC-297	0.945	Shell	Passed
N4B- Shell out	WRC-297	0.875	Nozzle	Passed
K1- Pressure r	WRC-297	0.875	Nozzle	Passed
N3B- Shell inl	WRC-297	0.875	Nozzle	Passed

Foundation/Support Summary

Saddle Parameters:

Saddle Width	150.000 mm
Saddle Bearing Angle	120.000 deg.
Centerline Dimension	720.000 mm
Wear Pad Width	200.000 mm
Wear Pad Thickness	10.000 mm
Wear Pad Bearing Angle	132.000 deg.
Distance from Saddle to Tangent	620.000 mm

Baseplate Length	930.000 mm
Baseplate Thickness	15.000 mm
Baseplate Width	180.000 mm
Number of Ribs (including outside ribs)	5
Rib Thickness	15.000 mm
Web Thickness	15.000 mm
Height of Center Web	306.000 mm
Number of Bolts in Baseplate	2

Baseplate Sketch

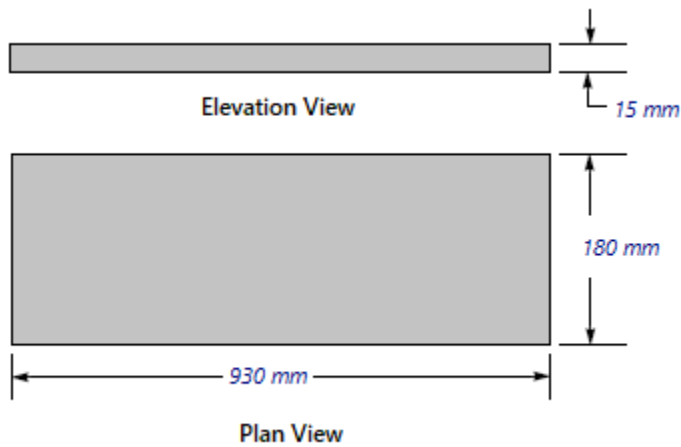
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Vessel Design Summary:

Step: 31 9:43am Aug 12, 2026



Maximum Tensile Bolt Load	147. kgf
---------------------------	----------

Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	7185.54	kgf
Maximum Transverse Saddle Shear Load	431.93	kgf
Maximum Longitudinal Saddle Shear Load	538.04	kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	6944.41	kgf
Maximum Transverse Saddle Shear Load	142.54	kgf
Maximum Longitudinal Saddle Shear Load	43.06	kgf

Weights:

Fabricated	- Bare w/o removable internals	5084.1	kg
Shop Test	- Fabricated + water (full)	6755.7	kg
Shipping	- Fab. + rem. intls.+ shipping app.	5084.1	kg
Erected	- Fab. + rem. intls.+ insul. (etc)	5084.1	kg
Empty	- Fab. + intls. + details + wghts.	5124.4	kg
Operating	- Empty + operating liquid (no ca)	5972.9	kg
Field Test	- Empty weight + test medium (full)	6767.5	kg



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Nozzle Summary: Step: 29 9:43am Aug 12,2026

Nozzle Calculation Summary (kgf/cm² & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N1- Channel in1	61	...	...	OK	9.60	OK	Passed
N2B- channel ou	61	...	...	OK	9.60	OK	Passed
N4B- Shell out1	32	...	...	OK	3.07	OK	Passed
K1- Pressure re	32	...	...	OK	3.07	OK	Passed
N3B- Shell inle	32	...	...	OK	3.09	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 32 Nozzle : N3B- Shell inle [Shellside]
 Minimum MAWP Nozzles : 61 Nozzle : N2B- channel ou [Tubeside]

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
30	N1- Channel in1	735.000	270.000	491.750
30	N2B- channel ou	735.000	90.000	491.750
50	N4B- Shell out1	1725.499	270.000	285.451
50	K1- Pressure re	4425.499	270.000	285.451
50	N3B- Shell inle	1725.499	90.000	285.451

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Step: 28 9:43am Aug 12,2026

Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Diameter	Pad Thk mm	Cut Length mm	Flg Class
N4B- Shell outl	6.000 in	120	WNF	6.625	14.275	368.27	8.00	325.26	600
K1- Pressure re	6.000 in	120	WNF	6.625	14.275	368.27	8.00	325.26	300
N3B- Shell inle	6.000 in	120	WNF	6.625	14.275	368.27	8.00	325.26	300
N1- Channel inl	10.000 in	80	WNF	10.750	15.088	...	...	350.73	600
N2B- channel ou	10.000 in	80	WNF	10.750	15.088	...	...	350.73	600

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
N4B- Shell	SA-312 TP304L	6.000	10.000	8.000	8.000	...
K1- Pressur	SA-312 TP304L	6.000	10.000	8.000	8.000	...
N3B- Shell	SA-312 TP304L	6.000	10.000	8.000	8.000	...
N1- Channel	SA-266 4	15.000	30.000	...	...	...
N2B- channe	SA-266 4	15.000	30.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum cm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
N4B- Shell outl	172.550	270.0	305.00	0.00	Shell
K1- Pressure re	442.550	270.0	305.00	0.00	Shell
N3B- Shell inle	172.550	90.0	305.00	0.00	Shell
N1- Channel inl	73.500	270.0	305.00	0.00	Chnl. rear
N2B- channel ou	73.500	90.0	305.00	0.00	Chnl. rear

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Bill of Material:

Step: 27 9:43am Aug 12,2026

Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	BLIND FLANGE: 95.0mm THK X 940.0mm OD	SA-266 4
1	WELD NECK FLANGE: 108.0mm THK X 940.0mm OD	SA-266 4
1	WELD NECK FLANGE: 72.0mm THK X 860.0mm OD	SA-965 F304L
1	ELLIPTICAL HEAD: 2.0 X 1, 6.0mm THK X 700.0mm ID X 5.0cm	SA-240 304L
1	INSULATION: 1000mm X 45mm THK	Hot
9	BAFFLES: 680.0mm DIA X 6.0mm THK	SA-240 304L
1	TUBESHEET: 860.0mm X 105.0mm THK	SA-965 F304L
545	TUBES: 3300.0mm X 19.0mm DIA X 2.0mm THK	SA-213 TP304L
2	SADDLE: 150.0mm X 120 DEG	SA-516 70
1	INSULATION: 3469mm X 25mm THK	Hot
1	INSULATION: 225mm X 25mm THK	Hot
2	GASKET: 755.0mm OD X 729.0mm ID	...
24	BODY FLANGE BOLTS: 31.8mm DIA	SA-193 B7
48	NUTS FOR BODY FLANGE BOLTS: 31.8mm DIA	...
20	BODY FLANGE BOLTS: 25.4mm DIA	SA-193 B7
20	NUTS FOR BODY FLANGE BOLTS: 25.4mm DIA	...
2	CYLINDRICAL SEGMENT 18.0mm THK X 873.5mm X 100.0cm	SA-516 70
1	CYLINDRICAL SEGMENT 18.0mm THK X 452.0mm X 100.0cm	SA-516 70
1	CYLINDRICAL SEGMENT 10.0mm THK X 1765.4mm X 346.9cm	SA-240 304L
1	CYLINDRICAL SEGMENT 10.0mm THK X 433.7mm X 346.9cm	SA-240 304L
1	NAMEPLATE	...



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MDMT Summary:

Step: 30 9:43am Aug 12,2026

Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Notes								
Bolting [21]		-48						
Shell [16]		-196						
shell head [16]		-196						
N4B- Shell out[1]	B	-29						No
K1- Pressure r[1]	B	-29						No
N3B- Shell inl[1]	B	-29						No
Warmest MDMT:		-29						
Chnl rear cov[11]	B	-2	-2	-29	1.000	23.750	1.00	No
Chnl rear fla[11]	B	-16	-30	-29	0.752	15.000	1.00	No
Chnl. rear [8]	B	-11	-25	-29	0.746	18.000	1.00	No
N1- Channel in[1]	B	-11	-29	-29	0.725	18.000	1.00	No
Nozzle Flg [5]	A	-18	-48					
N2B- channel o[1]	B	-11	-29	-29	0.725	18.000	1.00	No
Nozzle Flg [5]	A	-18	-48					
Bolting [21]		-48						
Warmest MDMT:		-2	-2					

Exchanger Side	Computed MDMT	Required MDMT	Pass/Fail
Shell	-29.0	11.4	Pass
Channel/Tube	-29.0	11.4	Pass

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

- Impact test temps were not entered in and not considered in the analysis.
- UCS-66(i) applies to impact tested materials not by specification and
- UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).



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MDMT Summary:

Step: 30 9:43am Aug 12, 2026

The Basic MDMT includes the (30F) PWHT credit if applicable.

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Warnings and Errors:

Step: 0 9:43am Aug 12,2026

Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Input Echo:

Step: 1 9:43am Aug 12,2026

Units used in this Analysis (Kgfc²):

Name	System Unit	Constant	User Unit
Length	Feet	30.4800	cm
Force	Pounds	0.4536	kgf
Mass	Pounds	0.4536	kg
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	kgf-m.
Stress	lbs./sq.in.	0.0703	kgf/cm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0703	kgf/cm ²
Elast. Modulus	lbs./sq.in.	0.0703	kgf/cm ²
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	16.0183	kg/m ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kgf/cm ²
Inertia	in.4	41.6231	cm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8824	kgf/m ²
Elevation	Feet	0.3048	m
Volume	in.3	16.3871	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	10.197	kgf/cm ²
Channel Side Design Pressure	46.908	kgf/cm ²
Shell Side Design Temperature	155.0	°C
Channel Side Design Temperature	155.0	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	Air/Water/Steam	
Service Type, Channel Side	Sour	
MDMT (CET), Shell Side	11.4	°C
MDMT (CET), Tube Side	11.4	°C
User defined MAWP, Shell Side	0	kgf/cm ²
User defined MAWP, Channel Side	0	kgf/cm ²
User defined MAPnc, Shell Side	0	kgf/cm ²
User defined MAPnc, Channel Side	0	kgf/cm ²
User defined Test Pres., Shell Side	0	kgf/cm ²
User defined Test Pres., Channel Side	0	kgf/cm ²

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070



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Input Echo:

Step: 1 9:43am Aug 12,2026

0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Consider Vortex Shedding

N

Shop Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes

Field Pressure Test:

Type of Pressure Test	UG-99(b) Note [35]
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	km/hr
Wind Zone Number	5	
Base Elevation	30	cm
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1] 1.0	
Terrain Category	2	
Equipment Class	Class A	
Topography Factor	[k3] 1.0	
Importance Factor	[k4] 1.15	
Damping Factor (Beta) for Wind (Ope)	0.0100	



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Input Echo: Step: 1 9:43am Aug 12,2026

Damping Factor (Beta) for Wind (Empty) 0.0000
Damping Factor (Beta) for Wind (Filled) 0.0000

Seismic Design Code IS 1893-RSM
Importance Factor for IS-1893 2.000
Soil Factor 1.000
Zone Number 2.000
Percent Seismic for Hydrotest 0.000
Period of Vibration (User defined) Sec 0.000
Damping Factor 2.000

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Consider Vertical Acceleration Yes

Design Pressure + Static Head Y
Consider MAP New and Cold in Noz. Design N
Consider External Loads for Nozzle Des. Y
Use ASME VIII-1 Appendix 1-9 N

Perform Blast Load Analysis No
Material Database Year Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk) Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37 No
Use Table G instead of exact equation for "A" Yes
Shell Head Joints are Tapered Yes
Compute "K" in corroded condition Yes
Use Code Case 2286 No
Use the MAWP to compute the MDMT Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C) Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C) No

Using Metric Material Databases, ASME II D Yes
Calculate B31.3 type stress for Nozzles with Loads Yes
Reduce the MDMT due to lower membrane stress Yes
Consider Longitudinal Stress in MDMT Calculations Yes

Complete Listing of Vessel Elements and Details:



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Input Echo:

Step: 1 9:43am Aug 12,2026

Element#	1/6
Element From Node	10
Element To Node	20
Element Type	Flange
Description	Chnl rear cover
Distance "FROM" to "TO"	9.5 cm
Flange Outside Diameter	940 mm
Element Thickness	95 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	73 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	46.908 kgf/cm ²
Design Temperature Internal Pressure	155 °C
Design External Pressure	0 kgf/cm ²
Design Temperature External Pressure	40 °C
Effective Diameter Multiplier	1.2
Material Name	SA-266 4
Allowable Stress, Ambient	1407.1 kgf/cm ²
Allowable Stress, Operating	1407.1 kgf/cm ²
Allowable Stress, Hydrotest	2294.2 kgf/cm ²
Material Density	0.00775 kg/cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	2227 kgf/cm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K03017
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element#	2/6
Element From Node	20
Element To Node	30
Element Type	Flange
Description	Chnl rear flang
Distance "FROM" to "TO"	13.1 cm
Flange Inside Diameter	700 mm
Element Thickness	108 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	90 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	46.908 kgf/cm ²
Design Temperature Internal Pressure	155 °C
Design External Pressure	0 kgf/cm ²
Design Temperature External Pressure	40 °C
Effective Diameter Multiplier	1.2
Material Name	SA-266 4
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element#	3/6
Element From Node	30
Element To Node	40
Element Type	Cylinder
Description	Chnl. rear
Distance "FROM" to "TO"	100 cm
Inside Diameter	700 mm
Element Thickness	18 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	18 mm



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Input Echo:

Step: 1 9:43am Aug 12,2026

External Corrosion Allowance	0	mm
Design Internal Pressure	46.908	kgf/cm ²
Design Temperature Internal Pressure	155	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	40	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Allowable Stress, Ambient	1407.1	kgf/cm ²
Allowable Stress, Operating	1407.1	kgf/cm ²
Allowable Stress, Hydrotest	2386	kgf/cm ²
Material Density	0.00775	kg/cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	2357.5	kgf/cm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	30	
Detail Type	Liquid	
Detail ID	Gas	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	70	cm
Liquid Density	14.993	kg/m ³

Element From Node	30	
Detail Type	Insulation	
Detail ID	Hot insulation	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Insulation	100	cm
Thickness of Insulation	45	mm
Density	0.00012	kg/cm ³

Element From Node	30	
Detail Type	Nozzle	
Detail ID	N1- Channel inl	
Dist. from "FROM" Node / Offset dist	50	cm
Nozzle Diameter	10	in.
Nozzle Schedule	80	
Nozzle Class	600	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	135.76	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 4	

Element From Node	30	
Detail Type	Nozzle	
Detail ID	N2B- channel ou	
Dist. from "FROM" Node / Offset dist	50	cm
Nozzle Diameter	10	in.
Nozzle Schedule	80	
Nozzle Class	600	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	135.76	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 4	



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Input Echo:

Step: 1 9:43am Aug 12,2026

Element From Node	40
Element To Node	50
Element Type	Flange
Description	Shell flange
Distance "FROM" to "TO"	8.1 cm
Flange Inside Diameter	700 mm
Element Thickness	72 mm
Internal Corrosion Allowance	0 mm
Nominal Thickness	6 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	10.197 kgf/cm ²
Design Temperature Internal Pressure	155 °C
Design External Pressure	0 kgf/cm ²
Design Temperature External Pressure	40 °C
Effective Diameter Multiplier	1.2
Material Name	SA-965 F304L
Allowable Stress, Ambient	1172.6 kgf/cm ²
Allowable Stress, Operating	891.31 kgf/cm ²
Allowable Stress, Hydrotest	1560.1 kgf/cm ²
Material Density	0.008027 kg/cm ³
P Number Thickness	0 mm
Yield Stress, Operating	1333.8 kgf/cm ²
External Pressure Chart Name	HA-3
UNS Number	S30403
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element#	5/6
Element From Node	50
Element To Node	60
Element Type	Cylinder
Description	Shell
Distance "FROM" to "TO"	346.9 cm
Inside Diameter	700 mm
Element Thickness	10 mm
Internal Corrosion Allowance	0 mm
Nominal Thickness	10 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	10.197 kgf/cm ²
Design Temperature Internal Pressure	155 °C
Design External Pressure	0 kgf/cm ²
Design Temperature External Pressure	40 °C
Effective Diameter Multiplier	1.2
Material Name	SA-240 304L
Allowable Stress, Ambient	1172.6 kgf/cm ²
Allowable Stress, Operating	1167.5 kgf/cm ²
Allowable Stress, Hydrotest	1560.1 kgf/cm ²
Material Density	0.008027 kg/cm ³
P Number Thickness	0 mm
Yield Stress, Operating	1333.8 kgf/cm ²
External Pressure Chart Name	HA-3
UNS Number	S30403
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	50
Detail Type	Saddle
Detail ID	Left Saddle
Dist. from "FROM" Node / Offset dist	62 cm
Width of Saddle	150 mm



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Input Echo:

Step: 1 9:43am Aug 12,2026

Height of Saddle at Bottom	720	mm
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	200	mm
Thickness of Wear Plate	10	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.0	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	cm

Element From Node	50	
Detail Type	Saddle	
Detail ID	Right saddle	
Dist. from "FROM" Node / Offset dist	311	cm
Width of Saddle	150	mm
Height of Saddle at Bottom	720	mm
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	200	mm
Thickness of Wear Plate	10	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.30000001	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	cm

Element From Node	50	
Detail Type	Liquid	
Detail ID	Water	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	70	cm
Liquid Density	999.54	kg/m³

Element From Node	50	
Detail Type	Insulation	
Detail ID	Hot insulation	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Insulation	346.9	cm
Thickness of Insulation	25	mm
Density	0.00012	kg/cm³

Element From Node	50	
Detail Type	Nozzle	
Detail ID	N4B- Shell outl	
Dist. from "FROM" Node / Offset dist	30	cm
Nozzle Diameter	6	in.
Nozzle Schedule	120	
Nozzle Class	600	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	61.631	kgf
Grade of Attached Flange	GR 2.3	
Nozzle Matl	SA-312 TP304L	

Element From Node	50	
Detail Type	Nozzle	
Detail ID	K1- Pressure re	
Dist. from "FROM" Node / Offset dist	300	cm
Nozzle Diameter	6	in.
Nozzle Schedule	120	



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Input Echo:

Step: 1 9:43am Aug 12,2026

Nozzle Class	300	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	45.547	kgf
Grade of Attached Flange	GR 2.3	
Nozzle Matl	SA-312 TP304L	

Element From Node	50	
Detail Type	Nozzle	
Detail ID	N3B- Shell inle	
Dist. from "FROM" Node / Offset dist	30	cm
Nozzle Diameter	6	in.
Nozzle Schedule	120	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	45.547	kgf
Grade of Attached Flange	GR 2.3	
Nozzle Matl	SA-312 TP304L	

Element#	6/6	
Element From Node	60	
Element To Node	70	
Element Type	Elliptical	
Description	shell head	
Distance "FROM" to "TO"	5	cm
Inside Diameter	700	mm
Element Thickness	5.1	mm
Internal Corrosion Allowance	0	mm
Nominal Thickness	6	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	10.197	kgf/cm ²
Design Temperature Internal Pressure	155	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	40	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-240 304L	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	

Element From Node	60	
Detail Type	Liquid	
Detail ID	Water	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	70	cm
Liquid Density	999.54	kg/m ³

Element From Node	60	
Detail Type	Insulation	
Detail ID	Hot insulation	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Insulation	22.5	cm
Thickness of Insulation	25	mm
Density	0.00012	kg/cm ³



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XY Coordinate Calculations: Step: 2 9:43am Aug 12,2026

XY Coordinate Calculations:

From	To	X (Horiz.) cm	Y (Vert.) cm	DX (Horiz.) cm	DY (Vert.) cm
Chnl rear cover		9.95	...	9.5	...
Chnl rear flang		23.5	...	13.1	...
Chnl. rear		123.5	...	100	...
Shell flange		142.55	...	8.1	...
Shell		489.45	...	346.9	...
shell head		494.45	...	5	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Flg Calc [Int P]: New Flange

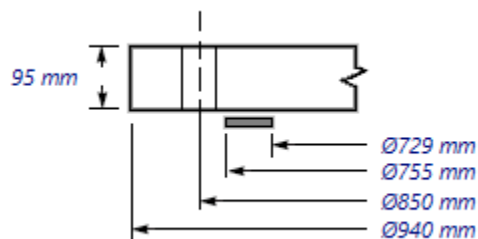
Flng: 7 9:43am Aug 12,2026

Flange Input Data Values

Description: New Flange :

Chnl rear cover

Flange Type		Blind	
Design Pressure	P	46.91	kgf/cm ²
Design Temperature		155	°C
Internal Corrosion Allowance	ci	1.5000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	940.000	mm
Flange Thickness	t	95.0000	mm
Flange Material		SA-266 4	
Flange Material UNS number		K03017	
Flange Allowable Stress At Temperature	Sfo	1407.13	kgf/cm ²
Flange Allowable Stress At Ambient	Sfa	1407.13	kgf/cm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	1753.82	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1753.82	kgf/cm ²
Perimeter along the Center of the Bolts	L	2670.354	mm
Diameter of Bolt Circle	C	850.000	mm
Nominal Bolt Diameter	a	31.7500	mm
Type of Threads		TEMA Thread Series	
Number of Bolts		24	
Flange Face Outside Diameter	Fod	766.000	mm
Flange Face Inside Diameter	Fid	700.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	755.000	mm
Gasket Inside Diameter	Gi	729.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	703.07	kgf/cm ²
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm



ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	$N = (Go - Gi)/2$	13.000 mm
Basic Gasket Width,	$bo = N/2$	6.500 mm
Effective Gasket Width,	$b = Cb \cdot \sqrt{bo}$	6.425 mm
Gasket Reaction Diameter,	$G = Go - 2 \cdot b$	742.151 mm

Basic Flange and Bolt Loads:



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Flg Calc [Int P]: New Flange Flng: 7 9:43am Aug 12,2026

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 742.1509^2 \cdot 46.908$$

$$= 202918.188 \text{ kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.4246 \cdot \pi \cdot 742.1509 \cdot 3.0 \cdot 46.91$$

$$= 42158.320 \text{ kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(202918 + 42158 + 0, 0)$$

$$= 245076.500 \text{ kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 703.07 \cdot 6.4246 \cdot \pi \cdot 742.151 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 105313.305 \text{ kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G)/S_b, Wg/S_a)$$

$$= \max((245077 + 0 + 4 \cdot 0/742.15)/1754, 105313/1754)$$

$$= 139.739 \text{ cm}^2$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	139.739	143.845	
Radial Distance between Bolts and Edge:	31.750	45.000	
Circ. Spacing between the Bolts:	71.450	110.947	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 143.845 \cdot 1753.82 / (703.07 \cdot \pi (755.0 + 729.0))$$

$$= 7.697 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a (A_m + A_b) / 2$$

$$= 1753.82 (139.7389 + 143.8449) / 2$$

$$= 248677.05 \text{ kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o$$

$$= 245076.50 \text{ kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (850.0 - 742.1509) / 2$$

$$= 53.9246 \text{ mm}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$= 1.9 \cdot W_{m1} \cdot hG \cdot B_{sc} / (t^2 \cdot G) + C \cdot Z \cdot P_{eq} \cdot G^2 / t^2$$

$$= 1.9 \cdot 245077 \cdot 53.9246 \cdot 1.0 / (93.5^2 \cdot 742.1509) + 0.3 \cdot 1.0 \cdot 46.91 \cdot 742.1509^2 / 93.5^2$$

$$= 1273.62 \text{ kgf/cm}^2$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

$$= 1.9 \cdot W \cdot hG \cdot B_{sc} / (t^2 \cdot G)$$

$$= 1.9 \cdot 248677 \cdot 53.9246 \cdot 1.0 / (93.5^2 \cdot 742.1509)$$



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Flg Calc [Int P]: New Flange

Flng: 7 9:43am Aug 12,2026

= 392.70 kgf/cm²

Bolt Stress, Operating [BSO]:

= Wm1 / Ab

= 245077/143.8449

= 1703.76 kgf/cm²

Bolt Stress, Seating [BSa]:

= (Wm2 / Ab)

= (105313/143.8449)

= 732.13 kgf/cm²

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	1273.62	1407.13	392.70	1407.13
Bolting	1703.76	1753.82	732.13	1753.82

Reqd. Blind Flange Thickness at Center

90.454 mm

Reqd. Blind Flange Thickness at Gasket

50.894 mm

Estimated M.A.W.P. (Operating)

48 kgf/cm²

Estimated Finished Weight of Flange at given Thk.

511.0 kg

Estimated Unfinished Weight of Forging at given Thk

511.0 kg

Minimum Design Metal Temperature Results:

Thickness Ratio = 1., Temperature Reduction per Fig. UCS 66.1 = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve B

-2 °C

Min Metal Temp. w/o impact per UG-20(f)

-29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	1233.72	1407.13	380.40	1407.13
Bolting	1703.76	1753.82	732.13	1753.82

Minimum Req'd Thickness, ambient, no Corrosion: 88.976 mm

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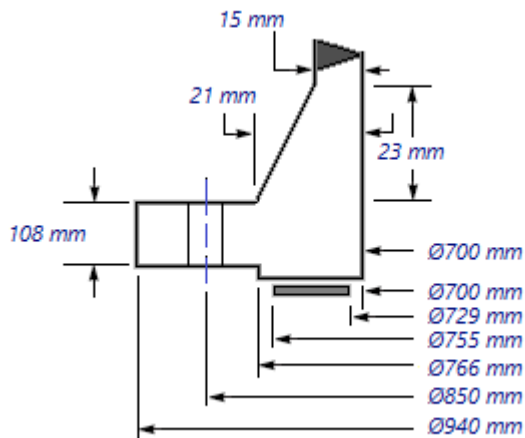
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Flg Calc [Int P]: FLANGE

Flng: 8 9:43am Aug 12,2026

Flange Input Data Values**Description: FLANGE :****Chnl rear flang**

Flange Type	Integral Weld Neck		
Design Pressure	P	46.91	kgf/cm ²
Design Temperature		155	°C
Internal Corrosion Allowance	ci	1.5000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	700.000	mm
Flange Outside Diameter	A	940.000	mm
Flange Thickness	t	108.0000	mm
Thickness of Hub at Small End	go	15.0000	mm
Thickness of Hub at Large End	gl	21.0000	mm
Length of Hub	h	23.0000	mm
Flange Material		SA-266 4	
Flange Material UNS number		K03017	
Flange Allowable Stress At Temperature	Sfo	1407.13	kgf/cm ²
Flange Allowable Stress At Ambient	Sfa	1407.13	kgf/cm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	1753.82	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1753.82	kgf/cm ²
Diameter of Bolt Circle	C	850.000	mm
Nominal Bolt Diameter	a	31.7500	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		24	
Flange Face Outside Diameter	Fod	766.000	mm
Flange Face Inside Diameter	Fid	700.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	755.000	mm
Gasket Inside Diameter	Gi	729.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	703.07	kgf/cm ²
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm





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 Flg Calc [Int P]: FLANGE Flng: 8 9:43am Aug 12,2026

ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (46.91 \cdot (700.0/2 + 1.5)) / (1407.13 \cdot 1.0 - 0.6 \cdot 46.91) + Ca$$

$$= 13.4567 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.13 \cdot 1.0 \cdot 13.5) / (351.5 + 0.6 \cdot 13.5)$$

$$= 52.826 \text{ kgf/cm}^2$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.13 \cdot 1.0 \cdot 15.0) / (350.0 + 0.6 \cdot 15.0)$$

$$= 58.794 \text{ kgf/cm}^2$$

Corroded Flange Thickness, $t_c = T - c_i$	106.500 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	703.000 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	19.500 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	13.500 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	54.000 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	13.000 mm
Basic Gasket Width, $bo = N/2$	6.500 mm
Effective Gasket Width, $b = Cb \cdot \sqrt{bo}$	6.425 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	742.151 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 742.1509^2 \cdot 46.908$$

$$= 202918.188 \text{ kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.4246 \cdot \pi \cdot 742.1509 \cdot 3.0 \cdot 46.91$$

$$= 42158.320 \text{ kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 703.0^2 \cdot 46.908$$

$$= 182073.719 \text{ kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 202918 - 182074$$

$$= 20844.467 \text{ kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(202918 + 42158 + 0, 0)$$

$$= 245076.500 \text{ kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 703.07 \cdot 6.4246 \cdot \pi \cdot 742.151 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 105313.305 \text{ kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G)/Sb, Wg/Sa)$$

$$= \max((245077 + 0 + 4 \cdot 0/742.15)/1754, 105313/1754)$$

$$= 139.739 \text{ cm}^2$$



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ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 31.75 + 6 \cdot 106.5/(3.0 + 0.5)$$

$$= 246.071 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 850.0 \cdot \sin(\pi/24)$$

$$= 110.947 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs/(2a + t)}, 1)$$

$$= \max(\sqrt{110.947/(2 \cdot 31.75 + 106.5)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	139.739	143.845	
Radial Distance between Hub and Bolts:	44.450	54.000	
Radial Distance between Bolts and Edge:	31.750	45.000	
Circ. Spacing between the Bolts:	71.450	110.947	246.071

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 143.845 \cdot 1753.82 / (703.07 \cdot \pi(755.0 + 729.0))$$

$$= 7.697 \text{ mm}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 16 mm
Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa(Am + Ab) / 2$$

$$= 1753.82(139.7389 + 143.8449) / 2$$

$$= 248677.05 \text{ kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 245077 - 202918$$

$$= 42158.32 \text{ kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (850.0 - 742.1509) / 2$$

$$= 53.9246 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((850.0 - 703.0) / 2 + 53.9246)$$

$$= 63.7123 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (850.0 - 703.0 - 19.5) / 2$$

$$= 63.7500 \text{ mm}$$

Summary of Moments for Internal Pressure: (kgf-m.)

Loading	Force	Distance	Bolt Corr	Moment
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Flg Calc [Int P]: FLANGE

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End Pressure, Md	182074.	63.7500	1.0000	11607.
Face Pressure, Mt	20844.	63.7123	1.0000	1328.
Gasket Load, Mg	42158.	53.9246	1.0000	2273.
Gasket Seating, Matm	248677.	53.9246	1.0000	13410.

Total Moment for Operation, Mo 15209. kgf-m.

Total Moment for Gasket seating, Mg 13410. kgf-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 97.419 mm

Hub Ratio, h/h0 = HL / H0 0.236

Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.444

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.895

Factor V per 4.16.5 0.406

Factor f per 4.16.5 1.280

Stress factor from Table 4.16.4 [K]:

= A / B

= 940.0/703.0

= 1.337

Stress factor from Table 4.16.4 [Y]:

= (0.66845 + (5.7169(K² • log10(K)) / (K² - 1)) / (K - 1))

= (0.66845 + (5.7169(1.337² • log10(1.337)) / (1.337² - 1)) / (1.337 - 1))

= 6.838

Stress factor from Table 4.16.4 [T]:

= (K²(1 + 8.55246 • log10(K)) - 1) / ((1.0472 + 1.9448 • K²) • (K - 1))

= (1.337²(1 + 8.55246 • log10(1.337)) - 1) / ((1.0472 + 1.9448 • 1.337²) • (1.337 - 1))

= 1.781

Stress factor from Table 4.16.4 [U]:

= (K²(1 + 8.55246 • log10(K)) - 1) / (1.36136(K² - 1) • (K - 1))

= (1.337²(1 + 8.55246 • log10(1.337)) - 1) / (1.36136(1.337² - 1) • (1.337 - 1))

= 7.514

Stress factor from Table 4.16.4 [Z]:

= (K² + 1) / (K² - 1)

= (1.337² + 1) / (1.337² - 1)

= 3.538

Stress factor from Table 4.16.4 for integral flanges [d]:

= U • g0² • h0 / V

= 7.514 • 13.5² • 97.419 / 0.406

= 328955.469 mm³

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h0

= 0.895 / 97.419

= 0.009 1/mm

Stress factor from Table 4.16.4 [L]:

= (t • e + 1) / T + t³/d

= (106.5 • 0.009 + 1) / 1.781 + 106.5³ / 328955.469

= 4.783

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= (f • Mop / Bcor) / (L • g1²)

= (1.2797 • 15209 / 703.0) / (4.7828 • 19.5²)

= 1522.26 kgf/cm²

Longitudinal Hub Stress, Seating [SHa]:



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Flg Calc [Int P]: FLANGE

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$$= (f \cdot \text{Matm} / \text{Bcor}) / (L \cdot g^{1^2})$$

$$= (1.2797 \cdot 13410 / 703.0) / (4.7828 \cdot 19.5^2)$$

$$= 1342.22 \text{ kgf/cm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot \text{Mop} / \text{Bcor}) / (L \cdot t^2)$$

$$= (2.305 \cdot 15209 / 703.0) / (4.7828 \cdot 106.5^2)$$

$$= 91.92 \text{ kgf/cm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot \text{Matm} / \text{Bcor}) / (L \cdot t^2)$$

$$= (2.305 \cdot 13410 / 703.0) / (4.7828 \cdot 106.5^2)$$

$$= 81.05 \text{ kgf/cm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot \text{Mo} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRo}$$

$$= (6.8379 \cdot 15209 / (106.5^2 \cdot 703.0)) - 3.5384 \cdot 92$$

$$= 979.00 \text{ kgf/cm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot \text{Matm} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRa}$$

$$= (6.8379 \cdot 13410 / (106.5^2 \cdot 703.0)) - 3.5384 \cdot 81$$

$$= 863.21 \text{ kgf/cm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (\text{SHo} + \max(\text{SRo}, \text{STo})) / 2$$

$$= (1522 + \max(92, 979)) / 2$$

$$= 1250.63 \text{ kgf/cm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (\text{SHa} + \max(\text{SRa}, \text{STa})) / 2$$

$$= (1342 + \max(81, 863)) / 2$$

$$= 1102.71 \text{ kgf/cm}^2$$

Bolt Stress, Operating [BSo]:

$$= \text{Wm1} / \text{Ab}$$

$$= 245077 / 143.8449$$

$$= 1703.76 \text{ kgf/cm}^2$$

Bolt Stress, Seating [BSa]:

$$= (\text{Wm2} / \text{Ab})$$

$$= (105313 / 143.8449)$$

$$= 732.13 \text{ kgf/cm}^2$$

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	1522.26	2110.70	1342.22	2110.70
Radial Flange	91.92	1407.13	81.05	1407.13
Tangential Flange	979.00	1407.13	863.21	1407.13
Maximum Average	1250.63	1407.13	1102.71	1407.13
Bolting	1703.76	1753.82	732.13	1753.82

Minimum Required Flange Thickness 101.879 mm

Estimated M.A.W.P. (Operating) 48 kgf/cm²
 Estimated Finished Weight of Flange at given Thk. 266.0 kg
 Estimated Unfinished Weight of Forging at given Thk 313.9 kg

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot \text{Ma} / \text{Bsc} \cdot \text{Cnvfac} \cdot V / (\lambda \cdot \text{Eamb} \cdot \text{go}^2 \cdot \text{ho} \cdot \text{Ki})$$



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$$= 52.14 \cdot 13409.8 / 1.0 \cdot 100000.023 \cdot 0.406 / (4.152 \cdot 2067026 \cdot 13.5^2 \cdot 97.419 \cdot 0.3)$$

$$= 0.620 \quad (\text{should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 15208.6 / 1.0 \cdot 100000.023 \cdot 0.406 / (4.152 \cdot 1986602 \cdot 13.5^2 \cdot 97.419 \cdot 0.3)$$

$$= 0.732 \quad (\text{should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 13409.8 / 1.0 \cdot 100000.023 \cdot 0.406 / (4.783 \cdot 2067026 \cdot 13.5^2 \cdot 97.419 \cdot 0.3)$$

$$= 0.538 \quad (\text{should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 15208.6 / 1.0 \cdot 100000.023 \cdot 0.406 / (4.783 \cdot 1986602 \cdot 13.5^2 \cdot 97.419 \cdot 0.3)$$

$$= 0.635 \quad (\text{should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.752, Temperature Reduction per Fig. UCS 66.1 = 14 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-16 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-30 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Note: UCS-66(b)(c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kgf/cm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	1473.55	2110.70	1286.40	2110.70
Radial Flange	101.79	1407.13	88.86	1407.13
Tangential Flange	913.19	1407.13	797.20	1407.13
Maximum Average	1193.37	1407.13	1041.80	1407.13
Bolting	1703.76	1753.82	732.13	1753.82

Minimum Req'd Thickness, ambient, no Corrosion: 99.035 mm

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Flg Calc [Int P]: New Flange

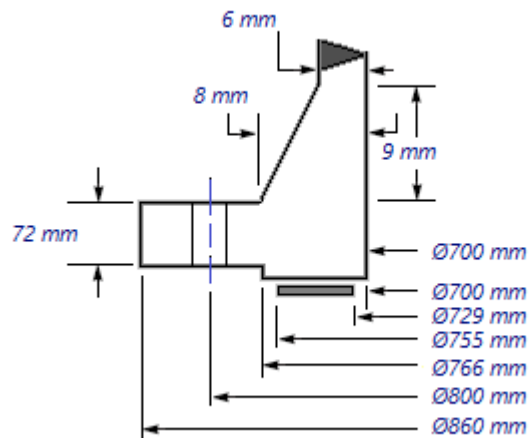
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Flange Input Data Values

Description: New Flange :

Shell flange

Flange Type	Integral Weld Neck		
Design Pressure	P	10.20	kgf/cm ²
Design Temperature		155	°C
Internal Corrosion Allowance	ci	0.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	700.000	mm
Flange Outside Diameter	A	860.000	mm
Flange Thickness	t	72.0000	mm
Thickness of Hub at Small End	go	6.0000	mm
Thickness of Hub at Large End	g1	8.0000	mm
Length of Hub	h	9.0000	mm
Flange Material	SA-965 F304L		
Flange Material UNS number	S30403		
Flange Allowable Stress At Temperature	Sfo	891.31	kgf/cm ²
Flange Allowable Stress At Ambient	Sfa	1172.61	kgf/cm ²
Bolt Material	SA-193 B7		
Bolt Allowable Stress At Temperature	Sb	1753.82	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1753.82	kgf/cm ²
Diameter of Bolt Circle	C	800.000	mm
Nominal Bolt Diameter	a	25.4000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts	20		
Flange Face Outside Diameter	Fod	766.000	mm
Flange Face Inside Diameter	Fid	700.000	mm
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	755.000	mm
Gasket Inside Diameter	Gi	729.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	703.07	kgf/cm ²
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm





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Flg Calc [Int P]: New Flange

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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (10.2 \cdot (700.0/2 + 0.0)) / (891.31 \cdot 1.0 - 0.6 \cdot 10.2) + Ca$$

$$= 4.0320 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (891.31 \cdot 1.0 \cdot 6.0) / (350.0 + 0.6 \cdot 6.0)$$

$$= 15.124 \text{ kgf/cm}^2$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1172.61 \cdot 1.0 \cdot 6.0) / (350.0 + 0.6 \cdot 6.0)$$

$$= 19.897 \text{ kgf/cm}^2$$

Corroded Flange ID,	Bcor = B+2*ci	700.000 mm
Corroded Large Hub,	g1Cor = g1-ci	8.000 mm
Corroded Small Hub,	g0Cor = go-ci	6.000 mm
Code R Dimension,	R = (C-B)/2 - g1	42.000 mm
Gasket Contact Width,	N = (Go - Gi)/2	13.000 mm
Basic Gasket Width,	bo = N/2	6.500 mm
Effective Gasket Width,	b = Cb*sqrt(bo)	6.425 mm
Gasket Reaction Diameter,	G = Go - 2 * b	742.151 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 742.1509^2 \cdot 10.197$$

$$= 44112.652 \text{ kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.4246 \cdot \pi \cdot 742.1509 \cdot 3.0 \cdot 10.2$$

$$= 9164.853 \text{ kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 700.0^2 \cdot 10.1974$$

$$= 39244.145 \text{ kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 44113 - 39244$$

$$= 4868.506 \text{ kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(44113 + 9165 + 0, 0)$$

$$= 53277.504 \text{ kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 703.07 \cdot 6.4246 \cdot \pi \cdot 742.151 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 105313.305 \text{ kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((53278 + 0 + 4 \cdot 0 / 742.15) / 1754, 105313 / 1754)$$

$$= 60.048 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmx]:

$$= 2a + 6t/(m + 0.5)$$



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Flg Calc [Int P]: New Flange

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$$= 2 \cdot 25.4 + 6 \cdot 72.0 / (3.0 + 0.5)$$

$$= 174.229 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 800.0 \cdot \sin(\pi / 20)$$

$$= 125.148 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{(Bs / (2a + t))}, 1)$$

$$= \max(\sqrt{(125.148 / (2 \cdot 25.4 + 72.0))}, 1)$$

$$= 1.0095$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	60.048	71.097	
Radial Distance between Hub and Bolts:	34.925	42.000	
Radial Distance between Bolts and Edge:	26.987	30.000	
Circ. Spacing between the Bolts:	57.150	125.148	174.229

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 71.097 \cdot 1753.82 / (703.07 \cdot \pi(755.0 + 729.0))$$

$$= 3.804 \text{ mm}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 16 mm
Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa(Am + Ab) / 2$$

$$= 1753.82(60.0481 + 71.0966) / 2$$

$$= 115001.90 \text{ kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 53278 - 44113$$

$$= 9164.85 \text{ kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (800.0 - 742.1509) / 2$$

$$= 28.9246 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((800.0 - 700.0) / 2 + 28.9246)$$

$$= 39.4623 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (800.0 - 700.0 - 8.0) / 2$$

$$= 46.0000 \text{ mm}$$

Summary of Moments for Internal Pressure: (kgf-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	39244.	46.0000	1.0095	1822.
Face Pressure, Mt	4869.	39.4623	1.0095	194.



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Flg Calc [Int P]: New Flange Flng: 9 9:43am Aug 12, 2026

Gasket Load, Mg	9165.	28.9246	1.0095	268.
Gasket Seating, Matm	115002.	28.9246	1.0095	3358.

Total Moment for Operation, Mo 2284. kgf-m.
Total Moment for Gasket seating, Mg 3358. kgf-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 64.807 mm
Hub Ratio, h/h0 = HL / H0 0.139
Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.333

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.907
Factor V per 4.16.5 0.468
Factor f per 4.16.5 1.369

Stress factor from Table 4.16.4 [K]:

= A / B
= 860.0/700.0
= 1.229

Stress factor from Table 4.16.4 [Y]:

= (0.66845 + (5.7169(K² • log10(K))) / (K² - 1))/(K - 1)
= (0.66845 + (5.7169(1.229² • log10(1.229)))/(1.229² - 1))/(1.229 - 1)
= 9.550

Stress factor from Table 4.16.4 [T]:

= (K²(1 + 8.55246 • log10(K)) - 1)/((1.0472 + 1.9448 • K²) • (K - 1))
= (1.229²(1 + 8.55246 • log10(1.229)) - 1)/((1.0472 + 1.9448 • 1.229²) • (1.229 - 1))
= 1.827

Stress factor from Table 4.16.4 [U]:

= (K²(1 + 8.55246 • log10(K)) - 1)/(1.36136(K² - 1) • (K - 1))
= (1.229²(1 + 8.55246 • log10(1.229)) - 1)/(1.36136(1.229² - 1) • (1.229 - 1))
= 10.495

Stress factor from Table 4.16.4 [Z]:

= (K² + 1)/(K² - 1)
= (1.229² + 1)/(1.229² - 1)
= 4.926

Stress factor from Table 4.16.4 for integral flanges [d]:

= U • g0² • h0 / V
= 10.495 • 6.0² • 64.807/0.468
= 52263.027 mm³

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h0
= 0.907/64.807
= 0.014 1/mm

Stress factor from Table 4.16.4 [L]:

= (t • e + 1) / T + t³/d
= (72.0 • 0.014 + 1)/1.827 + 72.0³/52263.027
= 8.240

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= (f • Mop / Bcor) / (L • g1²)
= (1.3691•2284/700.0)/(8.2404•8.0²)
= 847.05 kgf/cm²

Longitudinal Hub Stress, Seating [SHa]:

= (f • Matm / Bcor) / (L • g1²)
= (1.3691•3358/700.0)/(8.2404•8.0²)



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Flg Calc [Int P]: New Flange

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$$= 1245.38 \text{ kgf/cm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (2.3436 \cdot 2284 / 700.0) / (8.2404 \cdot 72.0^2)$$

$$= 17.90 \text{ kgf/cm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (2.3436 \cdot 3358 / 700.0) / (8.2404 \cdot 72.0^2)$$

$$= 26.32 \text{ kgf/cm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (9.5501 \cdot 2284 / (72.0^2 \cdot 700.0)) - 4.9263 \cdot 18$$

$$= 512.90 \text{ kgf/cm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (9.5501 \cdot 3358 / (72.0^2 \cdot 700.0)) - 4.9263 \cdot 26$$

$$= 754.10 \text{ kgf/cm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (847 + \max(18, 513)) / 2$$

$$= 679.97 \text{ kgf/cm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHA + \max(SRa, STa)) / 2$$

$$= (1245 + \max(26, 754)) / 2$$

$$= 999.74 \text{ kgf/cm}^2$$

Bolt Stress, Operating [BSo]:

$$= Wm1 / Ab$$

$$= 53278 / 71.0966$$

$$= 749.37 \text{ kgf/cm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (105313 / 71.0966)$$

$$= 1481.27 \text{ kgf/cm}^2$$

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	847.05	1336.96	1245.38	1758.92
Radial Flange	17.90	891.31	26.32	1172.61
Tangential Flange	512.90	891.31	754.10	1172.61
Maximum Average	679.97	891.31	999.74	1172.61
Bolting	749.37	1753.82	1481.27	1753.82

Minimum Required Flange Thickness

67.640 mm

Estimated M.A.W.P. (Operating)

13 kgf/cm²

Estimated Finished Weight of Flange at given Thk.

114.4 kg

Estimated Unfinished Weight of Forging at given Thk

127.5 kg

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 3419.3 / 1.0279 \cdot 100000.023 \cdot 0.468 / (6.986 \cdot 1989688 \cdot 6.0^2 \cdot 64.807 \cdot 0.3)$$

$$= 0.835 \text{ (should be } \leq 1)$$



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Flg Calc [Int P]: New Flange

Flng: 9 9:43am Aug 12, 2026

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 2325.6 / 1.0279 \cdot 100000.023 \cdot 0.468 / (6.986 \cdot 1893659 \cdot 6.0^2 \cdot 64.807 \cdot 0.3)$$

$$= 0.596 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 3358.0 / 1.0095 \cdot 100000.023 \cdot 0.468 / (8.24 \cdot 1989688 \cdot 6.0^2 \cdot 64.807 \cdot 0.3)$$

$$= 0.708 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 2284.0 / 1.0095 \cdot 100000.023 \cdot 0.468 / (8.24 \cdot 1893659 \cdot 6.0^2 \cdot 64.807 \cdot 0.3)$$

$$= 0.506 \text{ (should be } \leq 1)$$

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	847.05	1758.92	1245.38	1758.92
Radial Flange	17.90	1172.61	26.32	1172.61
Tangential Flange	512.90	1172.61	754.10	1172.61
Maximum Average	679.97	1172.61	999.74	1172.61
Bolting	749.37	1753.82	1481.27	1753.82

Minimum Req'd Thickness, ambient, no Corrosion: 67.640 mm

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Internal Pressure Calculations: Step: 6 9:43am Aug 12,2026

Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd kgf/cm ²	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) kgf/cm ²	Element#
Chnl rear cover		46.908	73	1.5	940	1407.1	1
Chnl rear flang		46.908	90	1.5	700	1407.1	2
Chnl. rear		46.909	18	1.5	700	1407.1	3
Shell flange		10.197	6	...	700	891.31	4
Shell		10.267	10	...	700	1167.5	5
shell head		10.267	6	...	700	1167.5	6

Element Required Thickness and MAWP :

From	To	Design Pressure kgf/cm ²	M.A.W.P. Corroded kgf/cm ²	M.A.P. New & Cold kgf/cm ²	Minimum Thickness mm	Required Thickness mm	Element#
Chnl rear cover		46.908	48.2861	48.2863	95	90.4536	1
Chnl rear flang		46.908	48.2864	48.2864	108	101.879	2
Chnl. rear		46.908	64.2428	70.2007	18	13.457	3
Shell flange		10.1974	13.3667	17.5854	72	67.6402	4
Shell		10.1974	32.7258	32.9385	10	3.09427	5
shell head		10.1974	16.9178	17.0618	5.1	3.08065	6

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

Shell Side MAWP = 13 kgf/cm²
 Shell Side MAPnc = 17 kgf/cm²
 Channel Side MAWP = 48 kgf/cm²
 Channel Side MAPnc = 48 kgf/cm²

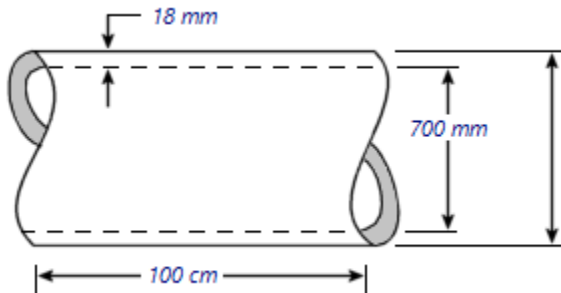
Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Chnl. rear

Cylindrical Shell Design Information:			
Design Pressure P	46.909 kgf/cm ²	Int. Design Temperature	155 °C
Material	SA-516 70	Ext. Design Temperature	40 °C
Length	100.000 cm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	1.5000 mm	Allowable Stress (ope) S	1407.1 kgf/cm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	1407.1 kgf/cm ²
Inside Diameter Di	700.000 mm	Actual Stress	1027.5 kgf/cm ²
Spec. Min. Thk t,e, or sa	18.0000 mm	Spec. Nominal Thk.	18.000 mm
Surface Area	23122.1 cm ²	SG of Contents	0.015
Element Volume	384846.0 cm ³	Weight of Contents	5.8 kgf
Empty Weight	314.7 kgf	Operating Weight	320.4 kgf
		Comp. All. Ope. Stress	1458.3 kgf/cm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (46.909 \cdot 351.5) / (1407.13 \cdot 1.0 - 0.6 \cdot 46.909) \\
 &= 11.9570 + 1.5000 = 13.4570 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$\begin{aligned}
 &\text{Less Operating Hydrostatic Head Pressure of } 0.001 \text{ kgf/cm}^2 \\
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (1407.13 \cdot 1.0 \cdot 16.5) / (351.5 + 0.6 \cdot 16.5) \\
 &= 64 - 0 = 64 \text{ kgf/cm}^2
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (1407.13 \cdot 1.0 \cdot 18.0) / (350.0 + 0.6 \cdot 18.0) \\
 &= 70 \text{ kgf/cm}^2
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (46.909 \cdot (351.5 + 0.6 \cdot 16.5)) / (1.0 \cdot 16.5) \\
 &= 1027.451 \text{ kgf/cm}^2
 \end{aligned}$$



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Element: Chnl. rear

Step: 7 9:43am Aug 12,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 2.507 %

Minimum Design Metal Temperature Results:

Govrn. thk, $t_g = 18.0$, $t_r = 12.315$, $c = 1.5$ mm, $E^* = 1.0$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.746$, Temp. Reduction = 14 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C

Min Metal Temp. at Required thickness (UCS 66.1) -25 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

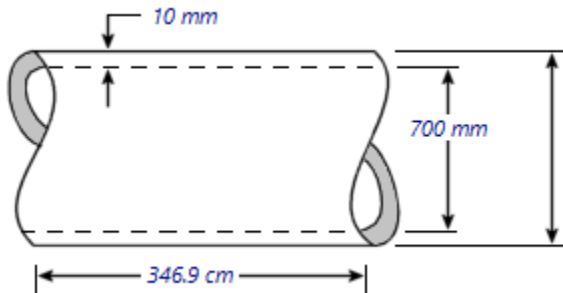
MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	48.286 kgf/cm ²
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	11 °C
Governing Thk.	18.000 mm	Basic MDMT	-11 °C
Stress/Thk. Ratio	0.746	Temperature Difference	14 °C
UCS-66(c) red.	No	Computed Min. Temp.	-25 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell

Cylindrical Shell Design Information:			
Design Pressure P	10.267 kgf/cm ²	Int. Design Temperature	155 °C
Material	SA-240 304L	Ext. Design Temperature	40 °C
Length	346.900 cm	External Pressure Chart	HA-3
		UNS Number	S30403
Int. Corr. All. c	0.0000 mm	Allowable Stress (ope) S	1167.5 kgf/cm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	1172.6 kgf/cm ²
Inside Diameter Di	700.000 mm	Actual Stress	365.5 kgf/cm ²
Spec. Min. Thk t,e, or sa	10.0000 mm	Spec. Nominal Thk.	10.000 mm
Surface Area	78466.9 cm ²	SG of Contents	1.000
Element Volume	773515.0 cm ³	Weight of Contents	773.2 kgf
Empty Weight	621.1 kgf	Operating Weight	1394.3 kgf
		Comp. All. Ope. Stress	687.8 kgf/cm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (10.267 \cdot 350.0) / (1167.53 \cdot 1.0 - 0.6 \cdot 10.267) \\
 &= 3.0943 + 0.0000 = 3.0943 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$\begin{aligned}
 &\text{Less Operating Hydrostatic Head Pressure of } 0.070 \text{ kgf/cm}^2 \\
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (1167.53 \cdot 1.0 \cdot 10.0) / (350.0 + 0.6 \cdot 10.0) \\
 &= 33 - 0.1 = 33 \text{ kgf/cm}^2
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (1172.61 \cdot 1.0 \cdot 10.0) / (350.0 + 0.6 \cdot 10.0) \\
 &= 33 \text{ kgf/cm}^2
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (10.267 \cdot (350.0 + 0.6 \cdot 10.0)) / (1.0 \cdot 10.0) \\
 &= 365.518 \text{ kgf/cm}^2
 \end{aligned}$$



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Element: Shell

Step: 8 9:43am Aug 12,2026

% Elongation per Table UG-79-1 $(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 1.408 %

Note: Please Check Requirements of Table UHA-44 for Elongation limits.

SA-240 304L, Min Metal Temp without impact per UHA-51: -196 °C

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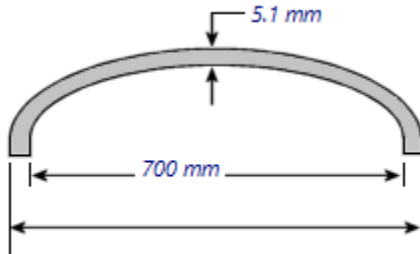
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Element: shell head

Step: 9 9:43am Aug 12,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: shell head

Elliptical Head Design Information:			
Design Pressure P	10.267 kgf/cm ²	Int. Design Temperature	155 °C
Material	SA-240 304L	Ext. Design Temperature	40 °C
Length	5.000 cm	External Pressure Chart	HA-3
		UNS Number	S30403
Int. Corr. All. c	0.0000 mm	Allowable Stress (ope) S	1167.5 kgf/cm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	1172.6 kgf/cm ²
Inside Diameter Di	700.000 mm	Actual Stress	705.7 kgf/cm ²
Spec. Min. Thk t,e, or sa	5.1000 mm	Spec. Nominal Thk.	6.000 mm
Surface Area	6613.6 cm ²	SG of Contents	1.000
Element Volume	64141.0 cm ³	Weight of Contents	64.1 kgf
Empty Weight	34.1 kgf	Operating Weight	98.2 kgf
		Comp. All. Ope. Stress	687.8 kgf/cm ²
		Aspect Ratio	2.00
Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 8.750 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 4.375 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (10.267 \cdot 700.0 \cdot 1.0) / (2 \cdot 1167.53 \cdot 1.0 - 0.2 \cdot 10.267)$$

$$= 3.0806 + 0.0000 = 3.0806 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.070 kgf/cm²

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 1167.53 \cdot 1.0 \cdot 5.1) / (1.0 \cdot 700.0 + 0.2 \cdot 5.1)$$

$$= 17 - 0.1 = 17 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 1172.61 \cdot 1.0 \cdot 5.1) / (1.0 \cdot 700.0 + 0.2 \cdot 5.1)$$

$$= 17 \text{ kgf/cm}^2$$



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Element: shell head

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Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned} &= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t) \\ &= (10.267 \cdot (1.0 \cdot 700.0 + 0.2 \cdot 5.1)) / (2 \cdot 1.0 \cdot 5.1) \\ &= 705.650 \text{ kgf/cm}^2 \end{aligned}$$

Straight Flange Required Thickness:

$$\begin{aligned} &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \quad \text{per UG-27 (c)(1)} \\ &= (10.267 \cdot 350.0) / (1167.53 \cdot 1.0 - 0.6 \cdot 10.267) + 0.0 \\ &= 3.094 \text{ mm} \end{aligned}$$

Straight Flange Maximum Allowable Working Pressure:

[Less Operating Hydrostatic Head Pressure of 0.070 kgf/cm²](#)

$$\begin{aligned} &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \quad \text{per UG-27 (c)(1)} \\ &= (1167.53 \cdot 1.0 \cdot 6.0) / (350.0 + 0.6 \cdot 6.0) \\ &= 20 - 0.1 = 20 \text{ kgf/cm}^2 \end{aligned}$$

% Elongation per Table UG-79-1 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 3.689 %

Note: Please Check Requirements of Table UHA-44 for Elongation limits.

SA-240 304L, Min Metal Temp without impact per UHA-51: -196 °C

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 8.750 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 4.375 mm

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Shop Pressure Test Results: Step: 10 9:43am Aug 12,2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	17.377 kgf/cm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	13.257 kgf/cm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	22.086 kgf/cm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	14.703 kgf/cm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	14.582 kgf/cm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	14.582 kgf/cm ²
App 27-4	= M.A.W.P.	13.367 kgf/cm ²

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	62.772 kgf/cm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	60.980 kgf/cm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	62.678 kgf/cm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	53.115 kgf/cm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	67.078 kgf/cm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	67.078 kgf/cm ²
App 27-4	= M.A.W.P.	48.286 kgf/cm ²

Horizontal Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (kgf/cm²):

From To	Stress	Allowable	Ratio	Pressure
Chnl. rear	1376.4	2386.0	0.577	62.84
Shell	621.1	1560.1	0.398	17.45
shell head	1199.1	1560.1	0.769	17.45

Stress ratios for Nozzle and Pad Materials (kgf/cm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N1- Channel inl	Nozzle	1407.13	1407.13	1.000
N2B- channel ou	Nozzle	1407.13	1407.13	1.000
N4B- Shell outl	Nozzle	1172.61	891.31	1.316
N4B- Shell outl	Pad	1407.13	1407.13	1.000
K1- Pressure re	Nozzle	1172.61	891.31	1.316
K1- Pressure re	Pad	1407.13	1407.13	1.000
N3B- Shell inle	Nozzle	1172.61	891.31	1.316
N3B- Shell inle	Pad	1407.13	1407.13	1.000

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (kgf/cm²):

Description	Ambient	Operating	Ratio
Chnl rear cover	1407.13	1407.13	1.000
Chnl rear flang	1407.13	1407.13	1.000
Chnl. rear	1407.13	1407.13	1.000
Shell flange	1172.61	891.31	1.316
Shell	1172.61	1167.53	1.004
shell head	1172.61	1167.53	1.004

Minimum

1.000



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Shop Pressure Test Results: Step: 10 9:43am Aug 12,2026

Stress ratios for Exchanger Materials (kgf/cm²):

Description	Ambient	Operating	Ratio
Tube Material	1172.61	891.31	1.316
Tubesheet Material	1172.61	891.31	1.316

Minimum 1.316

Hoop Stress in Nozzle Wall during Pressure Test (kgf/cm²):

Description	Ambient	Operating	Ratio
N1- Channel inl	606.28	2294.24	0.264
N2B- channel ou	606.28	2294.24	0.264
N4B- Shell outl	107.93	1560.08	0.069
K1- Pressure re	107.93	1560.08	0.069
N3B- Shell inle	107.93	1560.08	0.069

Elements Suitable for Test Pressure.

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Field Pressure Test Results: Step: 11 9:43am Aug 12,2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	17.377 kgf/cm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	13.257 kgf/cm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	22.086 kgf/cm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	14.703 kgf/cm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	14.582 kgf/cm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	14.582 kgf/cm ²
App 27-4	= M.A.W.P.	13.367 kgf/cm ²

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	62.772 kgf/cm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	60.980 kgf/cm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	62.678 kgf/cm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	53.115 kgf/cm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	67.078 kgf/cm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	67.078 kgf/cm ²
App 27-4	= M.A.W.P.	48.286 kgf/cm ²

UG-99(b)(35), Test Pressure Calculation [Shell Side]:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 10.197 • 1.0

= 13.257 kgf/cm²

UG-99(b) Note 35, Test Pressure Calculation [Channel Side]:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 46.908 • 1.0

= 60.980 kgf/cm²

Horizontal Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (kgf/cm²):

From To	Stress	Allowable	Ratio	Pressure
Chnl. rear	1337.2	2386.0	0.560	61.05
Shell	474.4	1560.1	0.304	13.33
shell head	915.9	1560.1	0.587	13.33

Stress ratios for Nozzle and Pad Materials (kgf/cm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N1- Channel inl	Nozzle	1407.13	1407.13	1.000
N2B- channel ou	Nozzle	1407.13	1407.13	1.000
N4B- Shell outl	Nozzle	1172.61	891.31	1.316
N4B- Shell outl	Pad	1407.13	1407.13	1.000
K1- Pressure re	Nozzle	1172.61	891.31	1.316
K1- Pressure re	Pad	1407.13	1407.13	1.000
N3B- Shell inle	Nozzle	1172.61	891.31	1.316
N3B- Shell inle	Pad	1407.13	1407.13	1.000

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (kgf/cm²):



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Field Pressure Test Results: Step: 11 9:43am Aug 12,2026

Description	Ambient	Operating	Ratio
Chnl rear cover	1407.13	1407.13	1.000
Chnl rear flang	1407.13	1407.13	1.000
Chnl. rear	1407.13	1407.13	1.000
Shell flange	1172.61	891.31	1.316
Shell	1172.61	1167.53	1.004
shell head	1172.61	1167.53	1.004

Minimum 1.000

Stress ratios for Exchanger Materials (kgf/cm²):

Description	Ambient	Operating	Ratio
Tube Material	1172.61	891.31	1.316
Tubesheet Material	1172.61	891.31	1.316

Minimum 1.316

Hoop Stress in Nozzle Wall during Pressure Test (kgf/cm²):

Description	Ambient	Operating	Ratio
N1- Channel inl	589.00	2294.24	0.257
N2B- channel ou	589.00	2294.24	0.257
N4B- Shell outl	82.44	1560.08	0.053
K1- Pressure re	82.44	1560.08	0.053
N3B- Shell inle	82.44	1560.08	0.053

Elements Suitable for Test Pressure.

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ASME TS Calc: Pressure Test

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Input Echo, Tubesheet Number 1, Description: Fixed tubesheet

Shell Data:

Main Shell Description: Shell

Shell Test Pressure		13.26 kgf/cm ²
Shell Thickness	ts	10.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	700.000 mm
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-240 304L

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		S30403
Shell Allowable Stress at Temperature	Ss	1172.61 kgf/cm ²
Shell Allowable Stress at Ambient		1172.61 kgf/cm ²

Channel Data:

Channel Description: Chnl. rear

Channel Type:		Cylinder
Channel Test Pressure		60.98 kgf/cm ²
Channel Thickness	tc	18.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	700.000 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	1407.13 kgf/cm ²
Channel Allowable Stress at Ambient		1407.13 kgf/cm ²

Tube Data:

Number of Tube Holes	Nt	545
Tube Wall Thickness	et	2.0000 mm
Tube Outside Diameter	D	19.0000 mm
Total Straight Tube Length	Lt	3300.00 mm
Straight Tube Length (bet. inner tubsht faces) L		3195.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-213 TP304L
Tube Material UNS Number		S30403
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		1172.61 kgf/cm ²
Tube Allowable Stress At Ambient		1172.61 kgf/cm ²
Tube Yield Stress At design Temperature	Syt	1733.43 kgf/cm ²
Tube Pitch (Center to Center Spacing)	P	25.0000 mm
Tube Layout Pattern		Triangular

Fillet Weld Leg	af	2.0000 mm
Groove Weld Leg	ag	2.0000 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No

Radius to Outermost Tube Hole Center	ro	330.948 mm
Largest Center-to-Center Tube Distance	U1	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	3121.6 cm ²
Perimeter of Tube Layout	Cp	2058.4 mm



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Tubesheet Data:

Tubesheet TYPE: U-tube, Integral with Channel, Conf. e

Tubesheet Design Metal Temperature	T	21.11 °C
Tubesheet Material	SA-965 F304L	
Tubesheet Material UNS Number	S30403	
Tubesheet Allowable Stress at Temperature	S	1172.61 kgf/cm ²
Tubesheet Allowable Stress at Ambient	Tt	1172.61 kgf/cm ²
Thickness of Tubesheet	h	105.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	0.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	0.0000 mm
Tubesheet Outside Diameter	A	860.000 mm

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped?		Yes
Is the Tubesheet Flat on Tubeside?		Yes
Is the Tubesheet Flat on Shellside?		No
Step 1 Diameter on the Shellside	ds1	763.00 mm
Step 1 Depth on the Shellside	hs1	8.00 mm
Step 2 Diameter on the Shellside	ds2	697.00 mm
Step 2 Depth on the Shellside	hs2	8.00 mm
Calculated Tubesheet Diameter per Div 2, 4.18.4		860.00 mm

Note: Tubesheet diameter is now: 860.000 mm Div 2, 4.18.4.

Area of the Untubed Lanes	AL	0.0 cm²
Junction Stress Reduction option	Perform Plastic Calculation	
Perform Differential Pressure Design	No	
Run Multiple Load Cases	Yes	

Additional Data for Gasketed Tubesheets:

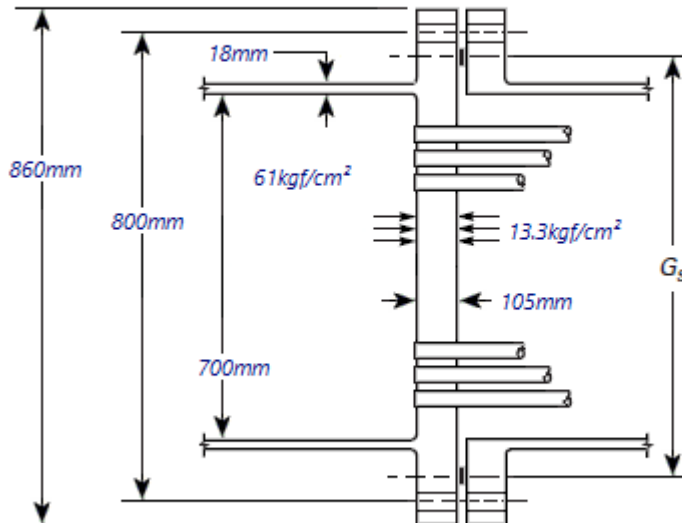
Tubesheet Gasket on which Side	Shell	
Flange Outside Diameter	A	860.000 mm
Flange Inside Diameter	B	700.000 mm
Flange Face Outside Diameter	Fod	766.000 mm
Flange Face Inside Diameter	Fid	700.000 mm
Gasket Outside Diameter	Go	755.000 mm
Gasket Inside Diameter	Gi	729.000 mm
Small end Hub thk.	g0	6.0000 mm
Large end Hub thk.	g1	8.0000 mm
Gasket Factor,	m	3.00
Gasket Design Seating Stress	y	703.07 kgf/cm ²
Flange Facing Sketch	Code Sketch 1a	
Gasket Thickness	tg	4.5000 mm
Full face Gasket Flange Option	Program Selects	

Bolting Information:

Diameter of Bolt Circle	C	800.000	mm
Nominal Bolt Diameter	dB	25.4000	mm
Thread Series	TEMA Thread Series		
Number of Bolts	n	20	
Bolt Material	SA-193	B7	
Bolt Allowable Stress At Temperature	Sb	1753.82	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1753.82	kgf/cm ²
Weld between Flange and Shell/Channel		0.0000	mm
Tubesheet Integral with		Channel	
Tubesheet Extended as Flange		Yes	
Thickness of Extended Portion of Tubesheet	Tf	89.0000	mm
Is Bolt Load Transferred to the Tubesheet		No	



Tubesheet Gasketed w/Shell and Integral w/Channel, Extended as a Flange



Configuration e:

Messages regarding the Tubesheet & Exchanger Input Data:

Warning - Bolt load is not transferred to tubesheet which is extended for bolting. Req. Tubesheet extension thk. can't be computed. Please check input.

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Material Group G

Elastic Mod. at Design Temperature	21.1 °C	0.19897E+07 kgf/cm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.19897E+07 kgf/cm²

Channel - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20670E+07 kgf/cm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm²

Tubes - TM-1 Material Group G

Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.19897E+07 kgf/cm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.19897E+07 kgf/cm²

TubeSheet - TM-1 Material Group G

Elastic Mod. at Design Temperature	21.1 °C	0.19897E+07 kgf/cm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.19897E+07 kgf/cm²

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

= $(P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)

= $(60.98 \cdot (19.0/2 - 0)) / (1172.61 \cdot 1.0 + 0.4 \cdot 60.98)$

= $0.4840 + 0.0000 = 0.4840$ mm



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Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart HA-3 at 21.11 °C
Elastic Modulus for Material 1968596.00 kgf/cm²

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.0000	19.00	3525.95	9.50	50.0000	0.0121884	946.27

EMAWP = (2.167/(D/T)-0.0833)*B = 137.0255 kgf/cm²

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.3979	19.00	3525.95	47.75	50.0000	0.0004824	474.80

EMAWP = (4*B)/(3*(D/T)) = (4 *474.8021)/(3 *47.7533) = 13.2571 kgf/cm²

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.4840 mm
Allowable Internal Pressure at Corroded thickness	269.57 kgf/cm ²
Required Internal Design Pressure	60.98 kgf/cm ²
Allowable External Pressure at Corroded thickness	137.03 kgf/cm ²
Required External Design Pressure	13.26 kgf/cm ²
Required Thickness due to Shell Side pressure	0.3979 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Gasketed Tubesheets:

ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	N = (Goc-Gic) / 2	13.000 mm
Basic Gasket Width,	b0 = N / 2	6.500 mm
Effective Gasket Width,	b = sqrt(b0) * 2.5	6.422 mm
Gasket Reaction Diameter,	G = Go-2.0*b	742.156 mm

Results for ASME U-tube Tubesheet Calculations for Configuration e:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

= 2 • ro + dt
= 2 • 330.948 + 19.0
= 680.896 mm

Basic ligament efficiency for shear [μ]:

= (p - dt) / p
= (25.0 - 19.0)/25.0
= 0.240

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

= Gs / Do (Configurations d, e, f)
= 742.1556/680.896
= 1.0900

Ratio [pc]:

= Dc / Do (Configurations a, e, f)
= 700.0/680.896
= 1.0281



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ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

$$= l_{tx} / h$$

$$= 0/105.0$$

$$= 0.000 \text{ (must be } 0 \leq \rho \leq 1 \text{)}$$

Effective tube hole Diameter [d*]:

$$= \max(dt - 2tt(E_t/E) \cdot (StT/S) \cdot (\rho); dt - 2tt)$$

$$= \max(19.0 - 2 \cdot 2.0 (1989688/1989688) \cdot$$

$$(1172/1172) \cdot (0); 19.0 - 2 \cdot 2.0)$$

$$= 19.0000 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{ (1 - 4 \cdot \min(A_l; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)) }$$

$$= 25.0 / \sqrt{ (1 - 4 \cdot \min(0; 4 \cdot 680.896 \cdot 25.0) / (\pi \cdot 680.896^2)) }$$

$$= 25.0000 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (25.0 - 19.0) / 25.0$$

$$= 0.2400$$

$$h/p = 4.200000 ; \mu^* = 0.240000 ; E^* = 405954 \text{ kgf/cm}^2$$

$$E^*/E = 0.204029 ; \nu^* = 0.407183$$

Note: As h/p (4.200) is > 2 , data values for $h/p = 2$ were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Intermediate parameters for Tubesheet Gasketed on the Shell Side:

β_s , k_s , λ_s , δ_s and ω_s are all = 0

Channel coefficient [β_c]:

$$= ((12(1 - \nu_c^2))^{1/4}) / \sqrt{ (D_c + t_c) t_c }$$

$$= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ (700.0 + 18.0) 18.0 }$$

$$= 0.0160 \text{ 1/mm}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_s^2))$$

$$= 0.016 \cdot 2067025 \cdot 18.0^3 / (6(1 - 0.3^2))$$

$$= 35304324.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta +$$

$$h \cdot \beta_c \cdot (Cm\Delta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot Cq\Delta)$$

$$= 6(700.0 + 18.0 / \cos(0)) \cdot 35304324 / 105.0^3 \cdot (1.0 +$$

$$105.0 \cdot 0.016 \cdot (1.0 + 1.0) / 2 + (105.0 \cdot 0.016)^2 / 2 \cdot 1.0)$$

$$= 537152.0 \text{ kgf/cm}^2$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot \nu_c / (2(D_c + t_c)))$$

$$= 700.0^2 / (4 \cdot 2067025 \cdot 18.0) \cdot (1 - 700.0 \cdot 0.3 / (2(700.0 + 18.0)))$$

$$= 0.28110E-02 \text{ mm/kgf/cm}^2$$

Channel coefficient [zetac]:

$$zetac = 0$$

Parameter [ω_c]:

$$= p_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + Cq\Delta \cdot h \cdot \beta_c^2) -$$

$$\zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2))$$



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$$= 1.028 \cdot 35304324 [0.002811 (1.0 + 0.016 \cdot 1.0 \cdot 105.0) - 0 (1.0/105.0 + 1.0 \cdot 0.016/2)]$$

$$= 4370.4482 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 860.0/680.896$$

$$= 1.2630$$

Coefficient [F]:

$$= (1 - v^*) / E^* \cdot (\lambda c + E \cdot \ln(K))$$

$$= (1 - 0.41) / 405954 \cdot (537152 + 1989688 \cdot \ln(1.26))$$

$$= 1.4629$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= D_o^2/16 \cdot [(p_s - 1) \cdot (p_s^2 + 1) \cdot P_s - (p_t - 1) \cdot (p_t^2 + 1) \cdot P_t]$$

$$= 680.896^2/16 \cdot [(1.09 - 1) \cdot (1.09^2 + 1) \cdot 13.257 - (1.028 - 1) \cdot (1.028^2 + 1) \cdot 60.98]$$

$$= -26356.4453 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= M_{ts} + (\omega c \cdot P_t) + W^* \cdot (C - G_s) / (2 \cdot \pi \cdot D_o)$$

$$= -26356 + (266511) + 69257 \cdot (800.0 - 742.156) / (2 \cdot \pi \cdot 680.896)$$

$$= 333796.5625 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= ((M^*) - D_o^2/32 \cdot F (P_s - P_t)) / (1 + F)$$

$$= ((333796.56) - 680.896^2/32 \cdot 1.463 (13.26 - 60.98)) / (1 + 1.46)$$

$$= 546221.6250 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= M_p + D_o^2/64 \cdot (3 + v^*) \cdot (P_s - P_t)$$

$$= 546221 + 680.896^2/64 \cdot (3 + 0.407) \cdot (13.26 - 60.98)$$

$$= -631689.1250 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(| M_p |; | M_o |)$$

$$= \max(| 546221 |; | -631689.12 |)$$

$$= 631689.1250 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu^* (h - h_g')^2)$$

$$= 6 \cdot 631689 / (0.24 (105.0 - 0)^2)$$

$$= 1432.4016 \text{ kgf/cm}^2$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot \beta_T \cdot S_y$$

$$= 2 \cdot 0.95 \cdot 1733.4$$

$$= 3293.51 \text{ kgf/cm}^2$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu^* (h - h_g')^2)$$

$$= 6 \cdot 602775 / (0.24 (67.7657 - 0)^2)$$

$$= 3281.5278 \text{ kgf/cm}^2$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$= h + C_{ats} + C_{atc}$$



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$$= 67.7657 + 0 + 0$$

$$= 67.7657 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$= \max(\text{HreqB}; \text{HreqS})$$

$$= \max(67.7657; 22.891)$$

$$= 67.7657 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [| Ps - Pt |]:

$$= | \text{Ps} - \text{Pt} |$$

$$= | 13.26 - 60.98 |$$

$$= 47.724 \text{ kgf/cm}^2$$

Shear Stress check [τ limit]:

$$= 4 \cdot \mu \cdot h / \text{Do} \cdot 0.8 \cdot \beta T \cdot \text{Sy}$$

$$= 4 \cdot 0.24 \cdot 105.0 / 680.9 \cdot 0.8 \cdot 0.95 \cdot 1733.426$$

$$= 195.029 \text{ kgf/cm}^2$$

| Ps - Pt | is $\leq$ the Shear Stress limit (τ limit). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$= 1 / (4 \cdot \mu \cdot (0.8 \cdot \beta T \cdot \text{Sy})) \cdot (4 \cdot \text{Ap} / \text{Cp}) \cdot | \text{Ps} - \text{Pt} | + \text{Cats} + \text{Catc}$$

$$= 1 / (4 \cdot 0.24 \cdot 1317.404) \cdot (4 \cdot 3121.6 / 2058.35) \cdot | 13.26 - 60.98 | + 0$$

$$= 22.8910 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Channel thickness of: 18.000 mm

Min. Channel len. adjacent to the tubesheet is: 202.049mm

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (19.0 - 2.0) 2.0$$

$$= 1.068 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 2.0 (19.0 - 2.0) 1646.755$$

$$= 1758.965 \text{ kgf}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > \text{ Ft)}$$

$$= 0.85 \cdot \pi \cdot 2.0 (19.0 + 0.67 \cdot 2.0) 1646.75$$

$$= 1758.9647 \text{ kgf}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 1646.755 / 1646.755$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > \text{ Ft)}$$

$$= 0.55 \cdot \pi \cdot 2.0 (19.0 + 0.67 \cdot 2.0) 1646.75$$

$$= 1157.5023 \text{ kgf}$$

Minimum Length of Weld [ar]:



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$$= 2 \left(\sqrt{((0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) \cdot fw \cdot fd) - 0.75 \cdot do} \right)$$

$$= 2 \left(\sqrt{((0.75 \cdot 19.0)^2 + 1.07 \cdot 2.0 \cdot (19.0 - 2.0) \cdot 1.0 \cdot 1.0) - 0.75 \cdot 19.0} \right)$$

$$= 2.448 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.2239 mm

Minimum Required Groove Weld Leg [agr]: 1.2239 mm

Tube-Tubesheet Jt allowable, 1758.96 is >= tube strength 1758.96 kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	1432.4 <=	3293.5 kgf/cm ²	Ok
Tubesheet shear stress	287.2 <=	1317.4 kgf/cm ²	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	67.766	105.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.224	2.000	Ok
Tube-Tubesheet Groove Weld Leg :	1.224	2.000	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend	Stresses Allwd	Shear	Allwd	Case Type	Pass/ Fail
D1uc	81.730	...	2071	3294	367	1317	Ps+Pt D1	Ok
D2uc	36.006	...	639	3294	80	1317	Ps+Pt D2	Ok
D3uc	67.766	...	1432	3294	287	1317	Ps+Pt D3	Ok
Max:	81.7300	... mm		0.629		0.279	(Str. Ratio)	

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

Case#	-- Integral Cyl. Shell	Reqd. Thk. Channel	Shell Stress	Stress Allwd	Channel Stress	Stress Allwd	Pass Fail
D1uc	...	...	...	...	...	3778	Ok
D2uc	...	...	...	...	...	3778	Ok
D3uc	...	...	...	...	...	3778	Ok



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Max

Note:

Because there was no net pressure differential across the tubesheet due to the specified pressures psd,min and ptd,min; load case D4 was not processed as the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	81.730	105.000	Ok
Tubesheet thickness flanged extension :	0.000	89.000	Ok
Tube thickness :	0.484	2.000	Ok
Tube-tubesheet fillet weld leg :	1.224	2.000	Ok
Tube-tubesheet groove weld leg :	1.224	2.000	Ok

Given Shell Thickness: 10.0000 mm

Given Channel Thickness: 18.0000 mm

Min Channel length of thk, (18.000) adj. to tubesheet: 202.049 mm

Note: This is a full strength Tube to Tubesheet Joint.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length cm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B kgf/cm ²
10	20	No Calc	...	93.5	No Calc	No Calc
20	30	No Calc	...	106.5	No Calc	No Calc
30	40	100	736	16.5	0.0033062	1158.04
40	50	No Calc	...	72	No Calc	No Calc
50	60	357.733	720	10	0.00039435	387.768
60	70	No Calc	710.2	5.1	0.00099737	617.924

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	Design Pressure kgf/cm ²	External M.A.W.P. kgf/cm ²
10	20	95	43.3827	...	No Calc
20	30	108	79.4512	...	No Calc
30	40	18	No Calc	...	34.6154
40	50	72	67.6402	...	No Calc
50	60	10	No Calc	...	7.18088
60	70	5.1	No Calc	...	4.9304

Minimum

5

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners cm	Allowable Length Bet. Stiffeners cm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	40	100	No Calc	No Calc	No Calc
40	50	No Calc	No Calc	No Calc	No Calc
50	60	357.733	No Calc	No Calc	No Calc
60	70	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg	Element ID Volume cm ³	Corroded Metal Wgt. kg	Corroded ID Volume cm ³	Extra due Misc % kg
10	20	510.965	...	510.965	...	51.0965
20	30	265.994	50771.8	262.638	50847.9	26.5994
30	40	314.68	384846	289.059	388152	31.468
40	50	114.423	31172.5	114.423	31172.5	11.4423
50	60	621.119	773515	621.119	773515	62.1119
60	70	34.073	64141	34.073	64141	3.4073
Total		1861	1304446.38	1832	1307828.12	186



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg	X Offset, Dtl. Cent. cm	Y Offset, Dtl. Cent. cm	Z Offset Dtl. Cent. cm	Description
30	Liqd	5.77009	50	...	...	Gas
30	Insl	13.2646	50	...	...	Hot insulation
30	Nozl	149.331	50	48.6525	...	N1- Channel inl
30	Nozl	149.331	50	-48.6525	...	N2B- channel ou
50	Sadl	120.157	62	52.5	...	Left Saddle
50	Sadl	120.157	311	52.5	...	Right saddle
50	Liqd	773.166	173.45	...	...	Water
50	Insl	24.3855	173.45	...	...	Hot insulation
50	Nozl	67.7938	30	43.4137	...	N4B- Shell outl
50	Nozl	50.1019	300	43.4137	...	K1- Pressure re
50	Nozl	50.1019	30	-43.4137	...	N3B- Shell inle
60	Liqd	64.1121	10.8333	...	...	Water
60	Insl	2.60648	11.25	...	...	Hot insulation
30	FTsh	395.292	105.25	...	...	Fixed tubesheet
30	Tube	1883.85	265	...	...	Tubes
40	Baf1	5.62451	35	...	...	Baffles
40	Baf1	5.62451	70	...	...	Baffles
40	Baf1	5.62451	105	...	...	Baffles
40	Baf1	5.62451	140	...	...	Baffles
40	Baf1	5.62451	175	...	...	Baffles
40	Baf1	5.62451	210	...	...	Baffles
40	Baf1	5.62451	245	...	...	Baffles
40	Baf1	5.62451	280	...	...	Baffles
40	Baf1	5.62451	315	...	...	Baffles

Total Weight of Each Detail Type:

Saddles	240.3
Liquid	843.0
Insulation	40.3
Nozzles	466.7
Exchanger Components	2329.8
Liquid in Tubes	5.5
Sum of the Detail Weights	3925.6 kg

Weight Summation Results: (kg)



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Element and Detail Weights:

Step: 13 9:43am Aug 12,2026

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	2047.4	2047.4	2047.4	2047.4	2047.4	2047.4
Saddles	240.3	240.3	240.3	240.3	240.3	240.3
Nozzles	466.7	466.7	466.7	466.7	466.7	466.7
Exchanger	2329.8	2329.8	2329.8	2329.8	2329.8	2329.8
Insulation	...	...	...	...	40.3	40.3 *
Ope. Liquid	...	...	...	...	...	843.0
Tube Ope Lqd	...	...	...	...	...	5.5
Test Liquid	...	1303.9	...	...	...	...
Tube Tst Lqd	...	367.7	...	...	...	...
Totals	5084.1	6755.7	5084.1	5084.1	5124.4	5972.9

Field Installation Options:

* Insulation installed after lifting.

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	5084.1 kg
Shop Test Wt.	- Fabricated weight + water (full)	6755.7 kg
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	5084.1 kg
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	5084.1 kg
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	5124.4 kg
Operating Wt.	- Empty weight + operating liq. uncorroded	5972.9 kg
Oper. Wt. + CA	- Corr wt. + operating liquid	5941.1 kg
Field Test Wt.	- Empty weight + test medium (full)	6767.5 kg

Exchanger Tube Data

Volume of Exchanger tubes :	367888.7 cm ³
Weight of Ope Liq in tubes :	5.5 kg
Weight of Water in tubes :	367.7 kg

Field Test weight is computed in the corroded condition.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	9745.22	...
20	30	6816.82	2880.84
30	40	23122.1	21991.2
40	50	4108.07	1781.28
50	60	78466.9	76287.3
60	70	6613.61	6411.09

Total	128872.8	109351.7 cm ²
Total	138.7	117.7 ft ²

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Nozzle Flange MAWP:

Step: 14 9:43am Aug 12,2026

Nozzle Flange MAWP Results: (kgf/cm² & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
N1- Channel in	91.62	104.01	155	600	GR 1.1	...	...	...	...
N2B- channel o	91.62	104.01	155	600	GR 1.1	...	...	...	...
N4B- Shell out	63.65	83.92	155	600	GR 2.3	...	...	...	...
K1- Pressure r	31.78	41.91	155	300	GR 2.3	...	...	...	...
N3B- Shell inl	31.78	41.91	155	300	GR 2.3	...	...	...	...

Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 31.776 kgf/cm²

Lowest Flange Pressure Rating was (Amb)[ShellSide]: 41.911 kgf/cm²

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 91.624 kgf/cm²

Lowest Flange Pressure Rating was (Amb)[TubeSide]: 104.014 kgf/cm²

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Warning:

There are nozzles in this model, but no flange MAWPs were de-rated. Be sure this is what you intended. There is a check box in the nozzle dialog that instructs PV Elite to perform the de-rating for each nozzle flange. See ASME VIII-1, UG-44(b) for more information.

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Wind Load Calculation:

Step: 15 9:43am Aug 12,2026

Wind Load Calculation Results:

Wind Load Calculations per India Std. IS 875 (PART 3):2015, Amd. 1&2 (2020):

Input Values:

Wind Design Code	IS 875 (PART 3):2015-2020
Basic Wind Speed for IS-875	180 km/hr
Wind Zone Number	5
Base Elevation	30 cm
Percent Wind for Hydrotest	33.0
Risk Factor	[k1] 1.0
Terrain Category	2
Equipment Class	Class A
Topography Factor	[k3] 1.0
Importance Factor	[k4] 1.15
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Use Gust Response Factor (Dynamic Analysis)	No

Calculated Values:

Vessel Length to Diameter ratio: 6.749

Design Wind Speed [Vz]:

= Basic Wind Speed • k1 • k2,1 • k3 • k4 (for the 1st element)

= 50.0 • 1.0 • 1.0 • 1.0 • 1.15

= 57.496 meters/s

Design Wind Pressure per 4.5.2 [pd]:

= max(0.7 • pz, Kd • Ka • Kc • 0.6 • Vz²)

= max(0.7 • 1983.503, 1.0 • 1.0 • 1.0 • 0.6 • 57.5²)

= 1983.5 N/m²

Element Wind Load (operating condition):

= Wind Area • Cf • pd

= 0.1 • 0.835 • 1983.503

= 177.5 N

From	Height m	k1	k2	k3	Vz m/sec	Cf
10	1.02	1.0000	1.0000	1.0000	57.50	0.8350
20	1.02	1.0000	1.0000	1.0000	57.50	0.8350
30	1.02	1.0000	1.0000	1.0000	57.50	0.8350
40	1.02	1.0000	1.0000	1.0000	57.50	0.8350
50	1.02	1.0000	1.0000	1.0000	57.50	0.8350
60	1.02	1.0000	1.0000	1.0000	57.50	0.8350

Wind Load Calculation:

From	To	Wind Height cm	Wind Diameter cm	Wind Area cm²	Wind Pressure kgf/m²	Element Wind Load kgf
10	20	102	112.8	1071.6	202.344	19.9082
20	30	102	84	1100.4	202.344	20.4433
30	40	102	99.12	9912	202.344	184.146
40	50	102	84	680.4	202.344	12.6405
50	60	102	92.4	32053.6	202.344	595.493
60	70	102	91.224	1680.69	202.344	31.2239



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Wind Load Calculation:

Step: 15 9:43am Aug 12,2026

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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 FileName : 401-E-30001B- Top- RP Page 61 of 186
 Earthquake Load Calculation: Step: 16 9:43am Aug 12,2026

Earthquake Load Calculation Results:

Seismic Analysis Results per IS 1893 (Part 4): 2024

Input Data:

Importance Factor as Entered by User I 2.0000
 Zone Number 2
 Soil Type Rocky or Hard Soil
 Response reduction factor (Force factor) R 2.0000
 Time Period of Vibration (program or user) 0.0303 Seconds
 Value of Sa/g from user Spectrum Data 0.0939
 Percent Damping 2.0 %
 Considering MCE for load determination No

Calculated results:

Horizontal Seismic Coefficient per 7.3.1 [Ah]:
 $= (Sa/g)_{ss} \cdot DampVal \cdot (I/R)$
 $= (0.0939 \cdot 1.0) \cdot 1.0$
 $= 0.094$

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Actual Time Period = 0.0303 secs
 Computed Sa/g = 0.0939

Note:

It was indicated to ignore the damping factor, assuming that the Site Spectrum specified is per the actual damping value of this vessel.
 The value of "DampVal" has been set to 1 (not used).

Earthquake Element Load, for the first Element:
 $= \text{Earthquake Weight} \cdot Ah \cdot \text{Scalar}$
 $= 716.58 \cdot 0.0939 \cdot 1.0$
 $= 67.3 \text{ kgf}$

Vertical load per 8.4:
 $= 2/3 \cdot Ah \cdot W \cdot \text{Scalar}$
 $= 2/3 \cdot 0.0939 \cdot 5733 \cdot 1.0$



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Earthquake Load Calculation:

Step: 16 9:43am Aug 12, 2026

= 358.7 kgf

Earthquake Load Calculation:

From	To	Earthquake Height cm	Earthquake Weight kgf	Element Ope Load kgf
10	20	47	716.577	67.2551
20	30	35	716.577	67.2551
30	40	35	716.577	67.2551
40	50	35	716.577	67.2551
50	Sad1	35	716.577	67.2551
Sad1	60	35	716.577	67.2551
50	60	35	716.577	67.2551
60	70	35	716.577	67.2551

The Earthquake Weight does not include the weight of the saddles.

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Center of Gravity Calculation: Step: 17 9:43am Aug 12,2026

Shop/Field Installation Options :

Insulation is installed in the Field after being lifted.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	329.050 cm
Center of Gravity of the Liquid	328.355 cm
Center of Gravity of the Insulation	248.055 cm
Center of Gravity of the Nozzles	138.146 cm
Center of Gravity of the Tubesheet(s)	128.750 cm
Center of Gravity of the Tubes	288.500 cm
Center of Gravity of the Baffles	309.450 cm
Center of Gravity of Bare Shell New and Cold	139.197 cm
Center of Gravity of Bare Shell Corroded	140.340 cm
Vessel CG in the Operating Condition	222.998 cm
Vessel CG in the Fabricated (Shop/Empty) Condition	204.279 cm
Vessel CG in the Test Condition	215.301 cm

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Saddle Calcs: Bundle Pull New: Step: 18 9:43am Aug 12,2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Bundle Pull, New Condition

Note:

Wear Pad Width (200.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

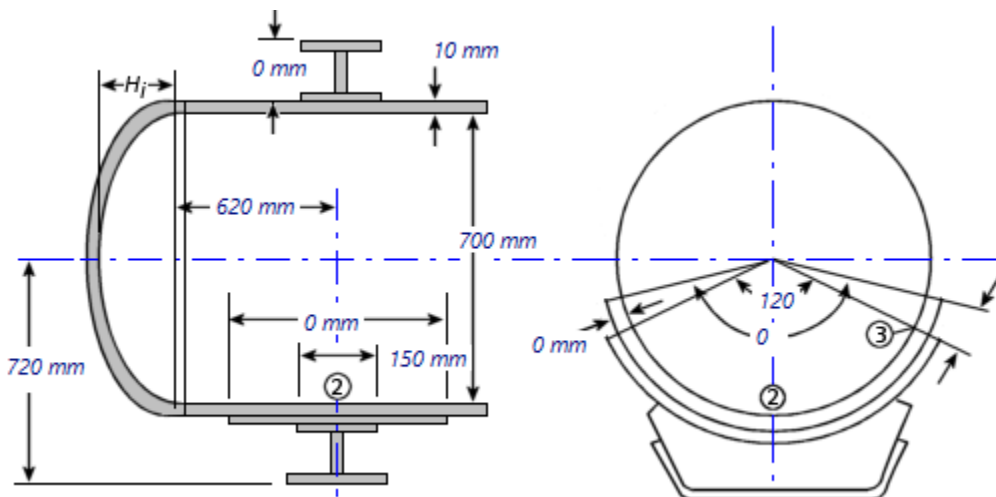
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(150.0 + 1.56 \cdot \sqrt{355.0 \cdot 10.0}, 2 \cdot 620.0)$$

$$= 242.9477 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	355.00	mm
Shell Thickness used in this Case	t	10.000	mm
Stiffened Vessel Length per 4.15.6	L	360.00	cm
Distance from Saddle to Vessel tangent	a1 or a	620.00	mm
Saddle Width	b1 or b	150.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		1167.53	kgf/cm ²
Head Allowable Stress used in Calculation		891.31	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	720.00	mm
Coefficient of Friction	mu	0.00	
Tube Bundle Pull out Force		2499.96	kgf
Tube Bundle Pull out Force Impact Factor		1.500	
Apply Bundle Pull out Force to Fixed Saddle only		Yes	



Results for Horizontal Vessel Analysis:

Saddle Force Q, + Bundle Pull, Force		4969.48	kgf
Pressure used in Analysis	P	0.000	kgf/cm ²

Horizontal Vessel Analysis Results:	Actual	Allowable
	kgf/cm ²	kgf/cm ²



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Saddle Calcs: Bundle Pull New: Step: 18 9:43am Aug 12,2026

Long. Stress at Top of Midspan	-37.34	-1560.08
Long. Stress at Top of Midspan	37.34	1167.53
Long. Stress at Bottom of Midspan	37.34	1167.53
Long. Stress at Top of Saddles	105.11	1167.53
Long. Stress at Bottom of Saddles	-58.26	-1560.08
Long. Stress at Bottom of Saddles	58.26	1167.53
Tangential Shear in Shell	107.43	710.89
Circ. Stress at Horn of Saddle	445.11	1751.29
Circ. Compressive Stress in Shell	15.55	1167.53

Intermediate Results: Saddle Reaction Q due to Bundle pull out Force:

Saddle Reaction Force due to Bundle Pull out Force [Fwl]:

= Bundle Pull Out Force • Impact Factor • B / Ls

= 2500 • 1.5 • 720.0/2490.0002

= 1084.3 kgf

Load Combination Results for Tube Bundle Pull Out [Q]:

= Saddle Load + max(Fwl, Fwt, Fsl, Fst)

= 3885 + max(1084, 0, 0, 0)

= 4969.5 kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 5089.64 kgf

Transverse Shear Load 0.00 kgf

Longitudinal Shear Load 3749.93 kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= min(2(ShellID/2 + t + WearPadThickness)sin(θ/2), 2•Rm)

= min(2(700.0/2 + 10.0 + 10.0)sin(120.0/2), 2•355.0)

= 640.859 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011

K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405

K9 = 0.2711 K10 = 0.0581 K1* = 0.1923

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= -Q•a [1 - (1- a/L + (Rm²-hm²)/(2a•L))/(1+(4hm)/3L)]

= -4969•62.0[1-(1-62.0/360.0+(35.5²-0.0²)/

(2•62.0•360.0))/(1+(4•0.0)/(3•360.0))]

= -443.6 kgf-m.

Moment per Equation 4.15.2 [M2]:

= Q•L/4((1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L))-4a/L)

= 4969•360/4((1+2(36²-0²)/(360²))/(1+(4•0)/

(3•360))-4•62/360)

= 1478.4 kgf-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ1]:

= P • Rm/(2t) - M2/(π•Rm²t)

= 0.0 • 355.0/(2•10.0) - 1478.4/(π•355.0²•10.0)

= -37.34 kgf/cm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ2]:

= P • Rm/(2t) + M2/(π • Rm² • t)

= 0.0 • 355.0/(2 • 10.0) + 1478.4/(π • 355.0² • 10.0)

= 37.34 kgf/cm²



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Saddle Calcs: Bundle Pull New: Step: 18 9:43am Aug 12,2026

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 t)$$

$$= 0.0 \cdot 355.0 / (2 \cdot 10.0) - 443.6 / (0.1066 \cdot \pi \cdot 355.0^2 \cdot 10.0)$$

$$= 105.11 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0.0 \cdot 355.0 / (2 \cdot 10.0) + 443.6 / (0.1923 \cdot \pi \cdot 355.0^2 \cdot 10.0)$$

$$= -58.26 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 4969(360.0 - 2 \cdot 62.0) / (360.0 + (4 \cdot 0.0 / 3))$$

$$= 3257.8 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 3257.77 / (355.0 \cdot 10.0)$$

$$= 107.43 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(355.0 \cdot 10.0)}$$

$$= 46.474 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.7603 \cdot 4969 \cdot 0.1 / (10.0(150.0 + 46.47 + 46.47))$$

$$= -15.55 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2)$$

$$= -4969 / (4 \cdot 10.0(150.0 + 46.474 + 46.474)) -$$

$$3 \cdot 0.0529 \cdot 4969 / (2 \cdot 10.0^2)$$

$$= -445.11 \text{ kgf/cm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(150.0 + 1.56 \cdot \sqrt{(355.0 \cdot 10.0)}, 2 \cdot 62.0)$$

$$= 242.95 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	930.0000	mm
Baseplate Thickness	Bpthk	15.0000	mm
Baseplate Width	Bpwid	180.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	15.0000	mm
Web Thickness	Webtk	15.0000	mm
Web Location	Webloc	Side	
Saddle Yield Stress	Sy	2651.1	kgf/cm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(350.0 \cdot 10.0)} + 200.0)^3) / 12$$

$$= 2.436 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 2.436 + 29.229(155.405 - 5.0)^2$$

$$= 6614.558 \text{ cm}^4$$



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	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	292.3	10.0	5.0	29.2	14614.5	0.661E+04
Wearplate	200.0	10.0	15.0	20.0	30000.0	0.394E+04
Web	15.0	335.0	187.5	50.2	942187.1	0.522E+04
BasePlate	180.0	15.0	362.5	27.0	978749.8	0.116E+05
Totals	...	...	...	126.5	1965551.5	0.274E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 1965552/126.479$$

$$= 155.405 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 4969.482$$

$$= 1011.3975 \text{ kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 10.4000 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 1590.6736 \text{ kgf/cm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 720.0 - 350.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 430.552 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 435.4600 \text{ kgf-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 24.7334 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 1767.4150 \text{ kgf/cm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(4969 + 120)180.0) / (2 \cdot 930.0 \cdot 1767.415)}$$

$$= 9.144 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (350.0 + 10.0 + 10.0)$$

$$= 640.859 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 640.8588 / (5 - 1)$$

$$= 160.215 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 160.2147 \cdot 180.0/2$$

$$= 144.193 \text{ cm}^2$$

Bearing Pressure [Bp]:



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$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 5089.639 / (930.0 \cdot 180.0)$$

$$= 3.040 \text{ kgf/cm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 144.2 \cdot 3.04$$

$$= 438.406 \text{ kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 20.25 + 12.016$$

$$= 32.266 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 438.4 / 32.2661$$

$$= 13.587 \text{ kgf/cm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	15.0	135.0	82.5	20.2	167062.5	466.
Web	80.1	15.0	7.5	12.0	9012.1	268.
Totals	...	...	...	32.3	176074.6	734.

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 150.0 - 15.0$$

$$= 135.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 176074.562 / 32.266$$

$$= 54.570 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(\text{ytot}, \text{Saddle Width} - \text{ytot})$$

$$= \max(54.57, 150.0 - 54.57)$$

$$= 95.430 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(734.0 / 32.266)}$$

$$= 47.695 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 720.0 - \cos(120.0/2) (350.0 + 10.0 + 10.0) - 15.0$$

$$= 520.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 2038903 / 2651.1)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 520.0 / 47.695$$

$$= 10.903$$

Bending Moment [Rm]:



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$$= F_l / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 3749.9 / (2 \cdot 930.0) \cdot 160.215 \cdot 520.0 / 2$$

$$= 83.982 \text{ kgf-m.}$$

Compressive Allowable, $KL/r < C_c$ (10.9026 < 123.2107) per AISC E2-1 [Sca]:

$$= (1 - (Kl/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (Kl/r) / (8 \cdot C_c) - (Kl/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (10.9)^2 / (2 \cdot 123.21^2)) 2651 / (5/3 + 3 \cdot (10.9) / (8 \cdot 123.21) - (10.9^3) / (8 \cdot 123.21^3))$$

$$= 1554 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 13.59 / 1553.6 + (83.98 \cdot 95.43 / 7339941) / 1767.42$$

$$= 0.071$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	15.0	135.0	82.5	20.2	167062.5	643.
Web	160.2	15.0	7.5	24.0	18024.2	287.
Totals	...	...	...	44.3	185086.6	930.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 185086.641 / 44.282$$

$$= 41.797 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(y_{tot}, \text{abs}(\text{Saddle Width} - y_{tot}))$$

$$= \max(41.797, \text{abs}(150.0 - 41.797))$$

$$= 108.203 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 720.0 - \cos(370.0 / 2) (15.0 + 0.0 + 0.0) - 0.0$$

$$= 335.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 930.2 / 44.282 }$$

$$= 45.833 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 335.0 / 45.833$$

$$= 7.309$$

Unit Force [Force,u]:

$$= F_l / (2 \cdot \text{Baseplate Length})$$

$$= 3749.934 / (2 \cdot 930.0)$$

$$= 2.016 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 2.016 \cdot 160.215 \cdot 335.0$$

$$= 108.208 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 108.208 / 85970.75$$

$$= 125.866 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ (7.3091 < 123.2107) per AISC E2-1 [Sca]:



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$$= (1 - (K_l r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (K_l r) / (8 \cdot C_c) - (K_l r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (7.31)^2 / (2 \cdot 123.21^2)) 2651 / (5/3 + 3 \cdot (7.31) / (8 \cdot 123.21) - (7.31^3) / (8 \cdot 123.21^3))$$

$$= 1567 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 19.5 / 1566.98 + (108.21 \cdot 108.203 / 930.228) / 1767.42$$

$$= 0.084$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1753.82 kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
Nominal Bolt Diameter	Bnd	36.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	1407.13 kgf/cm ²
Area Available in a Single Bolt	BltArea	6.0075 cm ²
Saddle Load QO (Weight)	QO	5089.6 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	3749.9 kgf
Maximum Transverse Force	Ft	0.0 kgf
Maximum Longitudinal Force	F1	3749.9 kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 3749.9 / (6.008 \cdot 2)$$

$$= 312.1 \text{ kgf/cm}^2. \text{ Must be less than } 1052.3 \text{ kgf/cm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 0.0 / (6.008 \cdot 2)$$

$$= 0.0 \text{ kgf/cm}^2. \text{ Must be less than } 1052.3 \text{ kgf/cm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (QO > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 3749.93 / (1052.29 \cdot 2.0)$$

$$= 1.7818 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot F_t + \text{Sum of X Moments}$$

$$= 720.0 \cdot 0.0 + 0.0$$

$$= 0.00 \text{ kgf-m.}$$

Eccentricity (e):

$$= R_{mom} / QO$$

$$= 0.0 / 5089.64$$

$$= 0.00 \text{ mm} < B_{plen} / 6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$



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= max[0.0, 1.7818, 0.0]
= 1.7818 cm²

ASME Horizontal Vessel Analysis: Bundle pullout, Right Saddle

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

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Saddle Calcs: Bundle Pull Cor: Step: 19 9:43am Aug 12,2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Bundle Pull, Corroded Condition

Note:

Wear Pad Width (200.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

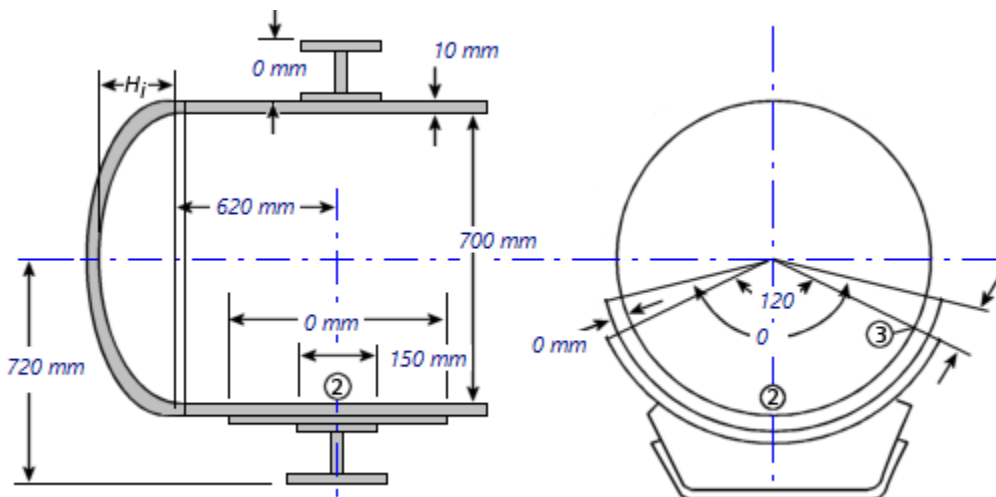
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(150.0 + 1.56 \cdot \sqrt{355.0 \cdot 10.0}, 2 \cdot 620.0)$$

$$= 242.9477 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	355.00	mm
Shell Thickness used in this Case	t	10.000	mm
Stiffened Vessel Length per 4.15.6	L	360.00	cm
Distance from Saddle to Vessel tangent	a1 or a	620.00	mm
Saddle Width	b1 or b	150.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		1167.53	kgf/cm ²
Head Allowable Stress used in Calculation		891.31	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	720.00	mm
Coefficient of Friction	mu	0.00	
Tube Bundle Pull out Force		2499.96	kgf
Tube Bundle Pull out Force Impact Factor		1.500	
Apply Bundle Pull out Force to Fixed Saddle only		Yes	



Results for Horizontal Vessel Analysis:

Saddle Force Q, + Bundle Pull, Force		4953.55	kgf
Pressure used in Analysis	P	0.000	kgf/cm ²

Horizontal Vessel Analysis Results:	Actual	Allowable
	kgf/cm ²	kgf/cm ²



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Long. Stress at Top of Midspan	-37.22	-1560.08
Long. Stress at Top of Midspan	37.22	1167.53
Long. Stress at Bottom of Midspan	37.22	1167.53
Long. Stress at Top of Saddles	104.77	1167.53
Long. Stress at Bottom of Saddles	-58.07	-1560.08
Long. Stress at Bottom of Saddles	58.07	1167.53
Tangential Shear in Shell	107.09	710.89
Circ. Stress at Horn of Saddle	443.68	1751.29
Circ. Compressive Stress in Shell	15.50	1167.53

Intermediate Results: Saddle Reaction Q due to Bundle pull out Force:

Saddle Reaction Force due to Bundle Pull out Force [Fwl]:

= Bundle Pull Out Force • Impact Factor • B / Ls

= 2500 • 1.5 • 720.0/2490.0002

= 1084.3 kgf

Load Combination Results for Tube Bundle Pull Out [Q]:

= Saddle Load + max(Fwl, Fwt, Fsl, Fst)

= 3869 + max(1084, 0, 0, 0)

= 4953.5 kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 5073.70 kgf

Transverse Shear Load 0.00 kgf

Longitudinal Shear Load 3749.93 kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= min(2(ShellID/2 + t + WearPadThickness)sin(θ/2), 2•Rm)

= min(2(700.0/2 + 10.0 + 10.0)sin(120.0/2), 2•355.0)

= 640.859 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011

K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405

K9 = 0.2711 K10 = 0.0581 K1* = 0.1923

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= -Q•a [1 - (1- a/L + (Rm²-hm²)/(2a•L))/(1+(4hm)/3L)]

= -4954•62.0[1-(1-62.0/360.0+(35.5²-0.0²)/

(2•62.0•360.0))/(1+(4•0.0)/(3•360.0))]

= -442.2 kgf-m.

Moment per Equation 4.15.2 [M2]:

= Q•L/4((1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L))-4a/L)

= 4954•360/4((1+2(36²-0²)/(360²))/(1+(4•0)/

(3•360))-4•62/360)

= 1473.7 kgf-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ1]:

= P • Rm/(2t) - M2/(π•Rm²t)

= 0.0 • 355.0/(2•10.0) - 1473.7/(π•355.0²•10.0)

= -37.22 kgf/cm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ2]:

= P • Rm/(2t) + M2/(π • Rm² • t)

= 0.0 • 355.0/(2 • 10.0) + 1473.7/(π • 355.0² • 10.0)

= 37.22 kgf/cm²



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Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 t)$$

$$= 0.0 \cdot 355.0 / (2 \cdot 10.0) - 442.2 / (0.1066 \cdot \pi \cdot 355.0^2 \cdot 10.0)$$

$$= 104.77 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0.0 \cdot 355.0 / (2 \cdot 10.0) + 442.2 / (0.1923 \cdot \pi \cdot 355.0^2 \cdot 10.0)$$

$$= -58.07 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 4954(360.0 - 2 \cdot 62.0) / (360.0 + (4 \cdot 0.0 / 3))$$

$$= 3247.3 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 3247.32 / (355.0 \cdot 10.0)$$

$$= 107.09 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(355.0 \cdot 10.0)}$$

$$= 46.474 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.7603 \cdot 4954 \cdot 0.1 / (10.0(150.0 + 46.47 + 46.47))$$

$$= -15.50 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2)$$

$$= -4954 / (4 \cdot 10.0(150.0 + 46.474 + 46.474)) -$$

$$3 \cdot 0.0529 \cdot 4954 / (2 \cdot 10.0^2)$$

$$= -443.68 \text{ kgf/cm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(150.0 + 1.56 \cdot \sqrt{(355.0 \cdot 10.0)}, 2 \cdot 62.0)$$

$$= 242.95 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	930.0000	mm
Baseplate Thickness	Bpthk	15.0000	mm
Baseplate Width	Bpwid	180.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	15.0000	mm
Web Thickness	Webtk	15.0000	mm
Web Location	Webloc	Side	
Saddle Yield Stress	Sy	2651.1	kgf/cm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(350.0 \cdot 10.0)} + 200.0)^3) / 12$$

$$= 2.436 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 2.436 + 29.229(155.405 - 5.0)^2$$

$$= 6614.558 \text{ cm}^4$$



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	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	292.3	10.0	5.0	29.2	14614.5	0.661E+04
Wearplate	200.0	10.0	15.0	20.0	30000.0	0.394E+04
Web	15.0	335.0	187.5	50.2	942187.1	0.522E+04
BasePlate	180.0	15.0	362.5	27.0	978749.8	0.116E+05
Totals	...	...	...	126.5	1965551.5	0.274E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 1965552/126.479$$

$$= 155.405 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 4953.545$$

$$= 1008.1539 \text{ kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 10.3666 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 1590.6736 \text{ kgf/cm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 720.0 - 350.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 430.552 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 434.0634 \text{ kgf-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 24.6541 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 1767.4150 \text{ kgf/cm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt[3]{(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt[3]{(3(4954 + 120)180.0) / (2 \cdot 930.0 \cdot 1767.415)}$$

$$= 9.129 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (350.0 + 10.0 + 10.0)$$

$$= 640.859 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 640.8588 / (5 - 1)$$

$$= 160.215 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 160.2147 \cdot 180.0/2$$

$$= 144.193 \text{ cm}^2$$

Bearing Pressure [Bp]:



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$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 5073.702 / (930.0 \cdot 180.0)$$

$$= 3.031 \text{ kgf/cm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 144.2 \cdot 3.03$$

$$= 437.033 \text{ kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 20.25 + 12.016$$

$$= 32.266 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 437.0 / 32.2661$$

$$= 13.545 \text{ kgf/cm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	15.0	135.0	82.5	20.2	167062.5	466.
Web	80.1	15.0	7.5	12.0	9012.1	268.
Totals	...	...	...	32.3	176074.6	734.

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 150.0 - 15.0$$

$$= 135.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 176074.562 / 32.266$$

$$= 54.570 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(\text{ytot}, \text{Saddle Width} - \text{ytot})$$

$$= \max(54.57, 150.0 - 54.57)$$

$$= 95.430 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(734.0 / 32.266)}$$

$$= 47.695 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 720.0 - \cos(120.0/2) (350.0 + 10.0 + 10.0) - 15.0$$

$$= 520.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 2038903 / 2651.1)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 520.0 / 47.695$$

$$= 10.903$$

Bending Moment [Rm]:



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$$= F_l / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 3749.9 / (2 \cdot 930.0) \cdot 160.215 \cdot 520.0 / 2$$

$$= 83.982 \text{ kgf-m.}$$

Compressive Allowable, $KL/r < C_c$ (10.9026 < 123.2107) per AISC E2-1 [Sca]:

$$= (1 - (Kl/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (Kl/r) / (8 \cdot C_c) - (Kl/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (10.9)^2 / (2 \cdot 123.21^2)) 2651 / (5/3 + 3 \cdot (10.9) / (8 \cdot 123.21) - (10.9^3) / (8 \cdot 123.21^3))$$

$$= 1554 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 13.54 / 1553.6 + (83.98 \cdot 95.43 / 7339941) / 1767.42$$

$$= 0.070$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	15.0	135.0	82.5	20.2	167062.5	643.
Web	160.2	15.0	7.5	24.0	18024.2	287.
Totals	...	...	...	44.3	185086.6	930.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 185086.641 / 44.282$$

$$= 41.797 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(y_{tot}, \text{abs}(\text{Saddle Width} - y_{tot}))$$

$$= \max(41.797, \text{abs}(150.0 - 41.797))$$

$$= 108.203 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 720.0 - \cos(370.0 / 2) (15.0 + 0.0 + 0.0) - 0.0$$

$$= 335.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 930.2 / 44.282 }$$

$$= 45.833 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 335.0 / 45.833$$

$$= 7.309$$

Unit Force [Force,u]:

$$= F_l / (2 \cdot \text{Baseplate Length})$$

$$= 3749.934 / (2 \cdot 930.0)$$

$$= 2.016 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 2.016 \cdot 160.215 \cdot 335.0$$

$$= 108.208 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 108.208 / 85970.75$$

$$= 125.866 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ (7.3091 < 123.2107) per AISC E2-1 [Sca]:



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$$= (1 - (Klr)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (Klr) / (8 \cdot Cc) - (Klr^3) / (8 \cdot Cc^3))$$

$$= (1 - (7.31)^2 / (2 \cdot 123.21^2)) 2651 /$$

$$(5/3 + 3 \cdot (7.31) / (8 \cdot 123.21) - (7.31^3) / (8 \cdot 123.21^3))$$

$$= 1567 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I) / Sba$$

$$= 19.44/1566.98 + (108.21 \cdot 108.203/930.228) / 1767.42$$

$$= 0.084$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1753.82 kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
Nominal Bolt Diameter	Bnd	36.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	1407.13 kgf/cm ²
Area Available in a Single Bolt	BltArea	6.0075 cm ²
Saddle Load QO (Weight)	QO	5073.7 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	3749.9 kgf
Maximum Transverse Force	Ft	0.0 kgf
Maximum Longitudinal Force	F1	3749.9 kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= F1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 3749.9 / (6.008 \cdot 2)$$

$$= 312.1 \text{ kgf/cm}^2. \text{ Must be less than } 1052.3 \text{ kgf/cm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 0.0 / (6.008 \cdot 2)$$

$$= 0.0 \text{ kgf/cm}^2. \text{ Must be less than } 1052.3 \text{ kgf/cm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (QO > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F1 / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 3749.93 / (1052.29 \cdot 2.0)$$

$$= 1.7818 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 720.0 \cdot 0.0 + 0.0$$

$$= 0.00 \text{ kgf-m.}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 0.0/5073.7$$

$$= 0.00 \text{ mm} < Bplen/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$



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= max[0.0, 1.7818, 0.0]
= 1.7818 cm²

ASME Horizontal Vessel Analysis: Bundle pullout, Right Saddle

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Note:

Wear Pad Width (200.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

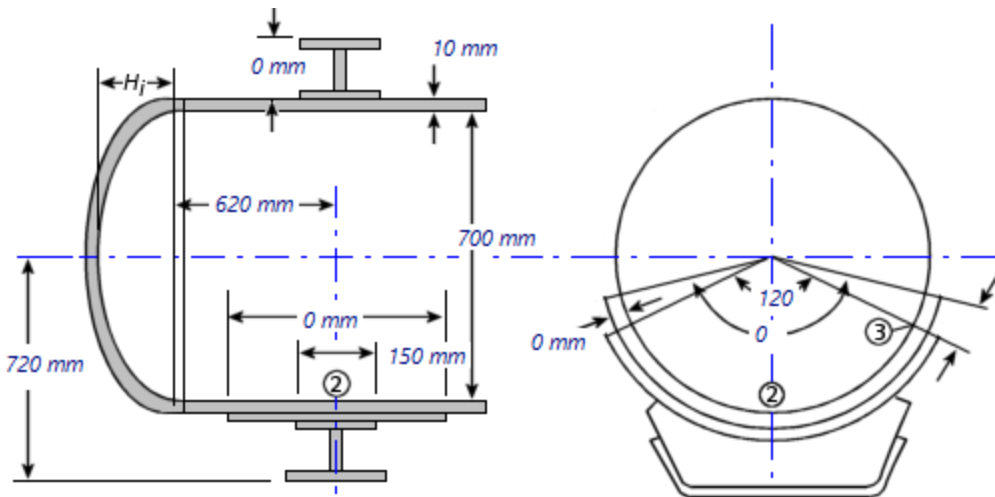
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(150.0 + 1.56 \cdot \sqrt{355.0 \cdot 10.0}, 2 \cdot 620.0)$$

$$= 242.9477 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	355.00	mm
Shell Thickness used in this Case	t	10.000	mm
Stiffened Vessel Length per 4.15.6	L	360.00	cm
Distance from Saddle to Vessel tangent	a1 or a	620.00	mm
Saddle Width	b1 or b	150.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		1167.53	kgf/cm ²
Head Allowable Stress used in Calculation		891.31	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	720.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		7065.39	kgf
Pressure used in Analysis	P	10.232	kgf/cm ²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	128.53	1167.53
Long. Stress at Bottom of Midspan	234.72	1167.53
Long. Stress at Top of Saddles	331.06	1167.53
Long. Stress at Bottom of Saddles	98.80	1167.53



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Tangential Shear in Shell	152.74	710.89
Circ. Stress at Horn of Saddle	632.83	1459.41
Circ. Compressive Stress in Shell	22.11	1167.53

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= Ftr(Ft/Num of Saddles + Z Force Load) \cdot B / E$$

$$= 3.0(863.9/2 + 0) \cdot 720.0/640.8588$$

$$= 1455.8 \text{ kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(Fl,wind; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(130.5, 0) \cdot 720.0/2490.0002$$

$$= 37.7 \text{ kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(Fl,eq; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(538.04, 0) \cdot 720.0/2490.0002$$

$$= 155.6 \text{ kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= Ftr(Ft/Num of Saddles + Z Force Load) \cdot B / E$$

$$= 3.0(538/2 + 0) \cdot 720.0/640.8588$$

$$= 906.7 \text{ kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(Fwl, Fwt, Fsl, Fst)$$

$$= 5610 + \max(38, 1456, 156, 907)$$

$$= 7065.4 \text{ kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(OD \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1984.306(0.0 + \pi/4(0.72 \cdot 1.2)^2)$$

$$= 1279.740 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 7185.54 kgf

Transverse Shear Load 431.93 kgf

Longitudinal Shear Load 538.04 kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot Rm)$$

$$= \min(2(700.0/2 + 10.0 + 10.0) \sin(120.0/2), 2 \cdot 355.0)$$

$$= 640.859 \text{ mm}$$

The Computed K values from Table 4.15.1:

$$K1 = 0.1066 \quad K2 = 1.1707 \quad K3 = 0.8799 \quad K4 = 0.4011$$

$$K5 = 0.7603 \quad K6 = 0.0529 \quad K7 = 0.0529 \quad K8 = 0.3405$$

$$K9 = 0.2711 \quad K10 = 0.0581 \quad K1* = 0.1923$$

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$$

$$= -7065 \cdot 62.0 [1 - (1 - 62.0/360.0 + (35.5^2 - 0.0^2) /$$

$$(2 \cdot 62.0 \cdot 360.0) / (1 + (4 \cdot 0.0) / (3 \cdot 360.0))]$$

$$= -630.8 \text{ kgf-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 ((1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L)$$



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$$= 7065 \cdot 360 / 4 \cdot ((1 + 2(36^2 - 0^2) / (360^2)) / (1 + (4 \cdot 0) / (3 \cdot 360))) - 4 \cdot 62 / 360$$

$$= 2102.0 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 10.232 \cdot 355.0 / (2 \cdot 10.0) - 2102.0 / (\pi \cdot 355.0^2 \cdot 10.0)$$

$$= 128.53 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 10.232 \cdot 355.0 / (2 \cdot 10.0) + 2102.0 / (\pi \cdot 355.0^2 \cdot 10.0)$$

$$= 234.72 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 10.232 \cdot 355.0 / (2 \cdot 10.0) - 630.8 / (0.1066 \cdot \pi \cdot 355.0^2 \cdot 10.0)$$

$$= 331.06 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 10.232 \cdot 355.0 / (2 \cdot 10.0) + 630.8 / (0.1923 \cdot \pi \cdot 355.0^2 \cdot 10.0)$$

$$= 98.80 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3))$$

$$= 7065(360.0 - 2 \cdot 62.0) / (360.0 + (4 \cdot 0.0 / 3))$$

$$= 4631.8 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 4631.75 / (355.0 \cdot 10.0)$$

$$= 152.74 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(355.0 \cdot 10.0)}$$

$$= 46.474 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.7603 \cdot 7065 \cdot 0.1 / (10.0(150.0 + 46.47 + 46.47))$$

$$= -22.11 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q / (2 \cdot t^2)$$

$$= -7065 / (4 \cdot 10.0(150.0 + 46.474 + 46.474)) - 3 \cdot 0.0529 \cdot 7065 / (2 \cdot 10.0^2)$$

$$= -632.83 \text{ kgf/cm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(150.0 + 1.56 \cdot \sqrt{(355.0 \cdot 10.0)}, 2 \cdot 62.0)$$

$$= 242.95 \text{ mm}$$

Distance between Saddle Supports [Ls]:

$$= 2490.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s(\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.000017 \cdot 2490.0(155.0 - 21.1)$$

$$= 5.547 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length Bplen 930.0000 mm



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Baseplate Thickness	Bpthk	15.0000	mm
Baseplate Width	Bpwid	180.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	15.0000	mm
Web Thickness	Webtk	15.0000	mm
Web Location	Webloc	Side	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	2671.5	kgf/cm²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t) + \text{WearPlateWidth}})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(350.0 \cdot 10.0) + 200.0})^3) / 12$$

$$= 2.436 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 2.436 + 29.229(155.405 - 5.0)^2$$

$$= 6614.558 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm²	AY mm³	I + AD^(2) cm⁴
Shell	292.3	10.0	5.0	29.2	14614.5	0.661E+04
Wearplate	200.0	10.0	15.0	20.0	30000.0	0.394E+04
Web	15.0	335.0	187.5	50.2	942187.1	0.522E+04
BasePlate	180.0	15.0	362.5	27.0	978749.8	0.116E+05
Totals	...	...	...	126.5	1965551.5	0.274E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 1965552 / 126.479$$

$$= 155.405 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 7065.387$$

$$= 1437.9594 \text{ kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 14.7862 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 1602.9094 \text{ kgf/cm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta} / 2) / (\text{theta} / 2 \text{ in radians})$$

$$= 720.0 - 350.0 \cdot \sin(120.0 / 2) / 1.0472$$

$$= 430.552 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 619.1174 \text{ kgf-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 35.1649 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 1781.0104 \text{ kgf/cm}^2$$



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Minimum Thickness of Baseplate per Moss:

$$= \sqrt{\frac{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}}{2 \cdot \text{BasePlateLength} \cdot \text{AllStress}}}$$

$$= \sqrt{\frac{3(7065 + 120)180.0}{2 \cdot 930.0 \cdot 1781.01}}$$

$$= 10.823 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (350.0 + 10.0 + 10.0)$$

$$= 640.859 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 640.8588 / (5 - 1)$$

$$= 160.215 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 160.2147 \cdot 180.0/2$$

$$= 144.193 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 7185.543 / (930.0 \cdot 180.0)$$

$$= 4.292 \text{ kgf/cm}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 144.2 \cdot 4.29$$

$$= 618.941 \text{ kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 20.25 + 12.016$$

$$= 32.266 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/\text{Ar}$$

$$= 618.9/32.2661$$

$$= 19.182 \text{ kgf/cm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	15.0	135.0	82.5	20.2	167062.5	466.
Web	80.1	15.0	7.5	12.0	9012.1	268.
Totals	...	...	...	32.3	176074.6	734.

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 150.0 - 15.0$$

$$= 135.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 176074.562/32.266$$

$$= 54.570 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(\text{ytot}, \text{Saddle Width} - \text{ytot})$$

$$= \max(54.57, 150.0 - 54.57)$$



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= 95.430 mm

Radius of Gyration [r]:
 $= \sqrt{\text{Total Inertia} / \text{Total Area}}$
 $= \sqrt{734.0/32.266}$
 = 47.695 mm

Length of Outer Rib [L]:
 $= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$
 $= 720.0 - \cos(120.0/2) (350.0 + 10.0 + 10.0) - 15.0$
 = 520.000 mm

Intermediate Term [Cc]:
 $= \sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 $= \sqrt{2 \cdot \pi^2 \cdot 2038903/2671.5}$
 = 122.739

Slenderness ratio [KL/r]:
 $= KL/r$
 $= 1 \cdot 520.0/47.695$
 = 10.903

Bending Moment [Rm]:
 $= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$
 $= 538.0 / (2 \cdot 930.0) \cdot 160.215 \cdot 520.0/2$
 = 12.050 kgf-m.

Compressive Allowable, $KL/r < Cc$ (10.9026 < 122.7395) per AISC E2-1 [Sca]:
 $= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 $= (1 - (10.9)^2 / (2 \cdot 122.74^2)) 2672 / (5/3 + 3 \cdot (10.9) / (8 \cdot 122.74) - (10.9)^3 / (8 \cdot 122.74^3))$
 = 1565 kgf/cm²

AISC Unity Check of Outside Ribs (must be ≤ 1)
 $= Sc/Sca + (Rm \cdot C1 / I) / Sba$
 $= 19.18/1565.38 + (12.05 \cdot 95.43/7339941) / 1781.01$
 = 0.021

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	15.0	135.0	82.5	20.2	167062.5	643.
Web	160.2	15.0	7.5	24.0	18024.2	287.
Totals	...	...	...	44.3	185086.6	930.

Distance to Centroid from Datum [ytot]:
 $= AY / A$
 $= 185086.641/44.282$
 = 41.797 mm

Distance to Centroid [C1]:
 $= \max(\text{ytot}, \text{abs}(\text{Saddle Width} - \text{ytot}))$
 $= \max(41.797, \text{abs}(150.0 - 41.797))$
 = 108.203 mm

Length of Inner Rib [L]:
 $= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$
 $= 720.0 - \cos(370.0/2) (15.0 + 0.0 + 0.0) - 0.0$
 = 335.000 mm

Radius of Gyration [r]:
 $= \sqrt{\text{Total Inertia} / \text{Total Area}}$
 $= \sqrt{930.2/44.282}$



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= 45.833 mm

Slenderness ratio [KL/r]:

= KL/r

= 1 • 335.0/45.833

= 7.309

Unit Force [Force,u]:

= FI / (2 • Baseplate Length)

= 538.041/(2 • 930.0)

= 0.289 kgf/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L

= 0.289 • 160.215 • 335.0

= 15.526 kgf-m.

Bending Stress due to Transverse Force and Weight Load [σB,base,c]:

= Bending Moment / Section Modulus

= 15.526/85970.75

= 18.059 kgf/cm²

Compressive Allowable, KL/r < Cc (7.3091 < 122.7395) per AISC E2-1 [Sca]:

= (1-(KL/r)²/(2•Cc²))Fy/(5/3+3•(KL/r)/(8•Cc)-(KL/r)³/(8•Cc³))

= (1-(7.31)²/(2 • 122.74²))2672/

(5/3+3•(7.31)/(8• 122.74)-(7.31³)/(8•122.74³))

= 1579 kgf/cm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= Sc/Sca + (Mbase,c • C1/I)/Sba

= 27.53/1578.94 + (15.53 • 108.203/930.228)/1781.01

= 0.028

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1399.95 kgf/cm²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
Nominal Bolt Diameter	Bnd	36.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	1784.41 kgf/cm²
Area Available in a Single Bolt	BlArea	6.0075 cm²
Saddle Load Q0 (Weight)	Q0	5729.7 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	155.6 kgf
Maximum Transverse Force	Ft	431.9 kgf
Maximum Longitudinal Force	F1	538.0 kgf
Saddle Bolted to Steel Foundation		No

Friction force due to thermal growth, [Ff]:

= 180.6 kgf

Shear Stress in a Single Bolt, Longitudinal Direction [τb,l]:

= max(Ff; F1) / (Bolt Area • Number of Bolts)

= 538.0/(6.008 • 2)

= 44.8 kgf/cm². Must be less than 1100.0 kgf/cm².

Shear Stress in a Single Bolt, Transverse Direction [τb,t]:

= Ft / (Bolt Area • Number of Bolts)

= 431.9/(6.008 • 2)

= 35.9 kgf/cm². Must be less than 1100.0 kgf/cm².



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Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = $F_l / (BoltShearAllowable \cdot Nbolts)$
 = $538.04 / (1099.96 \cdot 2.0)$
 = 0.2446 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = $B \cdot F_t + \text{Sum of X Moments}$
 = $720.0 \cdot 431.93 + 0.0$
 = 310.99 kgf-m.

Eccentricity (e):
 = $Rmom / QO$
 = $310.99 / 5729.74$
 = 54.28 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = max[Bltarearl, Bltarears, Bltareart]
 = max[0.0, 0.2446, 0.0]
 = 0.2446 cm²

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:
 Wear Pad Width (200.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 = $\min(b + 1.56 \cdot \sqrt{Rm \cdot t}, 2a)$
 = $\min(150.0 + 1.56 \cdot \sqrt{355.0 \cdot 10.0}, 2 \cdot 490.0)$
 = 242.9477 mm

Input and Calculated Values:

Vessel Mean Radius	Rm	355.00	mm
Shell Thickness used in this Case	t	10.000	mm
Stiffened Vessel Length per 4.15.6	L	360.00	cm
Distance from Saddle to Vessel tangent	a1 or a	490.00	mm
Saddle Width	b1 or b	150.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	17.76	cm
Shell Allowable Stress used in Calculation		1167.53	kgf/cm ²
Head Allowable Stress used in Calculation		1167.53	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	720.00	mm
Coefficient of Friction	mu	0.30	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case 1937.53 kgf



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Pressure used in Analysis P 10.232 kgf/cm²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	163.68	1167.53
Long. Stress at Bottom of Midspan	199.57	1167.53
Long. Stress at Top of Saddles	218.58	1167.53
Long. Stress at Bottom of Saddles	161.14	1167.53
Tangential Shear in Shell	43.63	710.89
Circ. Stress at Horn of Saddle	173.54	1459.41
Circ. Compressive Stress in Shell	6.06	1167.53

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (863.9/2 + 0) \cdot 720.0/640.8588$
 $= 1455.8 \text{ kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(130.5, 0) \cdot 720.0/2490.0002$
 $= 37.7 \text{ kgf}$

Saddle Reaction Force due to Earthquake F_l [Fsl]:
 $= \max(F_{l,eq}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(538.04, 0) \cdot 720.0/2490.0002$
 $= 155.6 \text{ kgf}$

Saddle Reaction Force due to Earthquake F_t [Fst]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (538/2 + 0) \cdot 720.0/640.8588$
 $= 906.7 \text{ kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 482 + \max(38, 1456, 156, 907)$
 $= 1937.5 \text{ kgf}$

Longitudinal Wind Force [F_l]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1984.306 (0.0 + \pi/4 (0.72 \cdot 1.2)^2)$
 $= 1279.740 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	2057.68	kgf
Transverse Shear Load	431.93	kgf
Longitudinal Shear Load	538.04	kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
 $= \min(2 (\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$
 $= \min(2 (700.0/2 + 10.0 + 10.0) \sin(120.0/2), 2 \cdot 355.0)$
 $= 640.859 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to R_m/2.



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Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 \cdot hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$$

$$= -1938 \cdot 49.0 [1 - (1 - 49.0/360.0 + (35.5^2 \cdot 17.755^2)/(2 \cdot 49.0 \cdot 360.0))/(1 + (4 \cdot 17.76)/(3 \cdot 360.0))]$$

$$= -156.0 \text{ kgf-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 ((1 + 2(Rm^2 \cdot hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 1938 \cdot 360/4 ((1 + 2(36^2 \cdot 18^2)/(360^2))/(1 + (4 \cdot 18)/(3 \cdot 360)) - 4 \cdot 49/360)$$

$$= 710.7 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 t)$$

$$= 10.232 \cdot 355.0/(2 \cdot 10.0) - 710.7/(\pi \cdot 355.0^2 \cdot 10.0)$$

$$= 163.68 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$$

$$= 10.232 \cdot 355.0/(2 \cdot 10.0) + 710.7/(\pi \cdot 355.0^2 \cdot 10.0)$$

$$= 199.57 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot Rm/(2t) - M1/(K1 \cdot \pi \cdot Rm^2 t)$$

$$= 10.232 \cdot 355.0/(2 \cdot 10.0) - 156.0/(0.1066 \cdot \pi \cdot 355.0^2 \cdot 10.0)$$

$$= 218.58 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot Rm/(2t) + M1/(K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 10.232 \cdot 355.0/(2 \cdot 10.0) + 156.0/(0.1923 \cdot \pi \cdot 355.0^2 \cdot 10.0)$$

$$= 161.14 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot hm/3))$$

$$= 1938(360.0 - 2 \cdot 49.0)/(360.0 + (4 \cdot 17.76/3))$$

$$= 1323.1 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.1707 \cdot 1323.08 / (355.0 \cdot 10.0)$$

$$= 43.63 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(355.0 \cdot 10.0)}$$

$$= 46.474 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.7603 \cdot 1938 \cdot 0.1 / (10.0(150.0 + 46.47 + 46.47))$$

$$= -6.06 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.22) [σ_7]:

$$= -Q/(4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q/(2 \cdot t^2)$$

$$= -1938/(4 \cdot 10.0(150.0 + 46.474 + 46.474)) - 3 \cdot 0.0529 \cdot 1938/(2 \cdot 10.0^2)$$

$$= -173.54 \text{ kgf/cm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(150.0 + 1.56 \cdot \sqrt{(355.0 \cdot 10.0)}, 2 \cdot 49.0)$$

$$= 242.95 \text{ mm}$$

Results for Vessel Ribs, Web and Base:



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Baseplate Length	Bplen	930.0000	mm
Baseplate Thickness	Bpthk	15.0000	mm
Baseplate Width	Bpwid	180.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	15.0000	mm
Web Thickness	Webtk	15.0000	mm
Web Location	Webloc	Side	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	2671.5	kgf/cm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(350.0 \cdot 10.0)} + 200.0)10.0^3)/12$$

$$= 2.436 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 2.436 + 29.229(155.405 - 5.0)^2$$

$$= 6614.558 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² (2) cm ⁴
Shell	292.3	10.0	5.0	29.2	14614.5	0.661E+04
Wearplate	200.0	10.0	15.0	20.0	30000.0	0.394E+04
Web	15.0	335.0	187.5	50.2	942187.1	0.522E+04
BasePlate	180.0	15.0	362.5	27.0	978749.8	0.116E+05
Totals	...	...	...	126.5	1965551.5	0.274E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 1965552/126.479$$

$$= 155.405 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0)\cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 1937.528$$

$$= 394.3289 \text{ kgf}$$

$$\text{Tension Stress, } St = (Fh/As) = 4.0548 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 1602.9094 \text{ kgf/cm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 720.0 - 350.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 430.552 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 169.7794 \text{ kgf-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 9.6432 \text{ kgf/cm}^2$$



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Allowed Stress, $S_a = 2/3 \cdot \text{Yield Str} = 1781.0104 \text{ kgf/cm}^2$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{\frac{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}}{2 \cdot \text{BasePlateLength} \cdot \text{AllStress}}}$$

$$= \sqrt{\frac{3(1938 + 120)180.0}{2 \cdot 930.0 \cdot 1781.01}}$$

$$= 5.792 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (350.0 + 10.0 + 10.0)$$

$$= 640.859 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 640.8588 / (5 - 1)$$

$$= 160.215 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 160.2147 \cdot 180.0 / 2$$

$$= 144.193 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 2057.684 / (930.0 \cdot 180.0)$$

$$= 1.229 \text{ kgf/cm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 144.2 \cdot 1.23$$

$$= 177.243 \text{ kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 20.25 + 12.016$$

$$= 32.266 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 177.2 / 32.2661$$

$$= 5.493 \text{ kgf/cm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	15.0	135.0	82.5	20.2	167062.5	466.
Web	80.1	15.0	7.5	12.0	9012.1	268.
Totals	...	...	...	32.3	176074.6	734.

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 150.0 - 15.0$$

$$= 135.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 176074.562 / 32.266$$

$$= 54.570 \text{ mm}$$

Distance to Centroid [C1]:



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= max(ytot, Saddle Width - ytot)
 = max(54.57, 150.0 - 54.57)
 = 95.430 mm

Radius of Gyration [r]:
 $= \sqrt{ \text{Total Inertia} / \text{Total Area} }$
 $= \sqrt{ 734.0 / 32.266 }$
 = 47.695 mm

Length of Outer Rib [L]:
 = Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 720.0 - cos(120.0/2)(350.0 + 10.0 + 10.0) - 15.0
 = 520.000 mm

Intermediate Term [Cc]:
 $= \sqrt{ 2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress} }$
 $= \sqrt{ 2 \cdot \pi^2 \cdot 2038903 / 2671.5 }$
 = 122.739

Slenderness ratio [KL/r]:
 = KL/r
 = 1 • 520.0 / 47.695
 = 10.903

Bending Moment [Rm]:
 = FI / (2 • Bplen) • e • L / 2
 = 538.0 / (2 • 930.0) • 160.215 • 520.0 / 2
 = 12.050 kgf-m.

Compressive Allowable, KL/r < Cc (10.9026 < 122.7395) per AISC E2-1 [Sca]:
 $= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 $= (1 - (10.9)^2 / (2 \cdot 122.74^2)) 2672 / (5/3 + 3 \cdot (10.9) / (8 \cdot 122.74) - (10.9)^3 / (8 \cdot 122.74^3))$
 = 1565 kgf/cm²

AISC Unity Check of Outside Ribs (must be ≤ 1)
 = Sc/Sca + (Rm • C1 / I) / Sba
 = 5.49 / 1565.38 + (12.05 • 95.43 / 7339941) / 1781.01
 = 0.012

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	15.0	135.0	82.5	20.2	167062.5	643.
Web	160.2	15.0	7.5	24.0	18024.2	287.
Totals	...	...	...	44.3	185086.6	930.

Distance to Centroid from Datum [ytot]:
 = AY / A
 = 185086.641 / 44.282
 = 41.797 mm

Distance to Centroid [C1]:
 = max(ytot, abs(Saddle Width - ytot))
 = max(41.797, abs(150.0 - 41.797))
 = 108.203 mm

Length of Inner Rib [L]:
 = Saddle Height - Outside Radius - Bpthk
 = 720.0 - cos(370.0/2)(15.0 + 0.0 + 0.0) - 0.0
 = 335.000 mm

Radius of Gyration [r]:



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$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{930.2/44.282}$$

$$= 45.833 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 335.0/45.833$$

$$= 7.309$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 538.041 / (2 \cdot 930.0)$$

$$= 0.289 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.289 \cdot 160.215 \cdot 335.0$$

$$= 15.526 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 15.526/85970.75$$

$$= 18.059 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ (7.3091 < 122.7395) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (7.31)^2 / (2 \cdot 122.74^2)) 2672 /$$

$$(5/3 + 3 \cdot (7.31) / (8 \cdot 122.74) - (7.31)^3 / (8 \cdot 122.74^3))$$

$$= 1579 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 7.88/1578.94 + (15.53 \cdot 108.203/930.228) / 1781.01$$

$$= 0.015$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1399.95 kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
Nominal Bolt Diameter	Bnd	36.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	1784.41 kgf/cm ²
Area Available in a Single Bolt	BltArea	6.0075 cm ²
Saddle Load Q0 (Weight)	Q0	601.9 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	155.6 kgf
Maximum Transverse Force	Ft	431.9 kgf
Maximum Longitudinal Force	F1	538.0 kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= FI / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 538.0 / (6.008 \cdot 2)$$

$$= 44.8 \text{ kgf/cm}^2. \text{ Must be less than } 1100.0 \text{ kgf/cm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 431.9 / (6.008 \cdot 2)$$

$$= 35.9 \text{ kgf/cm}^2. \text{ Must be less than } 1100.0 \text{ kgf/cm}^2.$$

Bolt Area Calculation per Dennis R. Moss



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Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= $F_l / (BoltShearAllowable \cdot Nbolts)$
= $538.04 / (1099.96 \cdot 2.0)$
= 0.2446 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= $B \cdot F_t + \text{Sum of X Moments}$
= $720.0 \cdot 431.93 + 0.0$
= 310.99 kgf-m.

Eccentricity (e):
= $Rmom / QO$
= $310.99 / 601.88$
= 516.69 mm > $Bplen / 6$ --> Uplift in Transverse direction

$f = Bplen / 2 - Edgedis$
= $930.0 / 2 - 50.01$
= 414.99 mm

Modular Ratio Of Steel/Concrete (n1):
= ES / EC
= $2074057 / 219499.77$
= 9.45

$K1 = 3 (e - 0.5 \cdot Bplen)$
= $3 (516.69 - 0.5 \cdot 930.0)$
= 155.07 mm

$K2 = 6 \cdot n1 \cdot At / Bpwid \cdot (f + e)$
= $6 \cdot 9.45 \cdot 6.01 / 180.0 \cdot (414.99 + 516.69)$
= 176289.33 mm²

$K3 = -K2 \cdot (0.5 \cdot Bplen + f)$
= $-176289.33 \cdot (0.5 \cdot 930.0 + 414.99)$
= -155132378.75 mm³

Iteratively Solving for the Effective Bearing Length:
 $Y^3 + K1 \cdot Y^2 + K2 \cdot Y + K3 = 0$
 $Y^3 + 155.07 \cdot Y^2 + 176289.33 \cdot Y + -0.2E+09 = 0$
 $Y = 394.57 \text{ mm}$

$Num = (Bplen / 2 - Y / 3 - e)$
= $(930.0 / 2 - 394.57 / 3 - 516.69)$
= -183.22

$Denom = (Bplen / 2 - Y / 3 + f)$
= $(930.0 / 2 - 394.57 / 3 + 414.99)$
= 748.46

Total Bolt Tension Force [Tforce]:
= $-QO \cdot Num / Denom$
= $-601.88 \cdot -183.22 / 748.46$
= 147.33 kgf

Bolt Area Required due to Transverse Load [Bltareart]:
= $Tforce / (Stba \cdot Nbt)$
= $147.33 / (1399.95 \cdot 1.0)$
= 0.1052 cm²



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Required Area of a Single Bolt [Bltarear]:
 = max[Bltarear1, Bltarears, Bltareart]
 = max[0.0, 0.2446, 0.1052]
 = 0.2446 cm²

Baseplate Thickness Calculation per D. Moss:

Bearing Pressure (fc)
 = $2(QO + Tforce) / (Y \cdot Bpwid)$
 = $2(601.88 + 147.33) / (394.57 \cdot 180.0)$
 = 2.11 kgf/cm²

Distance from Baseplate Edge to the Web [ADIST]:
 = $(Bplen - WebIngh) / 2$
 = $(930.0 - 879.2) / 2$
 = 25.4000 mm

Overturning Moment due To Bolt Tension [Mt]:
 = $Tforce \cdot Adist$
 = $147.33 \cdot 2.54$
 = 3.74 kgf-m.

Equivalent Bearing Pressure (f1):
 = $fc \cdot (Y - Adist) / Y$
 = $2.11 \cdot (394.57 - 25.4) / 394.57$
 = 1.97 kgf/cm²

Overturning Moment due to Bearing Pressure [Mc]:
 = $(Adist^2 \cdot Bpwid / 6) \cdot (f1 + 2 \cdot fc)$
 = $(25.4^2 \cdot 180.0 / 6) \cdot (1.97 + 2 \cdot 2.11)$
 = 1.20 kgf-m.

Baseplate Required Thickness [Treq]:
 = $\sqrt{(6 \cdot \max(Mt, Mc) / (Bpwid \cdot Sba))}$
 = $\sqrt{(6 \cdot \max(3.74, 1.2) / (180.0 \cdot 2676.61))}$
 = 2.1588 mm

Review notes about nozzle loadings and supports in the Warnings report.

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Saddle Calcs: Test Case:

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Note:

Wear Pad Width (200.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

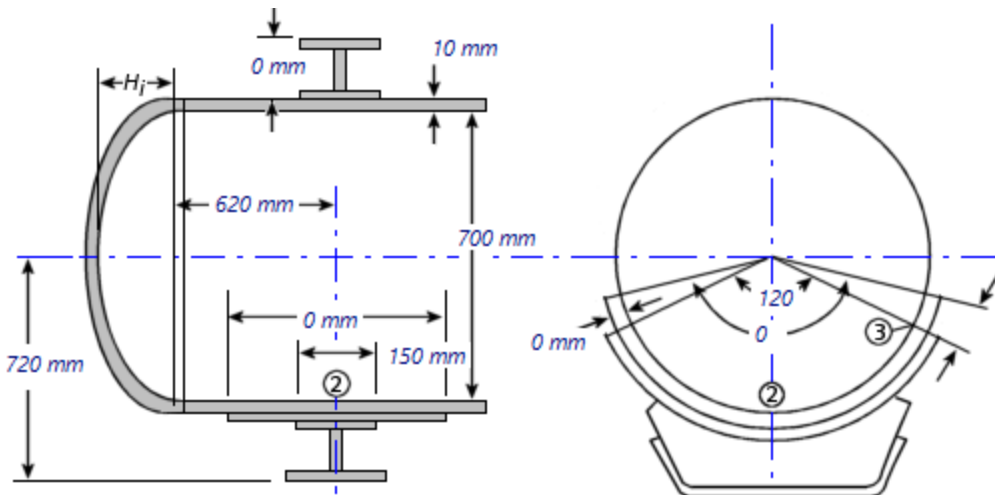
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(150.0 + 1.56 \cdot \sqrt{355.0 \cdot 10.0}, 2 \cdot 620.0)$$

$$= 242.9477 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	355.00	mm
Shell Thickness used in this Case	t	10.000	mm
Stiffened Vessel Length per 4.15.6	L	360.00	cm
Distance from Saddle to Vessel tangent	a1 or a	620.00	mm
Saddle Width	b1 or b	150.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		1560.08	kgf/cm ²
Head Allowable Stress used in Calculation		1560.08	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	720.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		6824.25	kgf
Pressure used in Analysis	P	13.292	kgf/cm ²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	184.66	1560.08
Long. Stress at Bottom of Midspan	287.21	1560.08
Long. Stress at Top of Saddles	380.27	1560.08
Long. Stress at Bottom of Saddles	155.94	1560.08



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Tangential Shear in Shell	147.53	923.92
Circ. Stress at Horn of Saddle	611.23	2340.13
Circ. Compressive Stress in Shell	21.36	1560.08

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3.0 (285.1/2 + 0) \cdot 720.0/640.8588$$

$$= 480.4 \text{ kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(43.06, 0) \cdot 720.0/2490.0002$$

$$= 12.5 \text{ kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$$

$$= 6344 + \max(12, 480, 0, 0)$$

$$= 6824.3 \text{ kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1984.306 (0.0 + \pi/4 (0.72 \cdot 1.2)^2)$$

$$= 1279.740 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	6944.41	kgf
Transverse Shear Load	142.54	kgf
Longitudinal Shear Load	43.06	kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(700.0/2 + 10.0 + 10.0) \sin(120.0/2), 2 \cdot 355.0)$$

$$= 640.859 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K11* = 0.1923	

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$$

$$= -6824 \cdot 62.0 [1 - (1 - 62.0/360.0 + (35.5^2 - 0.0^2) / (2 \cdot 62.0 \cdot 360.0)) / (1 + (4 \cdot 0.0)/(3 \cdot 360.0))]$$

$$= -609.2 \text{ kgf-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 6824 \cdot 360/4 ((1 + 2(36^2 - 0^2)/(360^2)) / (1 + (4 \cdot 0)/(3 \cdot 360)) - 4 \cdot 62/360)$$

$$= 2030.2 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 t)$$

$$= 13.292 \cdot 355.0/(2 \cdot 10.0) - 2030.2/(\pi \cdot 355.0^2 \cdot 10.0)$$

$$= 184.66 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:



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$$\begin{aligned}
 &= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t) \\
 &= 13.292 \cdot 355.0 / (2 \cdot 10.0) + 2030.2 / (\pi \cdot 355.0^2 \cdot 10.0) \\
 &= 287.21 \text{ kgf/cm}^2
 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\
 &= 13.292 \cdot 355.0 / (2 \cdot 10.0) - 609.2 / (0.1066 \cdot \pi \cdot 355.0^2 \cdot 10.0) \\
 &= 380.27 \text{ kgf/cm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\
 &= 13.292 \cdot 355.0 / (2 \cdot 10.0) + 609.2 / (0.1923 \cdot \pi \cdot 355.0^2 \cdot 10.0) \\
 &= 155.94 \text{ kgf/cm}^2
 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned}
 &= Q(L-2a) / (L + (4 \cdot hm / 3)) \\
 &= 6824(360.0 - 2 \cdot 62.0) / (360.0 + (4 \cdot 0.0 / 3)) \\
 &= 4473.7 \text{ kgf}
 \end{aligned}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$\begin{aligned}
 &= K2 \cdot T / (Rm \cdot t) \\
 &= 1.1707 \cdot 4473.68 / (355.0 \cdot 10.0) \\
 &= 147.53 \text{ kgf/cm}^2
 \end{aligned}$$

Decay Length (4.15.20) [x_1, x_2]:

$$\begin{aligned}
 &= 0.78 \cdot \sqrt{(Rm \cdot t)} \\
 &= 0.78 \cdot \sqrt{(355.0 \cdot 10.0)} \\
 &= 46.474 \text{ mm}
 \end{aligned}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$\begin{aligned}
 &= -K5 \cdot Q \cdot k / (t(b + X1 + X2)) \\
 &= -0.7603 \cdot 6824 \cdot 0.1 / (10.0(150.0 + 46.47 + 46.47)) \\
 &= -21.36 \text{ kgf/cm}^2
 \end{aligned}$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.22) [σ_7]:

$$\begin{aligned}
 &= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2) \\
 &= -6824 / (4 \cdot 10.0(150.0 + 46.474 + 46.474)) - \\
 &\quad 3 \cdot 0.0529 \cdot 6824 / (2 \cdot 10.0^2) \\
 &= -611.23 \text{ kgf/cm}^2
 \end{aligned}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$\begin{aligned}
 &= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a) \\
 &= \min(150.0 + 1.56 \cdot \sqrt{(355.0 \cdot 10.0)}, 2 \cdot 62.0) \\
 &= 242.95 \text{ mm}
 \end{aligned}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	930.0000	mm
Baseplate Thickness	Bpthk	15.0000	mm
Baseplate Width	Bpwid	180.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	15.0000	mm
Web Thickness	Webtk	15.0000	mm
Web Location	Webloc	Side	
Saddle Yield Stress	Sy	2651.1	kgf/cm ²
Height of Web at Center	Hw, c	306.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$\begin{aligned}
 &= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12 \\
 &= ((1.56 \cdot \sqrt{(350.0 \cdot 10.0)} + 200.0)^3) / 12 \\
 &= 2.436 \text{ cm}^4
 \end{aligned}$$

Resolved Inertia for the Shell [I,shell]:



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$$\begin{aligned}
 &= \text{ShellInertia} + A\text{Shell}(C1 - Y)^2 \\
 &= 2.436 + 29.229(155.405 - 5.0)^2 \\
 &= 6614.558 \text{ cm}^4
 \end{aligned}$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	292.3	10.0	5.0	29.2	14614.5	0.661E+04
Wearplate	200.0	10.0	15.0	20.0	30000.0	0.394E+04
Web	15.0	335.0	187.5	50.2	942187.1	0.522E+04
BasePlate	180.0	15.0	362.5	27.0	978749.8	0.116E+05
Totals	...	...	...	126.5	1965551.5	0.274E+05

Distance to Centroid [C1]:

$$\begin{aligned}
 &= AY / A \\
 &= 1965552 / 126.479 \\
 &= 155.405 \text{ mm}
 \end{aligned}$$

Angle [β]:

$$\begin{aligned}
 &= 180 - \text{Saddle Angle}/2 \\
 &= 180 - 120.0/2 \\
 &= 120.0
 \end{aligned}$$

Saddle Splitting Coefficient [K1]:

$$\begin{aligned}
 &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta)) \\
 &= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0)\cos(120.0)) \\
 &= 0.2035
 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$\begin{aligned}
 &= K1 \cdot Q \\
 &= 0.204 \cdot 6824.25 \\
 &= 1388.8829 \text{ kgf}
 \end{aligned}$$

$$\begin{aligned}
 \text{Tension Stress, } St &= (Fh / As) = 14.2816 \text{ kgf/cm}^2 \\
 \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 1590.6736 \text{ kgf/cm}^2
 \end{aligned}$$

Saddle Splitting Dimension [d]:

$$\begin{aligned}
 &= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians}) \\
 &= 720.0 - 350.0 \cdot \sin(120.0/2) / 1.0472 \\
 &= 430.552 \text{ mm}
 \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 597.9874 \text{ kgf-m.}$$

$$\begin{aligned}
 \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 33.9647 \text{ kgf/cm}^2 \\
 \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 1767.4150 \text{ kgf/cm}^2
 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned}
 &= \sqrt{(3(Q + \text{SaddleWt})\text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress}))} \\
 &= \sqrt{(3(6824 + 120)180.0 / (2 \cdot 930.0 \cdot 1767.415))} \\
 &= 10.680 \text{ mm}
 \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$\begin{aligned}
 &= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\
 &= 2 \cdot \cos(90 - 120.0/2) (350.0 + 10.0 + 10.0) \\
 &= 640.859 \text{ mm}
 \end{aligned}$$

Distance between Ribs [e]:

$$\begin{aligned}
 &= \text{Web Length} / (N_{\text{ribs}} - 1) \\
 &= 640.8588 / (5 - 1) \\
 &= 160.215 \text{ mm}
 \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$



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$$= 160.2147 \cdot 180.0/2$$

$$= 144.193 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 6944.407 / (930.0 \cdot 180.0)$$

$$= 4.148 \text{ kgf/cm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 144.2 \cdot 4.15$$

$$= 598.170 \text{ kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 20.25 + 12.016$$

$$= 32.266 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 598.2 / 32.2661$$

$$= 18.539 \text{ kgf/cm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	15.0	135.0	82.5	20.2	167062.5	466.
Web	80.1	15.0	7.5	12.0	9012.1	268.
Totals	...	...	...	32.3	176074.6	734.

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 150.0 - 15.0$$

$$= 135.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 176074.562 / 32.266$$

$$= 54.570 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(\text{ytot}, \text{Saddle Width} - \text{ytot})$$

$$= \max(54.57, 150.0 - 54.57)$$

$$= 95.430 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(734.0 / 32.266)}$$

$$= 47.695 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 720.0 - \cos(120.0/2) (350.0 + 10.0 + 10.0) - 15.0$$

$$= 520.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 2038903 / 2651.1)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL / r$$



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$$= 1 \cdot 520.0/47.695$$

$$= 10.903$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 43.1 / (2 \cdot 930.0) \cdot 160.215 \cdot 520.0/2$$

$$= 0.964 \text{ kgf-m.}$$

Compressive Allowable, $KL/r < Cc$ ($10.9026 < 123.2107$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (10.9)^2 / (2 \cdot 123.21^2)) 2651 /$$

$$(5/3 + 3 \cdot (10.9) / (8 \cdot 123.21) - (10.9^3) / (8 \cdot 123.21^3))$$

$$= 1554 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 18.54/1553.6 + (0.96 \cdot 95.43/7339941) / 1767.42$$

$$= 0.013$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	15.0	135.0	82.5	20.2	167062.5	643.
Web	160.2	15.0	7.5	24.0	18024.2	287.
Totals	...	...	...	44.3	185086.6	930.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 185086.641/44.282$$

$$= 41.797 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(ytot, \text{abs}(\text{Saddle Width} - ytot))$$

$$= \max(41.797, \text{abs}(150.0 - 41.797))$$

$$= 108.203 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - Bpthk$$

$$= 720.0 - \cos(370.0/2) (15.0 + 0.0 + 0.0) - 0.0$$

$$= 335.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (930.2/44.282) }$$

$$= 45.833 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 335.0/45.833$$

$$= 7.309$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 43.064 / (2 \cdot 930.0)$$

$$= 0.023 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.023 \cdot 160.215 \cdot 335.0$$

$$= 1.243 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [$\sigma B, \text{base}, c$]:

$$= \text{Bending Moment} / \text{Section Modulus}$$



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$$= 1.243/85970.75$$

$$= 1.445 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ ($7.3091 < 123.2107$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (7.31)^2 / (2 \cdot 123.21^2)) 2651 / (5/3 + 3 \cdot (7.31) / (8 \cdot 123.21) - (7.31^3) / (8 \cdot 123.21^3))$$

$$= 1567 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$

$$= 26.6 / 1566.98 + (1.24 \cdot 108.203 / 930.228) / 1767.42$$

$$= 0.018$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1753.82 kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
Nominal Bolt Diameter	Bnd	36.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	1407.13 kgf/cm ²
Area Available in a Single Bolt	BltArea	6.0075 cm ²
Saddle Load Q0 (Weight)	Q0	6464.0 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	12.5 kgf
Maximum Transverse Force	Ft	142.5 kgf
Maximum Longitudinal Force	Fl	43.1 kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 43.1 / (6.008 \cdot 2)$$

$$= 3.6 \text{ kgf/cm}^2. \text{ Must be less than } 1052.3 \text{ kgf/cm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 142.5 / (6.008 \cdot 2)$$

$$= 11.9 \text{ kgf/cm}^2. \text{ Must be less than } 1052.3 \text{ kgf/cm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} Q_0 > Q_L \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 43.06 / (1052.29 \cdot 2.0)$$

$$= 0.0205 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 720.0 \cdot 142.54 + 0.0$$

$$= 102.63 \text{ kgf-m.}$$

Eccentricity (e):

$$= Rmom / Q_0$$

$$= 102.63 / 6463.99$$

$$= 15.88 \text{ mm} < B_{plen} / 6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:



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= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:

= max[Bltarearl, Bltarears, Bltareart]

= max[0.0, 0.0205, 0.0]

= 0.0205 cm²

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:

Wear Pad Width (200.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$

and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

= $\min(b + 1.56 \cdot \sqrt{Rm \cdot t}, 2a)$

= $\min(150.0 + 1.56 \cdot \sqrt{355.0 \cdot 10.0}, 2 \cdot 490.0)$

= 242.9477 mm

Input and Calculated Values:

Vessel Mean Radius	Rm	355.00	mm
Shell Thickness used in this Case	t	10.000	mm
Stiffened Vessel Length per 4.15.6	L	360.00	cm
Distance from Saddle to Vessel tangent	a1 or a	490.00	mm
Saddle Width	b1 or b	150.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	17.76	cm
Shell Allowable Stress used in Calculation		1560.08	kgf/cm ²
Head Allowable Stress used in Calculation		1560.08	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	720.00	mm
Coefficient of Friction	mu	0.30	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		651.96	kgf
Pressure used in Analysis	P	13.292	kgf/cm ²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	229.90	1560.08
Long. Stress at Bottom of Midspan	241.97	1560.08
Long. Stress at Top of Saddles	248.37	1560.08
Long. Stress at Bottom of Saddles	229.04	1560.08
Tangential Shear in Shell	14.68	923.92
Circ. Stress at Horn of Saddle	58.39	2340.13
Circ. Compressive Stress in Shell	2.04	1560.08

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

= $Ftr(Ft/Num \text{ of Saddles} + Z \text{ Force Load}) \cdot B / E$

= $3.0(285.1/2 + 0) \cdot 720.0/640.8588$

= 480.4 kgf

Longitudinal Saddle Reaction Force [Fwl]:



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$$\begin{aligned}
 &= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s \\
 &= \max(43.06, 0) \cdot 720.0/2490.0002 \\
 &= 12.5 \text{ kgf}
 \end{aligned}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$\begin{aligned}
 &= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st}) \\
 &= 172 + \max(12, 480, 0, 0) \\
 &= 652.0 \text{ kgf}
 \end{aligned}$$

Longitudinal Wind Force [F_l]:

$$\begin{aligned}
 &= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2)) \\
 &= 1.1 \cdot 1984.306 (0.0 + \pi/4 (0.72 \cdot 1.2)^2) \\
 &= 1279.740 \text{ N}
 \end{aligned}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	772.11	kgf
Transverse Shear Load	142.54	kgf
Longitudinal Shear Load	43.06	kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$\begin{aligned}
 &= \min(2 (\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m) \\
 &= \min(2 (700.0/2 + 10.0 + 10.0) \sin(120.0/2), 2 \cdot 355.0) \\
 &= 640.859 \text{ mm}
 \end{aligned}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to R_m/2.

Moment per Equation 4.15.1 [M1]:

$$\begin{aligned}
 &= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)] \\
 &= -652 \cdot 49.0 [1 - (1 - 49.0/360.0 + (35.5^2 - 17.755^2) / (2 \cdot 49.0 \cdot 360.0)) / (1 + (4 \cdot 17.76)/(3 \cdot 360.0))] \\
 &= -52.5 \text{ kgf-m.}
 \end{aligned}$$

Moment per Equation 4.15.2 [M2]:

$$\begin{aligned}
 &= Q \cdot L / 4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L) \\
 &= 652 \cdot 360 / 4 ((1 + 2(36^2 - 18^2)/(360^2)) / (1 + (4 \cdot 18)/(3 \cdot 360)) - 4 \cdot 49/360) \\
 &= 239.1 \text{ kgf-m.}
 \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ₁]:

$$\begin{aligned}
 &= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 \cdot t) \\
 &= 13.292 \cdot 355.0 / (2 \cdot 10.0) - 239.1 / (\pi \cdot 355.0^2 \cdot 10.0) \\
 &= 229.90 \text{ kgf/cm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ₂]:

$$\begin{aligned}
 &= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t) \\
 &= 13.292 \cdot 355.0 / (2 \cdot 10.0) + 239.1 / (\pi \cdot 355.0^2 \cdot 10.0) \\
 &= 241.97 \text{ kgf/cm}^2
 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ*3]:

$$\begin{aligned}
 &= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t) \\
 &= 13.292 \cdot 355.0 / (2 \cdot 10.0) - 52.5 / (0.1066 \cdot \pi \cdot 355.0^2 \cdot 10.0) \\
 &= 248.37 \text{ kgf/cm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ*4]:

$$\begin{aligned}
 &= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t) \\
 &= 13.292 \cdot 355.0 / (2 \cdot 10.0) + 52.5 / (0.1923 \cdot \pi \cdot 355.0^2 \cdot 10.0) \\
 &= 229.04 \text{ kgf/cm}^2
 \end{aligned}$$



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Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a)/(L+(4 \cdot hm/3))$$

$$= 652(360.0 - 2 \cdot 49.0)/(360.0 + (4 \cdot 17.76/3))$$

$$= 445.2 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ₂]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 445.2 / (355.0 \cdot 10.0)$$

$$= 14.68 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x₁,x₂]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(355.0 \cdot 10.0)}$$

$$= 46.474 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ₆]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.7603 \cdot 652 \cdot 0.1 / (10.0(150.0 + 46.47 + 46.47))$$

$$= -2.04 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, L ≥ 8R_m (4.15.22) [σ₇]:

$$= -Q/(4 \cdot t(b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q / (2 \cdot t^2)$$

$$= -652 / (4 \cdot 10.0(150.0 + 46.474 + 46.474)) -$$

$$3 \cdot 0.0529 \cdot 652 / (2 \cdot 10.0^2)$$

$$= -58.39 \text{ kgf/cm}^2$$

Effective reinforcing plate width (4.15.24) [B₁]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(150.0 + 1.56 \cdot \sqrt{(355.0 \cdot 10.0)}, 2 \cdot 49.0)$$

$$= 242.95 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	930.0000	mm
Baseplate Thickness	Bpthk	15.0000	mm
Baseplate Width	Bpwid	180.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	15.0000	mm
Web Thickness	Webtk	15.0000	mm
Web Location	Webloc	Side	
Saddle Yield Stress	Sy	2651.1	kgf/cm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(350.0 \cdot 10.0)} + 200.0)10.0^3)/12$$

$$= 2.436 \text{ cm}^4$$

Resolved Inertia for the Shell [I₁,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 2.436 + 29.229(155.405 - 5.0)^2$$

$$= 6614.558 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	292.3	10.0	5.0	29.2	14614.5	0.661E+04
Wearplate	200.0	10.0	15.0	20.0	30000.0	0.394E+04
Web	15.0	335.0	187.5	50.2	942187.1	0.522E+04
BasePlate	180.0	15.0	362.5	27.0	978749.8	0.116E+05
Totals	...	...	...	126.5	1965551.5	0.274E+05



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Distance to Centroid [C1]:

$$= AY / A$$

$$= 1965552 / 126.479$$

$$= 155.405 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 651.957$$

$$= 132.6874 \text{ kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 1.3644 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 1590.6736 \text{ kgf/cm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 720.0 - 350.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 430.552 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 57.1289 \text{ kgf-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 3.2448 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 1767.4150 \text{ kgf/cm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(652 + 120)180.0 / (2 \cdot 930.0 \cdot 1767.415))}$$

$$= 3.561 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (350.0 + 10.0 + 10.0)$$

$$= 640.859 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 640.8588 / (5 - 1)$$

$$= 160.215 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 160.2147 \cdot 180.0 / 2$$

$$= 144.193 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 772.113 / (930.0 \cdot 180.0)$$

$$= 0.461 \text{ kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 144.2 \cdot 0.46$$

$$= 66.507 \text{ kgf}$$



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Area of the Rib and Web [Ar]:

$$\begin{aligned}
 &= \text{Rib Area} + \text{Web Area} \\
 &= 20.25 + 12.016 \\
 &= 32.266 \text{ cm}^2
 \end{aligned}$$

Compressive Stress [Sc]:

$$\begin{aligned}
 &= P/Ar \\
 &= 66.5/32.2661 \\
 &= 2.061 \text{ kgf/cm}^2
 \end{aligned}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	15.0	135.0	82.5	20.2	167062.5	466.
Web	80.1	15.0	7.5	12.0	9012.1	268.
Totals	...	...	...	32.3	176074.6	734.

Rib dimension [D]:

$$\begin{aligned}
 &= \text{Saddle Width} - \text{Web Thickness} \\
 &= 150.0 - 15.0 \\
 &= 135.000 \text{ mm}
 \end{aligned}$$

Distance to Centroid from Datum [ytot]:

$$\begin{aligned}
 &= AY / A \\
 &= 176074.562/32.266 \\
 &= 54.570 \text{ mm}
 \end{aligned}$$

Distance to Centroid [C1]:

$$\begin{aligned}
 &= \max(y_{tot}, \text{Saddle Width} - y_{tot}) \\
 &= \max(54.57, 150.0 - 54.57) \\
 &= 95.430 \text{ mm}
 \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned}
 &= \sqrt{ \text{Total Inertia} / \text{Total Area} } \\
 &= \sqrt{ 734.0/32.266 } \\
 &= 47.695 \text{ mm}
 \end{aligned}$$

Length of Outer Rib [L]:

$$\begin{aligned}
 &= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk} \\
 &= 720.0 - \cos(120.0/2) (350.0 + 10.0 + 10.0) - 15.0 \\
 &= 520.000 \text{ mm}
 \end{aligned}$$

Intermediate Term [Cc]:

$$\begin{aligned}
 &= \sqrt{ \frac{ 2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress} }{ } } \\
 &= \sqrt{ \frac{ 2 \cdot \pi^2 \cdot 2038903/2651.1 }{ } } \\
 &= 123.211
 \end{aligned}$$

Slenderness ratio [KL/r]:

$$\begin{aligned}
 &= KL/r \\
 &= 1 \cdot 520.0/47.695 \\
 &= 10.903
 \end{aligned}$$

Bending Moment [Rm]:

$$\begin{aligned}
 &= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2 \\
 &= 43.1 / (2 \cdot 930.0) \cdot 160.215 \cdot 520.0/2 \\
 &= 0.964 \text{ kgf-m.}
 \end{aligned}$$

Compressive Allowable, $KL/r < C_c$ ($10.9026 < 123.2107$) per AISC E2-1 [Sca]:

$$\begin{aligned}
 &= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3)) \\
 &= (1 - (10.9)^2 / (2 \cdot 123.21^2)) 2651 / \\
 &\quad (5/3 + 3 \cdot (10.9) / (8 \cdot 123.21) - (10.9)^3 / (8 \cdot 123.21^3)) \\
 &= 1554 \text{ kgf/cm}^2
 \end{aligned}$$



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Saddle Calcs: Test Case: Step: 21 9:43am Aug 12,2026

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c/S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 2.06/1553.6 + (0.96 \cdot 95.43/7339941) / 1767.42$$

$$= 0.002$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	15.0	135.0	82.5	20.2	167062.5	643.
Web	160.2	15.0	7.5	24.0	18024.2	287.
Totals	...	...	...	44.3	185086.6	930.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 185086.641/44.282$$

$$= 41.797 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(y_{tot}, \text{abs}(\text{Saddle Width} - y_{tot}))$$

$$= \max(41.797, \text{abs}(150.0 - 41.797))$$

$$= 108.203 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 720.0 - \cos(370.0/2) (15.0 + 0.0 + 0.0) - 0.0$$

$$= 335.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (930.2/44.282) }$$

$$= 45.833 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 335.0/45.833$$

$$= 7.309$$

Unit Force [Force,u]:

$$= F_l / (2 \cdot \text{Baseplate Length})$$

$$= 43.064 / (2 \cdot 930.0)$$

$$= 0.023 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.023 \cdot 160.215 \cdot 335.0$$

$$= 1.243 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 1.243/85970.75$$

$$= 1.445 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ (7.3091 < 123.2107) per AISC E2-1 [Sca]:

$$= (1 - (Kl/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (Kl/r) / (8 \cdot C_c) - (Kl/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (7.31)^2 / (2 \cdot 123.21^2)) 2651 /$$

$$(5/3 + 3 \cdot (7.31) / (8 \cdot 123.21) - (7.31^3) / (8 \cdot 123.21^3))$$

$$= 1567 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c/S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 2.96/1566.98 + (1.24 \cdot 108.203/930.228) / 1767.42$$

$$= 0.003$$



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Saddle Calcs: Test Case:

Step: 21 9:43am Aug 12,2026

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Material Specification	SA-193 B7	
Bolt Allowable Stress	Stba	1753.82 kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
Nominal Bolt Diameter	Bnd	36.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	1407.13 kgf/cm ²
Area Available in a Single Bolt	BltArea	6.0075 cm ²
Saddle Load QO (Weight)	QO	291.7 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	12.5 kgf
Maximum Transverse Force	Ft	142.5 kgf
Maximum Longitudinal Force	F1	43.1 kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= F1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 43.1 / (6.008 \cdot 2)$$

$$= 3.6 \text{ kgf/cm}^2. \text{ Must be less than } 1052.3 \text{ kgf/cm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 142.5 / (6.008 \cdot 2)$$

$$= 11.9 \text{ kgf/cm}^2. \text{ Must be less than } 1052.3 \text{ kgf/cm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (QO > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F1 / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 43.06 / (1052.29 \cdot 2.0)$$

$$= 0.0205 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 720.0 \cdot 142.54 + 0.0$$

$$= 102.63 \text{ kgf-m.}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 102.63 / 291.7$$

$$= 351.82 \text{ mm} > Bplen / 6 \text{ --> Uplift in Transverse direction}$$

f = Bplen / 2 - Edgedis

$$= 930.0 / 2 - 50.01$$

$$= 414.99 \text{ mm}$$

Modular Ratio Of Steel/Concrete (n1):

$$= ES / EC$$

$$= 2074057 / 219499.77$$

$$= 9.45$$

K1 = 3 (e - 0.5 • Bplen)

$$= 3 (351.82 - 0.5 \cdot 930.0)$$

$$= -339.54 \text{ mm}$$

K2 = 6 • n1 • At / Bpwid • (f + e)



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Saddle Calcs: Test Case: Step: 21 9:43am Aug 12, 2026

$$= 6 \cdot 9.45 \cdot 6.01 / 180.0 \cdot (414.99 + 351.82)$$

$$= 145093.18 \text{ mm}^2$$

$$K3 = -K2 \cdot (0.5 \cdot B_{plen} + f)$$

$$= -145093.17 \cdot (0.5 \cdot 930.0 + 414.99)$$

$$= -127680156.39 \text{ mm}^3$$

Iteratively Solving for the Effective Bearing Length:

$$Y^3 + K1 \cdot Y^2 + K2 \cdot Y + K3 = 0$$

$$Y^3 + -339.54 \cdot Y^2 + 145093.17 \cdot Y + -0.1E+09 = 0$$

$$Y = 525.63 \text{ mm}$$

$$\text{Num} = (B_{plen} / 2 - Y / 3 - e)$$

$$= (930.0 / 2 - 525.63 / 3 - 351.82)$$

$$= -62.03$$

$$\text{Denom} = (B_{plen} / 2 - Y / 3 + f)$$

$$= (930.0 / 2 - 525.63 / 3 + 414.99)$$

$$= 704.78$$

Total Bolt Tension Force [Tforce]:

$$= -QO \cdot \text{Num} / \text{Denom}$$

$$= -291.7 \cdot -62.03 / 704.78$$

$$= 25.67 \text{ kgf}$$

Bolt Area Required due to Transverse Load [Bltareart]:

$$= T_{\text{force}} / (S_{\text{tba}} \cdot N_{\text{bt}})$$

$$= 25.67 / (1753.82 \cdot 1.0)$$

$$= 0.0146 \text{ cm}^2$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[B_{\text{tarearl}}, B_{\text{tarears}}, B_{\text{tareart}}]$$

$$= \max[0.0, 0.0205, 0.0146]$$

$$= 0.0205 \text{ cm}^2$$

Baseplate Thickness Calculation per D. Moss:

Bearing Pressure (fc)

$$= 2(QO + T_{\text{force}}) / (Y \cdot B_{\text{pwid}})$$

$$= 2(291.7 + 25.67) / (525.63 \cdot 180.0)$$

$$= 0.67 \text{ kgf/cm}^2$$

Distance from Baseplate Edge to the Web [ADIST]:

$$= (B_{\text{plen}} - W_{\text{eblngth}}) / 2$$

$$= (930.0 - 879.2) / 2$$

$$= 25.4000 \text{ mm}$$

Overturning Moment due To Bolt Tension [Mt]:

$$= T_{\text{force}} \cdot A_{\text{dist}}$$

$$= 25.67 \cdot 2.54$$

$$= 0.65 \text{ kgf-m.}$$

Equivalent Bearing Pressure (f1):

$$= f_c \cdot (Y - A_{\text{dist}}) / Y$$

$$= 0.67 \cdot (525.63 - 25.4) / 525.63$$

$$= 0.64 \text{ kgf/cm}^2$$

Overturning Moment due to Bearing Pressure [Mc]:

$$= (A_{\text{dist}}^2 \cdot B_{\text{pwid}} / 6) \cdot (f1 + 2 \cdot f_c)$$

$$= (25.4^2 \cdot 180.0 / 6) \cdot (0.64 + 2 \cdot 0.67)$$

$$= 0.38 \text{ kgf-m.}$$

Baseplate Required Thickness [Treq]:

$$= \sqrt{(6 \cdot \max(M_t, M_c)) / (B_{\text{pwid}} \cdot S_{\text{ba}})}$$



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Saddle Calcs: Test Case:

Step: 21

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$$= \sqrt{6 \cdot \max(0.65, 0.38) / (180.0 \cdot 2110.7)}$$
$$= 1.0148 \text{ mm}$$

Review notes about nozzle loadings and supports in the Warnings report.

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Nozzle Calcs.: N1- Channel inl Nozl: 11 9:43am Aug 12,2026

Input, Nozzle Desc: N1- Channel inl

From: 30

Pressure for Reinforcement Calculations	P	46.908	kgf/cm ²
Temperature for Internal Pressure	Temp	155	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	700.00	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		73.50	cm
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 4	
Material UNS Number		K03017	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1407.13	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.13	kgf/cm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		10.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	44.0000	mm
Height of Beveled Transition	L'	14.3832	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		40.0000	mm
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	305.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	30.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	15.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

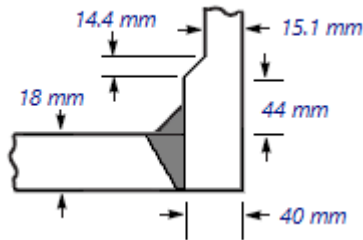


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Nozzle Calcs.: N1- Channel inl Nozl: 11 9:43am Aug 12,2026



Reinforcement CALCULATION, Description: N1- Channel inl

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	10.750 in.
Actual Thickness Used in Calculation	0.594 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 46.91 \cdot 351.5 / (1407 \cdot 1.0 - 0.6 \cdot 46.91)$
 $= 11.9567 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 46.91 \cdot 136.525 / (1407 \cdot 1.0 + 0.4 \cdot 46.91)$
 $= 4.4913 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 24.912 / 0.5773$
 $= 43.1496 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	491.7496	mm
Parallel to Vessel Wall, opening length	d	245.8748	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	41.2500	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1407.1 / 1407.1)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1407.1 / 1407.1)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$



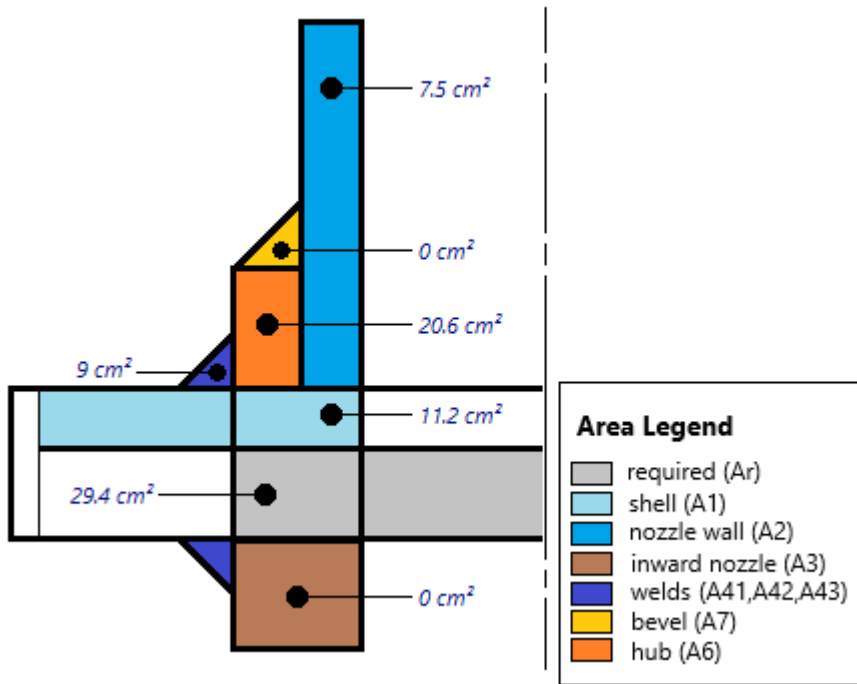
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Nozzle Calcs.: N1- Channel in1

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Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	29.399	NA	NA
Area in Shell	A1	11.171	NA	NA
Area in Nozzle Wall	A2	7.504	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	9.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	20.553	NA	NA
TOTAL AREA AVAILABLE	Atot	48.228	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (245.8748 \cdot 11.9567 \cdot 1.0 + 2 \cdot 38.5 \cdot 11.9567 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 29.399 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 245.875(1.0 \cdot 16.5 - 1.0 \cdot 11.957) - 2 \cdot 38.5$$

$$(1.0 \cdot 16.5 - 1.0 \cdot 11.9567) \cdot (1 - 1.0)$$

$$= 11.171 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 41.25)(13.59 - 4.49)1.0$$

$$= 7.504 \text{ cm}^2$$



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Nozzle Calcs.: N1- Channel inl Nozl: 11 9:43am Aug 12,2026

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2$$

$$= 30.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 9.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(41.2, 305.0, 44.0)) \cdot (40.0 - 15.1) \cdot 1.0$$

$$= 20.553 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 5.9913 mm
Wall Thickness per UG16(b),	tr16b = 3.0000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 13.4567 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 13.4567 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm
Wall Thickness per table UG-45	tb3 = 9.6026 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[9.603, \max(13.4567, 3.0)]$$

$$= 9.6026 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(5.9913, 9.6026)$$

$$= 9.6026 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 \cdot 15.088 = 13.202 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 443.6,	Allowable	: 1407.1 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 3074.2 kgf/cm ²	Passed
Occasional	: 201.1,	Allowable	: 1871.5 kgf/cm ²	Passed
Shear	: 158.2,	Allowable	: 985.0 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 13.202, tr = 4.491, c = 1.5 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.384$, Temp. Reduction = 55 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	0 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 18.0, tr = 11.957, c = 1.5 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.725$, Temp. Reduction = 15 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-26 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
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 Nozzle Calcs.: N1- Channel inl Nozl: 11 9:43am Aug 12,2026

Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 46.91/104.01 = 0.451

Weld Size Calculations, Description: N1- Channel inl

Intermediate Calc. for nozzle/shell Welds Tmin 16.5000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	21.2100 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t \cdot t_r)) \cdot S_v)$
 $= \max(0, (29.3985 - 11.1708 + 2 \cdot 38.5 \cdot 1.0 \cdot (1.0 \cdot 16.5 - 11.9567)) \cdot 1407)$
 $= 30571.45 \text{ kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i-\text{Can}/.707)^2 \cdot f_r2) \cdot S_v$
 $= (28.0572 + 9.0 - 0.0 \cdot 1.0) \cdot 1407$
 $= 52144.36 \text{ kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$
 $= (28.0572 + 0.0 + 9.0 + (12.705)) \cdot 1407$
 $= 70021.97 \text{ kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$
 $= (28.0572 + 0.0 + 9.0 + 0.0 + (12.705)) \cdot 1407$
 $= 70021.97 \text{ kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 273.05 \cdot 30.0 \cdot 0.49 \cdot 1407$
 $= 88719. \text{ kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 129.7312) \cdot (40.0 - 1.5) \cdot 0.7 \cdot 1407$
 $= 154557. \text{ kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 273.05 \cdot (15.0 - 0.0) \cdot 0.74 \cdot 1407$
 $= 66992. \text{ kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (88719 + 154557) = 243275 \text{ kgf}$

Failure Path [Path 2-2]:



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Nozzle Calcs.: N1- Channel inl Nozl: 11 9:43am Aug 12,2026

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (88719 + 0 + 66992 + 0) = 155710 \text{ kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (88719 + 66992 + 0) = 155710 \text{ kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 243275 kgf, must exceed W = 30571 kgf or W1 = 52144 kgf

Path 2-2 = 155710 kgf, must exceed W = 30571 kgf or W2 = 70021 kgf

Path 3-3 = 155710 kgf, must exceed W = 30571 kgf or W3 = 70021 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 61 kgf/cm²

The Drop for this Nozzle is : 27.7254 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 350.7254 mm

This nozzle or its diameter limit intersects weld seam # [3] on this element. The distance between the nozzle and seam is, 98.16 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 245.87 mm

Input Echo, WRC297 Item 1, Description: N1- Channel in:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	700.000	mm
Shell Thickness	18.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 1407.13	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1407.13	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	322.875	mm
Nozzle Thickness	40.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 4	
Nozzle Cold S.I. Allowable	SNmc 1407.13	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 1407.13	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	46.91	kgf/cm ²
Radial Load	P	-2294.47	kgf
Circumferential Shear	Vc	1657.12	kgf
Longitudinal Shear	Vl	1657.12	kgf
Circumferential Moment	Mc	1274.19	kgf-m.
Longitudinal Moment	Ml	1274.19	kgf-m.
Torsional Moment	Mt	1465.32	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No



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Nozzle Calcs.: N1- Channel inl Nozl: 11 9:43am Aug 12,2026

*Warning - The ratio, Dn/Tn (8.072) is < 10
check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 322.875 + 2 \cdot 1.65 \cdot \sqrt{359.75(18.0 - 1.5)}$$

$$= 577.122 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 719.500 mm
 Nozzle Outside Diameter (d) = 322.875 mm
 Vessel Thickness used (T) = 16.500 mm
 Nozzle Thickness used (t) = 38.500 mm
 T / t = 0.429
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.963$$

Nr/P = 0.060
 Mr/P = 0.105
 M0/P = 0.034
 N0/P = 0.080
 MrD/Mc = 0.263
 NrDL/Mc = 0.098
 M0d/Mc = 0.079
 N0DL/Mc = 0.086
 MrD/Ml = 0.092
 NrDL/Ml = 0.051
 M0D/Ml = 0.025
 N0DL/Ml = 0.089

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	67	67	67	67
Outplane Bending (P)	171	-171	171	-171
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-129	-129	129	129
Outplane Bending (ML)	-216	216	216	-216
Normal Pressure Stress	976	1023	976	1023
Outplane Stress Summary	869	1006	1561	831

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	50	50	50	50
Inplane Bending (P)	531	-531	531	-531
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-74	-74	74	74
Inplane Bending (ML)	-804	804	804	-804
Inplane Pressure Stress	488	488	488	488
Inplane Stress Summary	192	737	1948	-722

Vessel Stresses (kgf/cm²):



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Nozzle Calcs.: N1- Channel inl Nozl: 11 9:43am Aug 12,2026

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)			Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Shear (Vc)			19	19	-19	-19
Outplane Shear (Vl)			0	0	0	0
Outplane Shear (Mt)			54	54	54	54
Shear Stress Summary			73	73	34	34

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)			Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Two * Max Shear Stress			877	1025	1951	1556

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Outplane Membrane (P)			67	67	67	67
Outplane Bending (P)			171	-171	171	-171
Outplane Membrane (Mc)			-124	-124	124	124
Outplane Bending (Mc)			-688	688	688	-688
Outplane Membrane (ML)			0	0	0	0
Outplane Bending (ML)			0	0	0	0
Normal Pressure Stress			488	488	488	488
Outplane Stress Summary			-85	948	1539	-179

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Inplane Membrane (P)			50	50	50	50
Inplane Bending (P)			531	-531	531	-531
Inplane Membrane (Mc)			-142	-142	142	142
Inplane Bending (Mc)			-2286	2286	2286	-2286
Inplane Membrane (ML)			0	0	0	0
Inplane Bending (ML)			0	0	0	0
Inplane Pressure Stress			976	1023	976	1023
Inplane Stress Summary			-870	2686	3988	-1602

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			-19	-19	19	19
Torsional Shear (Mt)			54	54	54	54
Shear Stress Summary			34	34	73	73

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)			Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Two * Max Shear Stress			872	2687	3990	1605



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Nozzle Calcs.: N1- Channel inl Nozl: 11 9:43am Aug 12,2026

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	67	67	67	67
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-129	-129	129	129
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	155	211	155	211
Hoop Stress Summary	93	149	352	408

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	6	6	6	6
Axial Bending (P)	69	-69	69	-69
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-58	-58	58	58
Axial Bending (ML)	-106	106	106	-106
Axial Pressure Stress	77	77	77	77
Axial Stress Summary	-10	63	318	-34

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	8	8	-8	-8
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	23	23	23	23
Shear Stress Summary	31	31	14	14

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	122	160	357	443

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	67	67	67	67
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-124	-124	124	124
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	155	211	155	211
Hoop Stress Summary	98	155	347	403

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Nozzle Calcs.: N1- Channel inl Nozl: 11 9:43am Aug 12,2026

Axial Membrane (P)	6	6	6	6
Axial Bending (P)	69	-69	69	-69
Axial Membrane (Mc)	-58	-58	58	58
Axial Bending (Mc)	-341	341	341	-341
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	77	77	77	77

Axial Stress Summary	-245	298	553	-268
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-8	-8	8	8
Shear due to Torsion	23	23	23	23
Shear Stress Summary	14	14	31	31

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	345	299	558	675

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	976	1023	976	1023	976	1023	976	1023
Circ. Pl (SUS)	-62	-62	196	196	-92	-92	192	192
Circ. Q (SUS)	-45	45	387	-387	-1755	1755	2817	-2817
Long. Pm (SUS)	488	488	488	488	488	488	488	488
Long. Pl (SUS)	-24	-24	124	124	-57	-57	191	191
Long. Q (SUS)	-273	273	1335	-1335	-517	517	859	-859
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	19	19	-19	-19	-19	-19	19	19
Shear Q (SUS)	54	54	54	54	54	54	54	54
Pm (SUS)	976	1023	976	1023	976	1023	976	1023
Pm+Pl (SUS)	914	961	1172	1219	884	931	1168	1215
Pm+Pl+Q (Total)	876	1024	1950	1556	872	2686	3987	1605

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1023.00	1407.13	Passed
Pm+Pl (SUS)	1219.59	2110.70	Passed
Pm+Pl+Q (TOTAL)	3987.18	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



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WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	155	211	155	211	155	211	155	211
Circ. Pl (SUS)	-62	-62	196	196	-57	-57	191	191
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	77	77	77	77	77	77	77	77
Long. Pl (SUS)	-52	-52	64	64	-52	-52	64	64
Long. Q (SUS)	-37	37	175	-175	-272	272	410	-410
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	8	8	-8	-8	-8	-8	8	8
Shear Q (SUS)	23	23	23	23	23	23	23	23
Pm (SUS)	155.0	211.0	155.0	211.0	155.0	211.0	155.0	211.0
Pm+Pl (SUS)	93.9	149.5	351.3	407.2	98.9	154.5	346.3	402.2
Pm+Pl+Q (Total)	121.9	158.9	356.5	442.0	346.3	298.6	555.6	673.9

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	211.00	1407.13	Passed
Pm+Pl (SUS)	407.24	2110.70	Passed
Pm+Pl+Q (TOTAL)	673.86	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Nozzle Calcs.: N2B- channel ou Nozl: 12 9:43am Aug 12,2026

Input, Nozzle Desc: N2B- channel ou

From: 30

Pressure for Reinforcement Calculations	P	46.909	kgf/cm ²
Temperature for Internal Pressure	Temp	155	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	700.00	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		73.50	cm
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 4	
Material UNS Number		K03017	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1407.13	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.13	kgf/cm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		10.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	44.0000	mm
Height of Beveled Transition	L'	14.3832	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		40.0000	mm
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	305.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	30.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	15.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

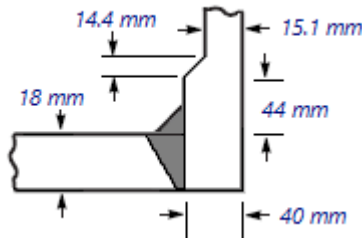


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Nozzle Calcs.: N2B- channel ou Nozl: 12 9:43am Aug 12,2026



Reinforcement CALCULATION, Description: N2B- channel ou

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	10.750 in.
Actual Thickness Used in Calculation	0.594 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 46.91 \cdot 351.5 / (1407 \cdot 1.0 - 0.6 \cdot 46.91)$
 $= 11.9570 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 46.91 \cdot 136.525 / (1407 \cdot 1.0 + 0.4 \cdot 46.91)$
 $= 4.4914 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 24.912 / 0.5773$
 $= 43.1496 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	491.7496	mm
Parallel to Vessel Wall, opening length	d	245.8748	mm
Normal to Vessel Wall (Thickness Limit), no pad	T1np	41.2500	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1407.1 / 1407.1)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1407.1 / 1407.1)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$



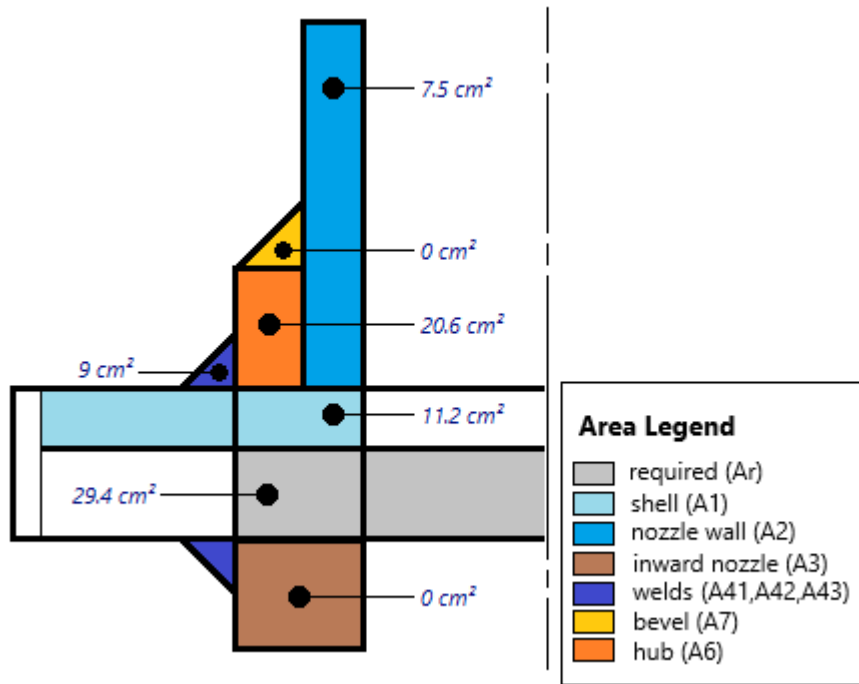
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Nozzle Calcs.: N2B- channel ou

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Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	29.399	NA	NA
Area in Shell	A1	11.170	NA	NA
Area in Nozzle Wall	A2	7.504	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	9.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	20.553	NA	NA
TOTAL AREA AVAILABLE	Atot	48.227	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (245.8748 \cdot 11.957 \cdot 1.0 + 2 \cdot 38.5 \cdot 11.957 \cdot 1.0 \cdot (1 - 1.0)) \\
 &= 29.399 \text{ cm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 245.875(1.0 \cdot 16.5 - 1.0 \cdot 11.957) - 2 \cdot 38.5 \\
 &\quad (1.0 \cdot 16.5 - 1.0 \cdot 11.957) \cdot (1 - 1.0) \\
 &= 11.170 \text{ cm}^2
 \end{aligned}$$

Area Available in Nozzle Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot tlnp)(tn - trn)fr2 \\
 &= (2 \cdot 41.25)(13.59 - 4.49)1.0 \\
 &= 7.504 \text{ cm}^2
 \end{aligned}$$



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Nozzle Calcs.: N2B- channel ou Nozl: 12 9:43am Aug 12,2026

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2$$

$$= 30.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 9.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(41.2, 305.0, 44.0)) \cdot (40.0 - 15.1) \cdot 1.0$$

$$= 20.553 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 5.9914 mm
Wall Thickness per UG16(b),	tr16b = 3.0000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 13.4570 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 13.4570 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm
Wall Thickness per table UG-45	tb3 = 9.6026 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[9.603, \max(13.457, 3.0)]$$

$$= 9.6026 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(5.9914, 9.6026)$$

$$= 9.6026 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 \cdot 15.088 = 13.202 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 443.6,	Allowable	: 1407.1 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 3074.2 kgf/cm ²	Passed
Occasional	: 201.1,	Allowable	: 1871.5 kgf/cm ²	Passed
Shear	: 158.2,	Allowable	: 985.0 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 13.202, tr = 4.491, c = 1.5 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.384$, Temp. Reduction = 55 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	0 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 18.0, tr = 11.957, c = 1.5 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.725$, Temp. Reduction = 15 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-26 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
--	--------



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Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 46.91/104.01 = 0.451

Weld Size Calculations, Description: N2B- channel ou

Intermediate Calc. for nozzle/shell Welds Tmin 16.5000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	21.2100 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t \cdot t_r)) \cdot S_v)$
 $= \max(0, (29.3992 - 11.1701 + 2 \cdot 38.5 \cdot 1.0 \cdot (1.0 \cdot 16.5 - 11.957)) \cdot 1407)$
 $= 30573.04 \text{ kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i-\text{Can}/.707)^2 \cdot f_r2) \cdot S_v$
 $= (28.0571 + 9.0 - 0.0 \cdot 1.0) \cdot 1407$
 $= 52144.25 \text{ kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$
 $= (28.0571 + 0.0 + 9.0 + (12.705)) \cdot 1407$
 $= 70021.85 \text{ kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$
 $= (28.0571 + 0.0 + 9.0 + 0.0 + (12.705)) \cdot 1407$
 $= 70021.85 \text{ kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 273.05 \cdot 30.0 \cdot 0.49 \cdot 1407$
 $= 88719. \text{ kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 129.7312) \cdot (40.0 - 1.5) \cdot 0.7 \cdot 1407$
 $= 154557. \text{ kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 273.05 \cdot (15.0 - 0.0) \cdot 0.74 \cdot 1407$
 $= 66992. \text{ kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (88719 + 154557) = 243275 \text{ kgf}$

Failure Path [Path 2-2]:



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$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (88719 + 0 + 66992 + 0) = 155710 \text{ kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (88719 + 66992 + 0) = 155710 \text{ kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 243275 kgf, must exceed W = 30573 kgf or W1 = 52144 kgf

Path 2-2 = 155710 kgf, must exceed W = 30573 kgf or W2 = 70021 kgf

Path 3-3 = 155710 kgf, must exceed W = 30573 kgf or W3 = 70021 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 61 kgf/cm²

The Drop for this Nozzle is : 27.7254 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 350.7254 mm

Input Echo, WRC297 Item 1, Description: N2B- channel o:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	700.000	mm
Shell Thickness	18.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 1407.13	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1407.13	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	322.875	mm
Nozzle Thickness	40.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 4	
Nozzle Cold S.I. Allowable	SNmc 1407.13	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 1407.13	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSTained loads.

Design Internal Pressure	Dp	46.91	kgf/cm ²
Radial Load	P	-2294.47	kgf
Circumferential Shear	Vc	1657.12	kgf
Longitudinal Shear	Vl	1657.12	kgf
Circumferential Moment	Mc	1274.19	kgf-m.
Longitudinal Moment	Ml	1274.19	kgf-m.
Torsional Moment	Mt	1465.32	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (8.072) is < 10
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:



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Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 322.875 + 2 \cdot 1.65 \cdot \sqrt{359.75(18.0 - 1.5)}$$

$$= 577.122 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 719.500 mm
 Nozzle Outside Diameter (d) = 322.875 mm
 Vessel Thickness used (T) = 16.500 mm
 Nozzle Thickness used (t) = 38.500 mm
 T / t = 0.429
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.963$$

Nr/P = 0.060
 Mr/P = 0.105
 M0/P = 0.034
 N0/P = 0.080
 MrD/Mc = 0.263
 NrDL/Mc = 0.098
 M0d/Mc = 0.079
 N0DL/Mc = 0.086
 MrD/Ml = 0.092
 NrDL/Ml = 0.051
 M0D/Ml = 0.025
 N0DL/Ml = 0.089

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	67	67	67	67
Outplane Bending (P)	171	-171	171	-171
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-129	-129	129	129
Outplane Bending (ML)	-216	216	216	-216
Normal Pressure Stress	976	1023	976	1023
Outplane Stress Summary	869	1006	1561	831

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	50	50	50	50
Inplane Bending (P)	531	-531	531	-531
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-74	-74	74	74
Inplane Bending (ML)	-804	804	804	-804
Inplane Pressure Stress	488	488	488	488
Inplane Stress Summary	192	737	1948	-722

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Nozzle Calcs.: N2B- channel ou Nozl: 12 9:43am Aug 12,2026

Outplane Shear (Vc)	19	19	-19	-19
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	54	54	54	54

Shear Stress Summary	73	73	34	34
----------------------	----	----	----	----

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	877	1025	1951	1556
------------------------	-----	------	------	------

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Membrane (P)	67	67	67	67
Outplane Bending (P)	171	-171	171	-171
Outplane Membrane (Mc)	-124	-124	124	124
Outplane Bending (Mc)	-688	688	688	-688
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	488	488	488	488

Outplane Stress Summary	-85	948	1539	-179
-------------------------	-----	-----	------	------

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	50	50	50	50
Inplane Bending (P)	531	-531	531	-531
Inplane Membrane (Mc)	-142	-142	142	142
Inplane Bending (Mc)	-2286	2286	2286	-2286
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	976	1023	976	1023

Inplane Stress Summary	-870	2686	3988	-1602
------------------------	------	------	------	-------

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-19	-19	19	19
Torsional Shear (Mt)	54	54	54	54

Shear Stress Summary	34	34	73	73
----------------------	----	----	----	----

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	872	2687	3990	1605
------------------------	-----	------	------	------

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the	Top	Top	Bottom	Bottom
hoop direction)	Outside	Inside	Outside	Inside



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Nozzle Calcs.: N2B- channel ou Nozl: 12 9:43am Aug 12,2026

Hoop Membrane (P)	67	67	67	67
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-129	-129	129	129
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	155	211	155	211

Hoop Stress Summary	93	149	352	408
---------------------	----	-----	-----	-----

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	6	6	6	6
Axial Bending (P)	69	-69	69	-69
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-58	-58	58	58
Axial Bending (ML)	-106	106	106	-106
Axial Pressure Stress	77	77	77	77

Axial Stress Summary	-10	63	318	-34
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	8	8	-8	-8
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	23	23	23	23

Shear Stress Summary	31	31	14	14
----------------------	----	----	----	----

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Outside	Inside	Outside	Inside

Two * Max Shear Stress	122	160	357	443
------------------------	-----	-----	-----	-----

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the hoop direction)	Left Outside	Left Inside	Right Outside	Right Inside

Hoop Membrane (P)	67	67	67	67
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-124	-124	124	124
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	155	211	155	211

Hoop Stress Summary	98	155	347	403
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to pipe cross-section)	Left Outside	Left Inside	Right Outside	Right Inside

Axial Membrane (P)	6	6	6	6
Axial Bending (P)	69	-69	69	-69
Axial Membrane (Mc)	-58	-58	58	58



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Nozzle Calcs.: N2B- channel ou Nozl: 12 9:43am Aug 12,2026

Axial Bending (Mc)	-341	341	341	-341
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	77	77	77	77

Axial Stress Summary	-245	298	553	-268
----------------------	------	-----	-----	------

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-8	-8	8	8
Shear due to Torsion	23	23	23	23
Shear Stress Summary	14	14	31	31

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	345	299	558	675

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	976	1023	976	1023	976	1023	976	1023
Circ. Pl (SUS)	-62	-62	196	196	-92	-92	192	192
Circ. Q (SUS)	-45	45	387	-387	-1755	1755	2817	-2817
Long. Pm (SUS)	488	488	488	488	488	488	488	488
Long. Pl (SUS)	-24	-24	124	124	-57	-57	191	191
Long. Q (SUS)	-273	273	1335	-1335	-517	517	859	-859
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	19	19	-19	-19	-19	-19	19	19
Shear Q (SUS)	54	54	54	54	54	54	54	54
Pm (SUS)	976	1023	976	1023	976	1023	976	1023
Pm+Pl (SUS)	914	961	1172	1219	884	931	1168	1215
Pm+Pl+Q (Total)	876	1024	1950	1556	872	2686	3987	1605

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1023.00	1407.13	Passed
Pm+Pl (SUS)	1219.59	2110.70	Passed
Pm+Pl+Q (TOTAL)	3987.18	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):



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Nozzle Calcs.: N2B- channel ou Nozl: 12 9:43am Aug 12,2026

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	155	211	155	211	155	211	155	211
Circ. Pl (SUS)	-62	-62	196	196	-57	-57	191	191
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	77	77	77	77	77	77	77	77
Long. Pl (SUS)	-52	-52	64	64	-52	-52	64	64
Long. Q (SUS)	-37	37	175	-175	-272	272	410	-410
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	8	8	-8	-8	-8	-8	8	8
Shear Q (SUS)	23	23	23	23	23	23	23	23
Pm (SUS)	155.0	211.0	155.0	211.0	155.0	211.0	155.0	211.0
Pm+Pl (SUS)	93.9	149.5	351.3	407.2	98.9	154.5	346.3	402.2
Pm+Pl+Q (Total)	121.9	158.9	356.5	442.0	346.3	298.6	555.6	673.9

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	211.00	1407.13	Passed
Pm+Pl (SUS)	407.24	2110.70	Passed
Pm+Pl+Q (TOTAL)	673.86	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Nozzle Calcs.: N4B- Shell outl

Nozl: 13 9:43am Aug 12,2026

Input, Nozzle Desc: N4B- Shell outl

From: 50

Pressure for Reinforcement Calculations	P	10.197	kgf/cm ²
Temperature for Internal Pressure	Temp	155	°C
Parent Material		SA-240 304L	
Parent Allowable Stress at Temperature	Sv	1167.53	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1172.61	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	700.00	mm
Shell Finished (Minimum) Thickness	t	10.0000	mm
Shell Internal Corrosion Allowance	c	0.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		172.55	cm
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-312 TP304L	
Material UNS Number		S30403	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	891.31	kgf/cm ²
Allowable Stress At Ambient	Sna	1172.61	kgf/cm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		6.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		120	
Flange Material		SA-182 F304L	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	305.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	6.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	1407.13	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1407.13	kgf/cm ²
Diameter of Pad along vessel surface	Dp	368.2750	mm
Thickness of Pad	te	8.0000	mm
Weld leg size between Pad and Shell	Wp	8.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	8.0000	mm
Reinforcing Pad Width		100.0000	mm
Flange Class		600	
Flange Grade		GR 2.3	

The Pressure Design option was Design Pressure + static head.



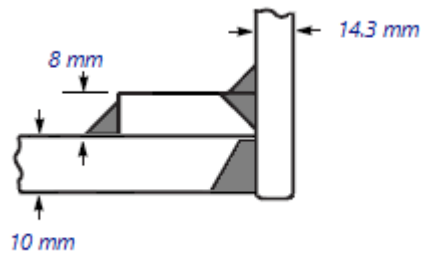
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Nozzle Calcs.: N4B- Shell outl

Nozl: 13 9:43am Aug 12,2026



Reinforcement CALCULATION, Description: N4B- Shell outl

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	6.625 in.
Actual Thickness Used in Calculation	0.562 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 10.2 \cdot 350.0 / (1168 \cdot 1.0 - 0.6 \cdot 10.2)$
 $= 3.0731 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 10.2 \cdot 84.1375 / (891 \cdot 1.0 + 0.4 \cdot 10.2)$
 $= 0.9582 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	285.4508 mm
Parallel to Vessel Wall, opening length	d	142.7254 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		25.0000 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 891.3 / 1167.5)$
 $= 0.763$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 891.3 / 1167.5)$
 $= 0.763$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 1407.1 / 1167.5)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.763, 1.0)$
 $= 0.763$



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Nozzle Calcs.: N4B- Shell outl

Nozl: 13 9:43am Aug 12,2026

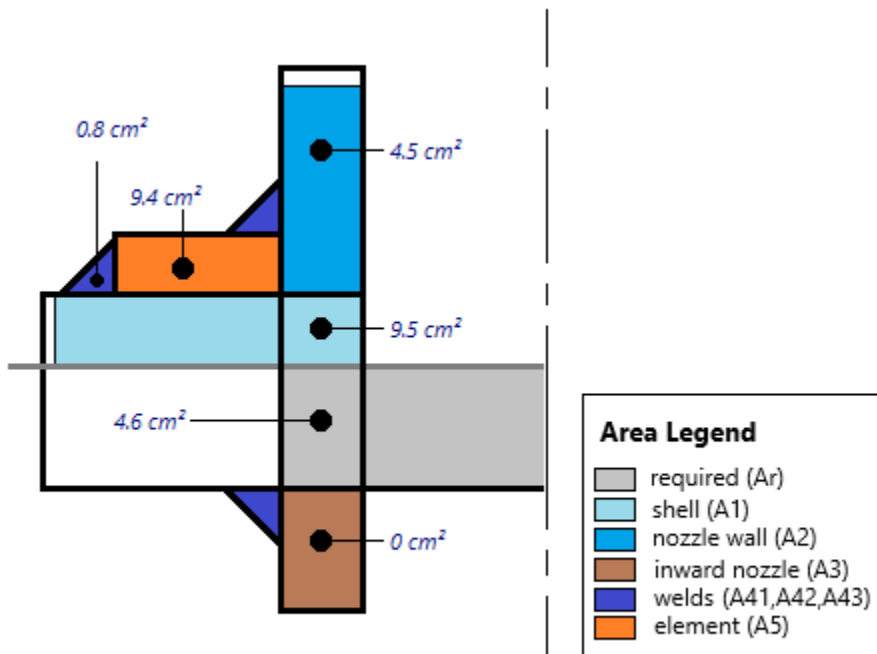


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	4.572	NA	NA
Area in Shell	A1	9.468	NA	NA
Area in Nozzle Wall	A2	4.510	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.763	NA	NA
Area in Element	A5	9.374	NA	NA
TOTAL AREA AVAILABLE	Atot	24.116	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (142.7254 \cdot 3.0731 \cdot 1.0 + 2 \cdot 12.7748 \cdot 3.0731 \cdot 1.0 \cdot (1 - 0.76)) \\
 &= 4.572 \text{ cm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 142.725(1.0 \cdot 10.0 - 1.0 \cdot 3.073) - 2 \cdot 12.775 \\
 &\quad (1.0 \cdot 10.0 - 1.0 \cdot 3.073) \cdot (1 - 0.763) \\
 &= 9.468 \text{ cm}^2
 \end{aligned}$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\
 &= (2 \cdot 25.0) \cdot (12.77 - 0.96) \cdot 0.7634
 \end{aligned}$$



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Nozzle Calcs.: N4B- Shell outl Nozl: 13 9:43am Aug 12,2026

$$= 4.510 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$$

$$= 10.0^2 \cdot 0.76 + (0.0)^2 \cdot 0.76 + 0.0^2 \cdot 1.0$$

$$= 0.763 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$$

$$= (285.4508 - 168.275) \cdot 8.0 \cdot 1.0$$

$$= 9.374 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 2.4582 \text{ mm}$

Wall Thickness per UG16(b), $tr16b = 3.0000 \text{ mm}$

Wall Thickness, shell/head, internal pressure $trb1 = 3.0731 \text{ mm}$

Wall Thickness $tb1 = \max(trb1, tr16b) = 3.0731 \text{ mm}$

Wall Thickness $tb2 = \max(trb2, tr16b) = 3.0000 \text{ mm}$

Wall Thickness per table UG-45 $tb3 = 7.7200 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[7.72, \max(3.0731, 3.0)]$$

$$= 3.0731 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(2.4582, 3.0731)$$

$$= 3.0731 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 \cdot 14.275 = 12.490 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	291.5,	Allowable	:	891.3 kgf/cm ²	Passed
Expansion	:	0.0,	Allowable	:	2288.4 kgf/cm ²	Passed
Occasional	:	26.1,	Allowable	:	1185.4 kgf/cm ²	Passed
Shear	:	178.1,	Allowable	:	623.9 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Pad Weld for Nozzle skipped, as Nozzle is not a Carbon Steel material.

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 8.0$, $c = 0.0 \text{ mm}$, $E^* = 1.0$

Thickness Ratio = $tr \cdot E^* / (t_g - c) = -1.0$, Temp. Reduction = 0°C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -29°C

Min Metal Temp. at Required thickness (UCS 66.1) -104°C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 8.0$, $c = 0.0 \text{ mm}$, $E^* = 1.0$

Thickness Ratio = $tr \cdot E^* / (t_g - c) = -1.0$, Temp. Reduction = 0°C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -29°C

Min Metal Temp. at Required thickness (UCS 66.1) -104°C



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Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Gov. MDMT of the Reinforcement Pad : -104 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: N4B- Shell outl

Intermediate Calc. for nozzle/shell Welds Tmin 8.0000 mm
 Intermediate Calc. for pad/shell Welds TminPad 8.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	5.6000 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm
Pad Weld	4.0000 = 0.5 * TminPad	5.6560 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$
 $= \max(0, (4.5718 - 9.4678 + 2 \cdot 12.7748 \cdot 0.7634 \cdot (1.0 \cdot 10.0 - 3.0731)) \cdot 1168)$
 $= \max(0, -4138.74) \text{ kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (4.5105 + 9.3741 + 0.7634 - 0.0 \cdot 0.76) \cdot 1168$
 $= 17101.85 \text{ kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (4.5105 + 0.0 + 0.7634 + (1.9505)) \cdot 1168$
 $= 8434.64 \text{ kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (4.5105 + 0.0 + 0.7634 + 9.3741 + (1.9505)) \cdot 1168$
 $= 19379.10 \text{ kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.1416/2.0) \cdot 168.275 \cdot 10.0 \cdot 0.49 \cdot 891$
 $= 11544. \text{ kgf}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.1416/2.0) \cdot 368.275 \cdot 8.0 \cdot 0.49 \cdot 1168$
 $= 26476. \text{ kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.1416 \cdot 77.7501) \cdot (14.2748 - 1.5) \cdot 0.7 \cdot 891$
 $= 19468. \text{ kgf}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$
 $= (3.1416/2) \cdot 168.275 \cdot 8.0 \cdot 0.74 \cdot 1407$
 $= 22019. \text{ kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$
 $= (3.1416/2.0) \cdot 168.275 \cdot (6.0 - 0.0) \cdot 0.74 \cdot 1168$



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= 13702. kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (26476 + 19468) = 45944 \text{ kgf}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ = (11544 + 22019 + 13702 + 0) = 47265 \text{ kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw}) \\ = (26476 + 13702 + 0) = 40178 \text{ kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 45943	kgf, must exceed W = 0 kgf	or W1 = 17101 kgf
Path 2-2 = 47265	kgf, must exceed W = 0 kgf	or W2 = 8434 kgf
Path 3-3 = 40177	kgf, must exceed W = 0 kgf	or W3 = 19379 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 32 kgf/cm²

The Drop for this Nozzle is : 10.2635 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 325.2635 mm

Input Echo, WRC297 Item 1, Description: N4B- Shell out:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	0.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	700.000	mm
Shell Thickness	10.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-240 304L	
Vessel Cold S.I. Allowable	Smc 1172.61	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1167.53	kgf/cm ²

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	168.275	mm
Nozzle Thickness	14.2748	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-312 TP304L	
Nozzle Cold S.I. Allowable	SNmc 1172.61	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 891.31	kgf/cm ²

Diameter of Reinforcing Pad	368.275	mm
Thickness of Reinforcing Pad	8.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSStained loads.

Design Internal Pressure	Dp	10.20	kgf/cm ²
Radial Load	P	-1376.68	kgf
Circumferential Shear	Vc	994.27	kgf
Longitudinal Shear	Vl	994.27	kgf
Circumferential Moment	Mc	458.71	kgf-m.
Longitudinal Moment	Ml	458.71	kgf-m.
Torsional Moment	Mt	527.52	kgf-m.

Include Axial Pressure Thrust	No
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Nozzle Calcs.: N4B- Shell outl Nozl: 13 9:43am Aug 12,2026

Include Pressure Stress Indices per Div. 2 No
Local Loads applied at end of Nozzle/Attachment No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{359.0(10.0 - 0.0)}$$

$$= 366.000 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 718.000 mm
Nozzle Outside Diameter (d) = 168.275 mm
Vessel Thickness used (T) = 18.000 mm
Nozzle Thickness used (t) = 12.775 mm
T / t = 1.409
d / t = 13.172

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.480$$

Nr/P = 0.104
Mr/P = 0.133
M0/P = 0.055
N0/P = 0.130
MrD/Mc = 0.241
NrDL/Mc = 0.150
M0d/Mc = 0.118
N0DL/Mc = 0.069
MrD/Ml = 0.153
NrDL/Ml = 0.105
M0D/Ml = 0.077
N0DL/Ml = 0.166

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	55	55	55	55
Outplane Bending (P)	140	-140	140	-140
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-139	-139	139	139
Outplane Bending (ML)	-386	386	386	-386
Normal Pressure Stress	193	203	193	203
Outplane Stress Summary	-137	366	914	-128

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	44	44	44	44
Inplane Bending (P)	338	-338	338	-338
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-87	-87	87	87
Inplane Bending (ML)	-773	773	773	-773
Inplane Pressure Stress	96	96	96	96
Inplane Stress Summary	-381	487	1340	-883



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Nozzle Calcs.: N4B- Shell outl Nozl: 13 9:43am Aug 12,2026

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE			Au	Al	Bu	B1
(Shear stress normal to longitudinal plane)			Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)			20	20	-20	-20
Outplane Shear (Vl)			0	0	0	0
Outplane Shear (Mt)			65	65	65	65
Shear Stress Summary			86	86	44	44

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE			Au	Al	Bu	B1
(Stress Intensities)			Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress			409	532	1345	885

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Stresses Normal to circumferential plane)			Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)			55	55	55	55
Outplane Bending (P)			140	-140	140	-140
Outplane Membrane (Mc)			-57	-57	57	57
Outplane Bending (Mc)			-595	595	595	-595
Outplane Membrane (ML)			0	0	0	0
Outplane Bending (ML)			0	0	0	0
Normal Pressure Stress			96	96	96	96
Outplane Stress Summary			-361	549	945	-525

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Stresses parallel to circumferential plane)			Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)			44	44	44	44
Inplane Bending (P)			338	-338	338	-338
Inplane Membrane (Mc)			-125	-125	125	125
Inplane Bending (Mc)			-1218	1218	1218	-1218
Inplane Membrane (ML)			0	0	0	0
Inplane Bending (ML)			0	0	0	0
Inplane Pressure Stress			193	203	193	203
Inplane Stress Summary			-768	1002	1920	-1183

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Shear stress normal to circumferential plane)			Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			-20	-20	20	20
Torsional Shear (Mt)			65	65	65	65
Shear Stress Summary			44	44	86	86

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Stress Intensities)			Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress			772	1006	1928	1194



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Nozzle Calcs.: N4B- Shell outl Nozl: 13 9:43am Aug 12,2026

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	55	55	55	55
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-139	-139	139	139
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	62	74	62	74
Hoop Stress Summary	-21	-9	257	269

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	22	22	22	22
Axial Bending (P)	407	-407	407	-407
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-203	-203	203	203
Axial Bending (ML)	-1011	1011	1011	-1011
Axial Pressure Stress	31	31	31	31
Axial Stress Summary	-753	454	1676	-1163

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	29	29	-29	-29
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	92	92	92	92
Shear Stress Summary	122	122	63	63

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	773	523	1679	1438

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	55	55	55	55
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-57	-57	57	57
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	62	74	62	74
Hoop Stress Summary	60	72	175	188

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to	Cu Left	Cl Left	Du Right	Dl Right
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Nozzle Calcs.: N4B- Shell outl Nozl: 13 9:43am Aug 12,2026

pipe cross-section)	Outside	Inside	Outside	Inside
Axial Membrane (P)	22	22	22	22
Axial Bending (P)	407	-407	407	-407
Axial Membrane (Mc)	-203	-203	203	203
Axial Bending (Mc)	-1670	1670	1670	-1670
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	31	31	31	31

Axial Stress Summary	-1412	1112	2334	-1821
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-29	-29	29	29
Shear due to Torsion	92	92	92	92

Shear Stress Summary	63	63	122	122
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Outside	Inside	Outside	Inside

Two * Max Shear Stress	1477	1116	2341	2024
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Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{355.0(10.0 - 0.0)}$$

$$= 364.895 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	710.000 mm
Nozzle Outside Diameter	(d) =	368.275 mm
Vessel Thickness used	(T) =	10.000 mm
Nozzle Thickness used	(t) =	112.775 mm
T / t	=	0.089
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.371$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.049
Mr/P	=	0.077
M0/P	=	0.022
N0/P	=	0.031
MrD/Mc	=	0.248
NrDL/Mc	=	0.102
M0d/Mc	=	0.074
N0DL/Mc	=	0.056
MrD/Ml	=	0.056
NrDL/Ml	=	0.055
M0D/Ml	=	0.017
N0DL/Ml	=	0.033

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom



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longitudinal plane)	Outside	Inside	Outside	Inside
Outplane Membrane (P)	42	42	42	42
Outplane Bending (P)	181	-181	181	-181
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-40	-40	40	40
Outplane Bending (ML)	-127	127	127	-127
Normal Pressure Stress	351	362	351	362
Outplane Stress Summary	406	310	744	135

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside
Inplane Membrane (P)	67	67	67	67
Inplane Bending (P)	636	-636	636	-636
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-68	-68	68	68
Inplane Bending (ML)	-420	420	420	-420
Inplane Pressure Stress	175	175	175	175
Inplane Stress Summary	391	-41	1368	-744

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	17	17	-17	-17
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	24	24	24	24
Shear Stress Summary	41	41	7	7

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	441	362	1368	880

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside
Outplane Membrane (P)	42	42	42	42
Outplane Bending (P)	181	-181	181	-181
Outplane Membrane (Mc)	-70	-70	70	70
Outplane Bending (Mc)	-552	552	552	-552
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	175	175	175	175
Outplane Stress Summary	-223	519	1022	-445

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside



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Inplane Membrane (P)	67	67	67	67
Inplane Bending (P)	636	-636	636	-636
Inplane Membrane (Mc)	-127	-127	127	127
Inplane Bending (Mc)	-1850	1850	1850	-1850
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	351	362	351	362
Inplane Stress Summary	-922	1516	3034	-1930

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0	0
Outplane Shear (Vl)	-17	-17	17	17	17
Torsional Shear (Mt)	24	24	24	24	24
Shear Stress Summary	7	7	41	41	

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stress Intensities)		Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	922	1516	3035	1931	

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	193	203	193	203	193	203	193	203
Circ. Pl (SUS)	-84	-84	194	194	-81	-81	169	169
Circ. Q (SUS)	-246	246	526	-526	-880	880	1556	-1556
Long. Pm (SUS)	96	96	96	96	96	96	96	96
Long. Pl (SUS)	-43	-43	131	131	-2	-2	112	112
Long. Q (SUS)	-435	435	1111	-1111	-455	455	735	-735
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	20	20	-20	-20	-20	-20	20	20
Shear Q (SUS)	65	65	65	65	65	65	65	65
Pm (SUS)	193.0	203.0	193.0	203.0	193.0	203.0	193.0	203.0
Pm+Pl (SUS)	115.4	124.6	389.5	399.3	124.9	132.4	364.6	374.4
Pm+Pl+Q (Total)	408	531	1342	886	772	1006	1925	1194

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	203.00	1167.53	Passed
Pm+Pl (SUS)	399.32	1751.29	Passed
Pm+Pl+Q (TOTAL)	1925.35	3502.58	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	351	362	351	362	351	362	351	362
Circ. Pl (SUS)	2	2	82	82	-60	-60	194	194
Circ. Q (SUS)	54	-54	308	-308	-1214	1214	2486	-2486
Long. Pm (SUS)	175	175	175	175	175	175	175	175
Long. Pl (SUS)	-1	-1	135	135	-28	-28	112	112
Long. Q (SUS)	216	-216	1056	-1056	-371	371	733	-733
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	17	17	-17	-17	-17	-17	17	17
Shear Q (SUS)	24	24	24	24	24	24	24	24
Pm (SUS)	351.0	362.0	351.0	362.0	351.0	362.0	351.0	362.0
Pm+Pl (SUS)	354.6	365.5	435.3	446.1	293.0	303.8	546.1	557.1
Pm+Pl+Q (Total)	440	361	1366	882	923	1516	3031	1931

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	362.00	1167.53	Passed
Pm+Pl (SUS)	557.07	1751.29	Passed
Pm+Pl+Q (TOTAL)	3031.84	3502.58	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	62	74	62	74	62	74	62	74
Circ. Pl (SUS)	-84	-84	194	194	-2	-2	112	112
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	31	31	31	31	31	31	31	31
Long. Pl (SUS)	-181	-181	225	225	-181	-181	225	225
Long. Q (SUS)	-604	604	1418	-1418	-1263	1263	2077	-2077
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	29	29	-29	-29	-29	-29	29	29
Shear Q (SUS)	92	92	92	92	92	92	92	92
Pm (SUS)	62.0	74.0	62.0	74.0	62.0	74.0	62.0	74.0
Pm+Pl (SUS)	156.3	155.8	285.0	291.6	217.9	229.5	265.2	266.5
Pm+Pl+Q (Total)	773	523	1676	1435	1478	1116	2339	2021



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Nozzle Calcs.: N4B- Shell outl

Nozl: 13 9:43am Aug 12,2026

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	74.00	891.31	Passed
Pm+Pl (SUS)	291.61	1336.96	Passed
Pm+Pl+Q (TOTAL)	2339.76	2673.92	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Nozzle Calcs.: K1- Pressure re

Nozl: 14 9:43am Aug 12,2026

Input, Nozzle Desc: K1- Pressure re

From: 50

Pressure for Reinforcement Calculations	P	10.197	kgf/cm ²
Temperature for Internal Pressure	Temp	155	°C
Parent Material		SA-240 304L	
Parent Allowable Stress at Temperature	Sv	1167.53	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1172.61	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	700.00	mm
Shell Finished (Minimum) Thickness	t	10.0000	mm
Shell Internal Corrosion Allowance	c	0.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		442.55	cm
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-312 TP304L	
Material UNS Number		S30403	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	891.31	kgf/cm ²
Allowable Stress At Ambient	Sna	1172.61	kgf/cm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		6.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		120	
Flange Material		SA-182 F304L	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	305.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	6.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	1407.13	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1407.13	kgf/cm ²
Diameter of Pad along vessel surface	Dp	368.2750	mm
Thickness of Pad	te	8.0000	mm
Weld leg size between Pad and Shell	Wp	8.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	8.0000	mm
Reinforcing Pad Width		100.0000	mm
Flange Class		300	
Flange Grade		GR 2.3	

The Pressure Design option was Design Pressure + static head.



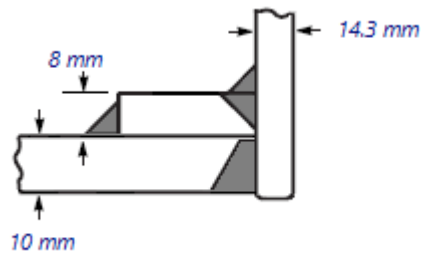
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Nozzle Calcs.: K1- Pressure re

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Reinforcement CALCULATION, Description: K1- Pressure re

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	6.625 in.
Actual Thickness Used in Calculation	0.562 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 10.2 \cdot 350.0 / (1168 \cdot 1.0 - 0.6 \cdot 10.2)$
 $= 3.0731 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 10.2 \cdot 84.1375 / (891 \cdot 1.0 + 0.4 \cdot 10.2)$
 $= 0.9582 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	285.4508	mm
Parallel to Vessel Wall, opening length	d	142.7254	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		25.0000	mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 891.3 / 1167.5)$
 $= 0.763$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 891.3 / 1167.5)$
 $= 0.763$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 1407.1 / 1167.5)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.763, 1.0)$
 $= 0.763$

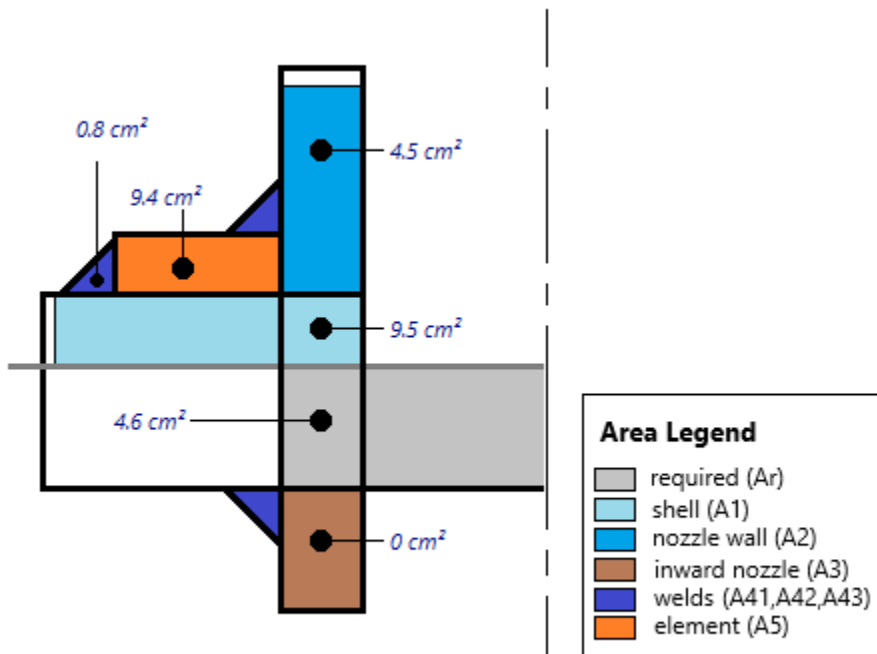


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	4.572	NA	NA
Area in Shell	A1	9.468	NA	NA
Area in Nozzle Wall	A2	4.510	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.763	NA	NA
Area in Element	A5	9.374	NA	NA
TOTAL AREA AVAILABLE	Atot	24.116	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (142.7254 \cdot 3.0731 \cdot 1.0 + 2 \cdot 12.7748 \cdot 3.0731 \cdot 1.0 \cdot (1 - 0.76)) \\
 &= 4.572 \text{ cm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 142.725(1.0 \cdot 10.0 - 1.0 \cdot 3.073) - 2 \cdot 12.775 \\
 &\quad (1.0 \cdot 10.0 - 1.0 \cdot 3.073) \cdot (1 - 0.763) \\
 &= 9.468 \text{ cm}^2
 \end{aligned}$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\
 &= (2 \cdot 25.0) \cdot (12.77 - 0.96) \cdot 0.7634
 \end{aligned}$$



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$$= 4.510 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:
 $= W_o^2 \cdot fr3 + (W_i - can/0.707)^2 \cdot fr2 + W_p^2 \cdot fr4$
 $= 10.0^2 \cdot 0.76 + (0.0)^2 \cdot 0.76 + 0.0^2 \cdot 1.0$
 $= 0.763 \text{ cm}^2$

Area Available in Element [A5]:
 $= (\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lp}, t_e)) \cdot fr4$
 $= (285.4508 - 168.275) \cdot 8.0 \cdot 1.0$
 $= 9.374 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 2.4582 \text{ mm}$
Wall Thickness per UG16(b),	$tr16b = 3.0000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$trb1 = 3.0731 \text{ mm}$
Wall Thickness	$tb1 = \max(trb1, tr16b) = 3.0731 \text{ mm}$
Wall Thickness	$tb2 = \max(trb2, tr16b) = 3.0000 \text{ mm}$
Wall Thickness per table UG-45	$tb3 = 7.7200 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.72, \max(3.0731, 3.0)]$
 $= 3.0731 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(t_a, t_b)$
 $= \max(2.4582, 3.0731)$
 $= 3.0731 \text{ mm}$

Available Nozzle Neck Thickness = $0.875 \cdot 14.275 = 12.490 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 291.5,	Allowable	: 891.3 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 2288.4 kgf/cm ²	Passed
Occasional	: 26.1,	Allowable	: 1185.4 kgf/cm ²	Passed
Shear	: 178.1,	Allowable	: 623.9 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle to Flange Weld skipped as Nozzle is not a Carbon Steel material.

Nozzle-Pad Weld for Nozzle skipped, as Nozzle is not a Carbon Steel material.

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govern. thk, $t_g = 8.0$, $c = 0.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $tr \cdot E^* / (t_g - c) = -1.0$, Temp. Reduction = 0°C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govern. thk, $t_g = 8.0$, $c = 0.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $tr \cdot E^* / (t_g - c) = -1.0$, Temp. Reduction = 0°C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
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Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Gov. MDMT of the Reinforcement Pad : -104 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: K1- Pressure re

Intermediate Calc. for nozzle/shell Welds Tmin 8.0000 mm
 Intermediate Calc. for pad/shell Welds TminPad 8.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	5.6000 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm
Pad Weld	4.0000 = 0.5 * TminPad	5.6560 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$
 $= \max(0, (4.5718 - 9.4678 + 2 \cdot 12.7748 \cdot 0.7634 \cdot (1.0 \cdot 10.0 - 3.0731)) \cdot 1168)$
 $= \max(0, -4138.74) \text{ kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (4.5105 + 9.3741 + 0.7634 - 0.0 \cdot 0.76) \cdot 1168$
 $= 17101.85 \text{ kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (4.5105 + 0.0 + 0.7634 + (1.9505)) \cdot 1168$
 $= 8434.64 \text{ kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (4.5105 + 0.0 + 0.7634 + 9.3741 + (1.9505)) \cdot 1168$
 $= 19379.10 \text{ kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.1416/2.0) \cdot 168.275 \cdot 10.0 \cdot 0.49 \cdot 891$
 $= 11544. \text{ kgf}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.1416/2.0) \cdot 368.275 \cdot 8.0 \cdot 0.49 \cdot 1168$
 $= 26476. \text{ kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.1416 \cdot 77.7501) \cdot (14.2748 - 1.5) \cdot 0.7 \cdot 891$
 $= 19468. \text{ kgf}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$
 $= (3.1416/2) \cdot 168.275 \cdot 8.0 \cdot 0.74 \cdot 1407$
 $= 22019. \text{ kgf}$

Tension, Shell Groove Weld [Tngw]:



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$$\begin{aligned}
 &= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng \\
 &= (3.1416/2.0) \cdot 168.275 \cdot (6.0 - 0.0) \cdot 0.74 \cdot 1168 \\
 &= 13702. \text{ kgf}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (26476 + 19468) = 45944 \text{ kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\
 &= (11544 + 22019 + 13702 + 0) = 47265 \text{ kgf}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (\text{Spew} + \text{Tngw} + \text{Sinw}) \\
 &= (26476 + 13702 + 0) = 40178 \text{ kgf}
 \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 45943	kgf, must exceed W = 0 kgf	or W1 = 17101 kgf
Path 2-2 = 47265	kgf, must exceed W = 0 kgf	or W2 = 8434 kgf
Path 3-3 = 40177	kgf, must exceed W = 0 kgf	or W3 = 19379 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 32 kgf/cm²

The Drop for this Nozzle is : 10.2635 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 325.2635 mm

Input Echo, WRC297 Item 1, Description: K1- Pressure r:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	0.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	700.000	mm
Shell Thickness	10.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-240 304L	
Vessel Cold S.I. Allowable	Smc 1172.61	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1167.53	kgf/cm ²
Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	168.275	mm
Nozzle Thickness	14.2748	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-312 TP304L	
Nozzle Cold S.I. Allowable	SNmc 1172.61	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 891.31	kgf/cm ²
Diameter of Reinforcing Pad	368.275	mm
Thickness of Reinforcing Pad	8.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	10.20	kgf/cm ²
Radial Load	P	-1376.68	kgf
Circumferential Shear	Vc	994.27	kgf
Longitudinal Shear	Vl	994.27	kgf
Circumferential Moment	Mc	458.71	kgf-m.
Longitudinal Moment	Ml	458.71	kgf-m.
Torsional Moment	Mt	527.52	kgf-m.



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Include Axial Pressure Thrust No
 Include Pressure Stress Indices per Div. 2 No
 Local Loads applied at end of Nozzle/Attachment No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{359.0(10.0 - 0.0)}$$

$$= 366.000 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 718.000 mm
 Nozzle Outside Diameter (d) = 168.275 mm
 Vessel Thickness used (T) = 18.000 mm
 Nozzle Thickness used (t) = 12.775 mm
 T / t = 1.409
 d / t = 13.172

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.480$$

Nr/P = 0.104
 Mr/P = 0.133
 M0/P = 0.055
 N0/P = 0.130
 MrD/Mc = 0.241
 NrDL/Mc = 0.150
 M0d/Mc = 0.118
 N0DL/Mc = 0.069
 MrD/M1 = 0.153
 NrDL/M1 = 0.105
 M0D/M1 = 0.077
 N0DL/M1 = 0.166

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Membrane (P)	55	55	55	55
Outplane Bending (P)	140	-140	140	-140
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-139	-139	139	139
Outplane Bending (ML)	-386	386	386	-386
Normal Pressure Stress	193	203	193	203
Outplane Stress Summary	-137	366	914	-128

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Inplane Membrane (P)	44	44	44	44
Inplane Bending (P)	338	-338	338	-338
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-87	-87	87	87
Inplane Bending (ML)	-773	773	773	-773
Inplane Pressure Stress	96	96	96	96



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Nozzle Calcs.: K1- Pressure re Nozl: 14 9:43am Aug 12,2026

Inplane Stress Summary	-381	487	1340	-883
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	20	20	-20	-20
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	65	65	65	65

Shear Stress Summary	86	86	44	44
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	409	532	1345	885
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	C1	Du	D1
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	55	55	55	55
Outplane Bending (P)	140	-140	140	-140
Outplane Membrane (Mc)	-57	-57	57	57
Outplane Bending (Mc)	-595	595	595	-595
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	96	96	96	96

Outplane Stress Summary	-361	549	945	-525
-------------------------	------	-----	-----	------

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	C1	Du	D1
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	44	44	44	44
Inplane Bending (P)	338	-338	338	-338
Inplane Membrane (Mc)	-125	-125	125	125
Inplane Bending (Mc)	-1218	1218	1218	-1218
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	193	203	193	203

Inplane Stress Summary	-768	1002	1920	-1183
------------------------	------	------	------	-------

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	C1	Du	D1
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-20	-20	20	20
Torsional Shear (Mt)	65	65	65	65

Shear Stress Summary	44	44	86	86
----------------------	----	----	----	----

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	C1	Du	D1
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside



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Nozzle Calcs.: K1- Pressure re Nozl: 14 9:43am Aug 12,2026

Two * Max Shear Stress	772	1006	1928	1194
------------------------	-----	------	------	------

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	55	55	55	55
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-139	-139	139	139
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	62	74	62	74
Hoop Stress Summary	-21	-9	257	269

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	22	22	22	22
Axial Bending (P)	407	-407	407	-407
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-203	-203	203	203
Axial Bending (ML)	-1011	1011	1011	-1011
Axial Pressure Stress	31	31	31	31
Axial Stress Summary	-753	454	1676	-1163

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	29	29	-29	-29
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	92	92	92	92
Shear Stress Summary	122	122	63	63

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	773	523	1679	1438

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	55	55	55	55
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-57	-57	57	57
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	62	74	62	74
Hoop Stress Summary	60	72	175	188

Nozzle Stresses (kgf/cm²):



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Nozzle Calcs.: K1- Pressure re Nozl: 14 9:43am Aug 12,2026

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	22	22	22	22
Axial Bending (P)	407	-407	407	-407
Axial Membrane (Mc)	-203	-203	203	203
Axial Bending (Mc)	-1670	1670	1670	-1670
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	31	31	31	31

Axial Stress Summary	-1412	1112	2334	-1821
----------------------	-------	------	------	-------

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-29	-29	29	29
Shear due to Torsion	92	92	92	92

Shear Stress Summary	63	63	122	122
----------------------	----	----	-----	-----

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	1477	1116	2341	2024

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{(355.0(10.0 - 0.0))}$$

$$= 364.895 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	710.000 mm
Nozzle Outside Diameter	(d) =	368.275 mm
Vessel Thickness used	(T) =	10.000 mm
Nozzle Thickness used	(t) =	112.775 mm
T / t	=	0.089
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.371$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.049
Mr/P	=	0.077
M0/P	=	0.022
N0/P	=	0.031
MrD/Mc	=	0.248
NrDL/Mc	=	0.102
M0d/Mc	=	0.074
N0DL/Mc	=	0.056
MrD/Ml	=	0.056
NrDL/Ml	=	0.055
M0D/Ml	=	0.017
N0DL/Ml	=	0.033

Vessel Stresses (kgf/cm²):



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Nozzle Calcs.: K1- Pressure re Nozl: 14 9:43am Aug 12,2026

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Membrane (P)	42	42	42	42
Outplane Bending (P)	181	-181	181	-181
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-40	-40	40	40
Outplane Bending (ML)	-127	127	127	-127
Normal Pressure Stress	351	362	351	362
Outplane Stress Summary	406	310	744	135

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Inplane Membrane (P)	67	67	67	67
Inplane Bending (P)	636	-636	636	-636
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-68	-68	68	68
Inplane Bending (ML)	-420	420	420	-420
Inplane Pressure Stress	175	175	175	175
Inplane Stress Summary	391	-41	1368	-744

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Shear (Vc)	17	17	-17	-17
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	24	24	24	24
Shear Stress Summary	41	41	7	7

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Two * Max Shear Stress	441	362	1368	880

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Outplane Membrane (P)	42	42	42	42
Outplane Bending (P)	181	-181	181	-181
Outplane Membrane (Mc)	-70	-70	70	70
Outplane Bending (Mc)	-552	552	552	-552
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	175	175	175	175
Outplane Stress Summary	-223	519	1022	-445

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to	Cu Left	Cl Left	Du Right	D1 Right
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Nozzle Calcs.: K1- Pressure re Nozl: 14 9:43am Aug 12,2026

circumferential plane)	Outside	Inside	Outside	Inside
Inplane Membrane (P)	67	67	67	67
Inplane Bending (P)	636	-636	636	-636
Inplane Membrane (Mc)	-127	-127	127	127
Inplane Bending (ML)	-1850	1850	1850	-1850
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	351	362	351	362
Inplane Stress Summary	-922	1516	3034	-1930

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)		Left	Left	Right	Right
		Outside	Inside	Outside	Inside
Outplane Shear (Vc)		0	0	0	0
Outplane Shear (Vl)		-17	-17	17	17
Torsional Shear (Mt)		24	24	24	24
Shear Stress Summary		7	7	41	41

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stress Intensities		Left	Left	Right	Right
		Outside	Inside	Outside	Inside
Two * Max Shear Stress		922	1516	3035	1931

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	193	203	193	203	193	203	193	203
Circ. Pl (SUS)	-84	-84	194	194	-81	-81	169	169
Circ. Q (SUS)	-246	246	526	-526	-880	880	1556	-1556
Long. Pm (SUS)	96	96	96	96	96	96	96	96
Long. Pl (SUS)	-43	-43	131	131	-2	-2	112	112
Long. Q (SUS)	-435	435	1111	-1111	-455	455	735	-735
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	20	20	-20	-20	-20	-20	20	20
Shear Q (SUS)	65	65	65	65	65	65	65	65
Pm (SUS)	193.0	203.0	193.0	203.0	193.0	203.0	193.0	203.0
Pm+Pl (SUS)	115.4	124.6	389.5	399.3	124.9	132.4	364.6	374.4
Pm+Pl+Q (Total)	408	531	1342	886	772	1006	1925	1194

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	203.00	1167.53	Passed
Pm+Pl (SUS)	399.32	1751.29	Passed
Pm+Pl+Q (TOTAL)	1925.35	3502.58	Passed



Because only sustained loads were specified, the P_m+P_l+Q allowable was $3 * S_mh$.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	351	362	351	362	351	362	351	362
Circ. Pl (SUS)	2	2	82	82	-60	-60	194	194
Circ. Q (SUS)	54	-54	308	-308	-1214	1214	2486	-2486
Long. Pm (SUS)	175	175	175	175	175	175	175	175
Long. Pl (SUS)	-1	-1	135	135	-28	-28	112	112
Long. Q (SUS)	216	-216	1056	-1056	-371	371	733	-733
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	17	17	-17	-17	-17	-17	17	17
Shear Q (SUS)	24	24	24	24	24	24	24	24
Pm (SUS)	351.0	362.0	351.0	362.0	351.0	362.0	351.0	362.0
Pm+Pl (SUS)	354.6	365.5	435.3	446.1	293.0	303.8	546.1	557.1
Pm+Pl+Q (Total)	440	361	1366	882	923	1516	3031	1931

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	362.00	1167.53	Passed
Pm+Pl (SUS)	557.07	1751.29	Passed
Pm+Pl+Q (TOTAL)	3031.84	3502.58	Passed

Because only sustained loads were specified, the P_m+P_l+Q allowable was $3 * S_mh$.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	62	74	62	74	62	74	62	74
Circ. Pl (SUS)	-84	-84	194	194	-2	-2	112	112
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	31	31	31	31	31	31	31	31
Long. Pl (SUS)	-181	-181	225	225	-181	-181	225	225
Long. Q (SUS)	-604	604	1418	-1418	-1263	1263	2077	-2077
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	29	29	-29	-29	-29	-29	29	29
Shear Q (SUS)	92	92	92	92	92	92	92	92
Pm (SUS)	62.0	74.0	62.0	74.0	62.0	74.0	62.0	74.0
Pm+Pl (SUS)	156.3	155.8	285.0	291.6	217.9	229.5	265.2	266.5
Pm+Pl+Q (Total)	773	523	1676	1435	1478	1116	2339	2021



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Nozzle Calcs.: K1- Pressure re Nozl: 14 9:43am Aug 12,2026

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	74.00	891.31	Passed
Pm+Pl (SUS)	291.61	1336.96	Passed
Pm+Pl+Q (TOTAL)	2339.76	2673.92	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Nozzle Calcs.: N3B- Shell inle

Nozl: 15 9:43am Aug 12,2026

Input, Nozzle Desc: N3B- Shell inle

From: 50

Pressure for Reinforcement Calculations	P	10.267	kgf/cm ²
Temperature for Internal Pressure	Temp	155	°C
Parent Material		SA-240 304L	
Parent Allowable Stress at Temperature	Sv	1167.53	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1172.61	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	700.00	mm
Shell Finished (Minimum) Thickness	t	10.0000	mm
Shell Internal Corrosion Allowance	c	0.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		172.55	cm
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-312 TP304L	
Material UNS Number		S30403	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	891.31	kgf/cm ²
Allowable Stress At Ambient	Sna	1172.61	kgf/cm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		6.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		120	
Flange Material		SA-182 F304L	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	305.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	6.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	1407.13	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1407.13	kgf/cm ²
Diameter of Pad along vessel surface	Dp	368.2750	mm
Thickness of Pad	te	8.0000	mm
Weld leg size between Pad and Shell	Wp	8.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	8.0000	mm
Reinforcing Pad Width		100.0000	mm
Flange Class		300	
Flange Grade		GR 2.3	

The Pressure Design option was Design Pressure + static head.



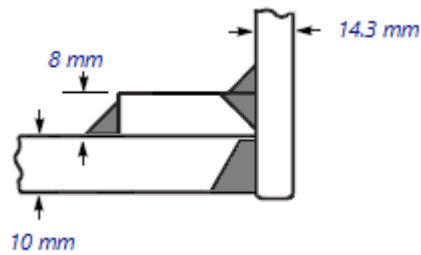
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Nozzle Calcs.: N3B- Shell inle

Nozl: 15 9:43am Aug 12,2026



Reinforcement CALCULATION, Description: N3B- Shell inle

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	6.625 in.
Actual Thickness Used in Calculation	0.562 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 10.27 \cdot 350.0 / (1168 \cdot 1.0 - 0.6 \cdot 10.27)$
 $= 3.0943 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 10.27 \cdot 84.1375 / (891 \cdot 1.0 + 0.4 \cdot 10.27)$
 $= 0.9648 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	285.4508 mm
Parallel to Vessel Wall, opening length	d	142.7254 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		25.0000 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 891.3 / 1167.5)$
 $= 0.763$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 891.3 / 1167.5)$
 $= 0.763$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 1407.1 / 1167.5)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.763, 1.0)$
 $= 0.763$



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Nozzle Calcs.: N3B- Shell inle

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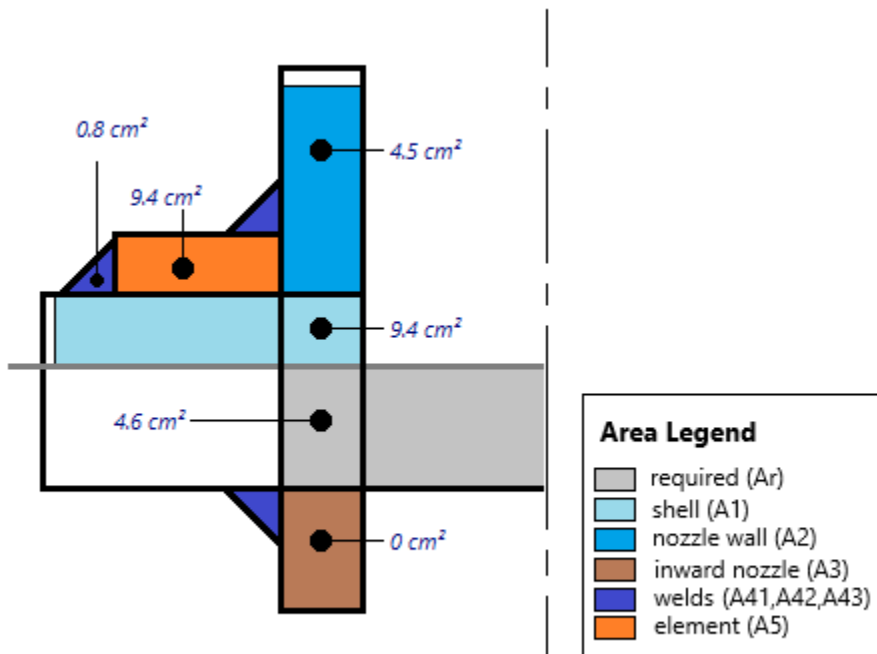


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	4.603	NA	NA
Area in Shell	A1	9.439	NA	NA
Area in Nozzle Wall	A2	4.508	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.763	NA	NA
Area in Element	A5	9.374	NA	NA
TOTAL AREA AVAILABLE	Atot	24.084	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (142.7254 \cdot 3.0943 \cdot 1.0 + 2 \cdot 12.7748 \cdot 3.0943 \cdot 1.0 \cdot (1 - 0.76)) \\
 &= 4.603 \text{ cm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 142.725(1.0 \cdot 10.0 - 1.0 \cdot 3.094) - 2 \cdot 12.775 \\
 &\quad (1.0 \cdot 10.0 - 1.0 \cdot 3.094) \cdot (1 - 0.763) \\
 &= 9.439 \text{ cm}^2
 \end{aligned}$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\
 &= (2 \cdot 25.0) \cdot (12.77 - 0.96) \cdot 0.7634
 \end{aligned}$$



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 Nozzle Calcs.: N3B- Shell inle Nozl: 15 9:43am Aug 12,2026

= 4.508 cm²

Area Available in Welds [A41 + A42 + A43]:
 = $W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$
 = $10.0^2 \cdot 0.76 + (0.0)^2 \cdot 0.76 + 0.0^2 \cdot 1.0$
 = 0.763 cm²

Area Available in Element [A5]:
 = $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lp}, t_e)) \cdot fr_4$
 = $(285.4508 - 168.275) \cdot 8.0 \cdot 1.0$
 = 9.374 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 2.4648 mm
Wall Thickness per UG16(b),	tr16b = 3.0000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 3.0943 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 3.0943 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm
Wall Thickness per table UG-45	tb3 = 7.7200 mm

Determine Nozzle Thickness candidate [tb]:
 = min[tb3, max(tb1, tb2)]
 = min[7.72, max(3.0943, 3.0)]
 = 3.0943 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = max(ta, tb)
 = max(2.4648, 3.0943)
 = 3.0943 mm

Available Nozzle Neck Thickness = $0.875 \cdot 14.275 = 12.490$ mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 291.7,	Allowable	: 891.3 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 2288.2 kgf/cm ²	Passed
Occasional	: 26.3,	Allowable	: 1185.4 kgf/cm ²	Passed
Shear	: 178.1,	Allowable	: 623.9 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle to Flange Weld skipped as Nozzle is not a Carbon Steel material.

Nozzle-Pad Weld for Nozzle skipped, as Nozzle is not a Carbon Steel material.

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 8.0, c = 0.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = -1.0$, Temp. Reduction = 0 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 8.0, c = 0.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = -1.0$, Temp. Reduction = 0 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
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Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Gov. MDMT of the Reinforcement Pad : -104 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: N3B- Shell inle

Intermediate Calc. for nozzle/shell Welds Tmin 8.0000 mm
 Intermediate Calc. for pad/shell Welds TminPad 8.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	5.6000 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm
Pad Weld	4.0000 = 0.5*TminPad	5.6560 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (4.6033 - 9.4388 + 2 \cdot 12.7748 \cdot 0.7634 \cdot (1.0 \cdot 10.0 - 3.0943)) \cdot 1168)$
 $= \max(0, -4072.92) \text{ kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_r2) \cdot S_v$
 $= (4.508 + 9.3741 + 0.7634 - 0.0 \cdot 0.76) \cdot 1168$
 $= 17098.94 \text{ kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$
 $= (4.508 + 0.0 + 0.7634 + (1.9505)) \cdot 1168$
 $= 8431.72 \text{ kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$
 $= (4.508 + 0.0 + 0.7634 + 9.3741 + (1.9505)) \cdot 1168$
 $= 19376.19 \text{ kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 168.275 \cdot 10.0 \cdot 0.49 \cdot 891$
 $= 11544. \text{ kgf}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.1416/2.0) \cdot 368.275 \cdot 8.0 \cdot 0.49 \cdot 1168$
 $= 26476. \text{ kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 77.7501) \cdot (14.2748 - 1.5) \cdot 0.7 \cdot 891$
 $= 19468. \text{ kgf}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$
 $= (3.1416/2) \cdot 168.275 \cdot 8.0 \cdot 0.74 \cdot 1407$
 $= 22019. \text{ kgf}$

Tension, Shell Groove Weld [Tngw]:



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$$\begin{aligned}
 &= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng \\
 &= (3.1416/2.0) \cdot 168.275 \cdot (6.0 - 0.0) \cdot 0.74 \cdot 1168 \\
 &= 13702. \text{ kgf}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SPEW} + \text{SNW}) = (26476 + 19468) = 45944 \text{ kgf}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (11544 + 22019 + 13702 + 0) = 47265 \text{ kgf}$

Failure Path [Path 3-3]:
 $= (\text{Spew} + \text{Tngw} + \text{Sinw})$
 $= (26476 + 13702 + 0) = 40178 \text{ kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 45943	kgf, must exceed W = 0 kgf	or W1 = 17098 kgf
Path 2-2 = 47265	kgf, must exceed W = 0 kgf	or W2 = 8431 kgf
Path 3-3 = 40177	kgf, must exceed W = 0 kgf	or W3 = 19376 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 32 kgf/cm²

The Drop for this Nozzle is : 10.2635 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 325.2635 mm

Input Echo, WRC297 Item 1, Description: N3B- Shell inl:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	0.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	700.000	mm
Shell Thickness	10.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-240 304L	
Vessel Cold S.I. Allowable	Smc 1172.61	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1167.53	kgf/cm ²

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	168.275	mm
Nozzle Thickness	14.2748	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-312 TP304L	
Nozzle Cold S.I. Allowable	SNmc 1172.61	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 891.31	kgf/cm ²

Diameter of Reinforcing Pad	368.275	mm
Thickness of Reinforcing Pad	8.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	10.27	kgf/cm ²
Radial Load	P	-1376.68	kgf
Circumferential Shear	Vc	994.27	kgf
Longitudinal Shear	Vl	994.27	kgf
Circumferential Moment	Mc	458.71	kgf-m.
Longitudinal Moment	Ml	458.71	kgf-m.
Torsional Moment	Mt	527.52	kgf-m.



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Include Axial Pressure Thrust No
 Include Pressure Stress Indices per Div. 2 No
 Local Loads applied at end of Nozzle/Attachment No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{359.0(10.0 - 0.0)}$$

$$= 366.000 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 718.000 mm
 Nozzle Outside Diameter (d) = 168.275 mm
 Vessel Thickness used (T) = 18.000 mm
 Nozzle Thickness used (t) = 12.775 mm
 T / t = 1.409
 d / t = 13.172

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.480$$

Nr/P = 0.104
 Mr/P = 0.133
 M0/P = 0.055
 N0/P = 0.130
 MrD/Mc = 0.241
 NrDL/Mc = 0.150
 M0d/Mc = 0.118
 N0DL/Mc = 0.069
 MrD/Ml = 0.153
 NrDL/Ml = 0.105
 M0D/Ml = 0.077
 N0DL/Ml = 0.166

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	55	55	55	55
Outplane Bending (P)	140	-140	140	-140
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-139	-139	139	139
Outplane Bending (ML)	-386	386	386	-386
Normal Pressure Stress	194	204	194	204
Outplane Stress Summary	-136	367	916	-127

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	44	44	44	44
Inplane Bending (P)	338	-338	338	-338
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-87	-87	87	87
Inplane Bending (ML)	-773	773	773	-773
Inplane Pressure Stress	97	97	97	97



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Inplane Stress Summary	-381	488	1341	-882
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	20	20	-20	-20
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	65	65	65	65

Shear Stress Summary	86	86	44	44
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	408	533	1346	885
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	C1	Du	D1
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	55	55	55	55
Outplane Bending (P)	140	-140	140	-140
Outplane Membrane (Mc)	-57	-57	57	57
Outplane Bending (Mc)	-595	595	595	-595
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	97	97	97	97

Outplane Stress Summary	-360	550	946	-525
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	C1	Du	D1
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	44	44	44	44
Inplane Bending (P)	338	-338	338	-338
Inplane Membrane (Mc)	-125	-125	125	125
Inplane Bending (Mc)	-1218	1218	1218	-1218
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	194	204	194	204

Inplane Stress Summary	-766	1003	1921	-1182
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	C1	Du	D1
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-20	-20	20	20
Torsional Shear (Mt)	65	65	65	65

Shear Stress Summary	44	44	86	86
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	C1	Du	D1
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside



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Two * Max Shear Stress	771	1007	1929	1193
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	55	55	55	55
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-139	-139	139	139
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	63	75	63	75
Hoop Stress Summary	-21	-8	257	270

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	22	22	22	22
Axial Bending (P)	407	-407	407	-407
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-203	-203	203	203
Axial Bending (ML)	-1011	1011	1011	-1011
Axial Pressure Stress	31	31	31	31
Axial Stress Summary	-753	454	1676	-1163

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	29	29	-29	-29
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	92	92	92	92
Shear Stress Summary	122	122	63	63

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	773	523	1679	1438

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Hoop Membrane (P)	55	55	55	55
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-57	-57	57	57
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	63	75	63	75
Hoop Stress Summary	60	72	176	188

Nozzle Stresses (kgf/cm²):



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CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	22	22	22	22
Axial Bending (P)	407	-407	407	-407
Axial Membrane (Mc)	-203	-203	203	203
Axial Bending (Mc)	-1670	1670	1670	-1670
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	31	31	31	31

Axial Stress Summary -1411 1112 2334 -1821

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-29	-29	29	29
Shear due to Torsion	92	92	92	92

Shear Stress Summary 63 63 122 122

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	1477	1116	2341	2024

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{355.0(10.0 - 0.0)}$$

$$= 364.895 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	710.000 mm
Nozzle Outside Diameter	(d) =	368.275 mm
Vessel Thickness used	(T) =	10.000 mm
Nozzle Thickness used	(t) =	112.775 mm
T / t	=	0.089
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.371$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.049
Mr/P	=	0.077
M0/P	=	0.022
N0/P	=	0.031
MrD/Mc	=	0.248
NrDL/Mc	=	0.102
M0d/Mc	=	0.074
N0DL/Mc	=	0.056
MrD/Ml	=	0.056
NrDL/Ml	=	0.055
M0D/Ml	=	0.017
N0DL/Ml	=	0.033

Vessel Stresses (kgf/cm²):



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Nozzle Calcs.: N3B- Shell inle Nozl: 15 9:43am Aug 12,2026

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	42	42	42	42
Outplane Bending (P)	181	-181	181	-181
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-40	-40	40	40
Outplane Bending (ML)	-127	127	127	-127
Normal Pressure Stress	354	364	354	364

Outplane Stress Summary	409	313	746	138
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	67	67	67	67
Inplane Bending (P)	636	-636	636	-636
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-68	-68	68	68
Inplane Bending (ML)	-420	420	420	-420
Inplane Pressure Stress	177	177	177	177

Inplane Stress Summary	392	-40	1370	-743
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	17	17	-17	-17
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	24	24	24	24

Shear Stress Summary	41	41	7	7
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	443	363	1370	881

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	42	42	42	42
Outplane Bending (P)	181	-181	181	-181
Outplane Membrane (Mc)	-70	-70	70	70
Outplane Bending (Mc)	-552	552	552	-552
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	177	177	177	177

Outplane Stress Summary	-222	520	1023	-444
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to	Cu Left	Cl Left	Du Right	Dl Right
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circumferential plane)	Outside	Inside	Outside	Inside
Inplane Membrane (P)	67	67	67	67
Inplane Bending (P)	636	-636	636	-636
Inplane Membrane (Mc)	-127	-127	127	127
Inplane Bending (ML)	-1850	1850	1850	-1850
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	354	364	354	364
Inplane Stress Summary	-919	1519	3036	-1927

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)		Left	Left	Right	Right
		Outside	Inside	Outside	Inside
Outplane Shear (Vc)		0	0	0	0
Outplane Shear (Vl)		-17	-17	17	17
Torsional Shear (Mt)		24	24	24	24
Shear Stress Summary		7	7	41	41

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stress Intensities		Left	Left	Right	Right
		Outside	Inside	Outside	Inside
Two * Max Shear Stress		919	1519	3037	1929

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	194	204	194	204	194	204	194	204
Circ. Pl (SUS)	-84	-84	194	194	-81	-81	169	169
Circ. Q (SUS)	-246	246	526	-526	-880	880	1556	-1556
Long. Pm (SUS)	97	97	97	97	97	97	97	97
Long. Pl (SUS)	-43	-43	131	131	-2	-2	112	112
Long. Q (SUS)	-435	435	1111	-1111	-455	455	735	-735
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	20	20	-20	-20	-20	-20	20	20
Shear Q (SUS)	65	65	65	65	65	65	65	65
Pm (SUS)	194.0	204.0	194.0	204.0	194.0	204.0	194.0	204.0
Pm+Pl (SUS)	116.4	125.6	390.5	400.3	125.9	133.4	365.6	375.4
Pm+Pl+Q (Total)	407	532	1343	885	771	1007	1926	1193

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	204.00	1167.53	Passed
Pm+Pl (SUS)	400.32	1751.29	Passed
Pm+Pl+Q (TOTAL)	1926.35	3502.58	Passed



Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	354	364	354	364	354	364	354	364
Circ. Pl (SUS)	2	2	82	82	-60	-60	194	194
Circ. Q (SUS)	54	-54	308	-308	-1214	1214	2486	-2486
Long. Pm (SUS)	177	177	177	177	177	177	177	177
Long. Pl (SUS)	-1	-1	135	135	-28	-28	112	112
Long. Q (SUS)	216	-216	1056	-1056	-371	371	733	-733
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	17	17	-17	-17	-17	-17	17	17
Shear Q (SUS)	24	24	24	24	24	24	24	24
Pm (SUS)	354.0	364.0	354.0	364.0	354.0	364.0	354.0	364.0
Pm+Pl (SUS)	357.6	367.5	438.3	448.1	296.0	305.8	549.1	559.1
Pm+Pl+Q (Total)	442	361	1368	882	920	1518	3034	1929

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	364.00	1167.53	Passed
Pm+Pl (SUS)	559.07	1751.29	Passed
Pm+Pl+Q (TOTAL)	3034.84	3502.58	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	63	75	63	75	63	75	63	75
Circ. Pl (SUS)	-84	-84	194	194	-2	-2	112	112
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	31	31	31	31	31	31	31	31
Long. Pl (SUS)	-181	-181	225	225	-181	-181	225	225
Long. Q (SUS)	-604	604	1418	-1418	-1263	1263	2077	-2077
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	29	29	-29	-29	-29	-29	29	29
Shear Q (SUS)	92	92	92	92	92	92	92	92
Pm (SUS)	63.0	75.0	63.0	75.0	63.0	75.0	63.0	75.0
Pm+Pl (SUS)	156.2	155.7	285.5	292.2	218.8	230.4	265.3	266.6
Pm+Pl+Q (Total)	773	522	1676	1436	1479	1116	2339	2022



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Nozzle Calcs.: N3B- Shell inle

Nozl: 15

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Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	75.00	891.31	Passed
Pm+Pl (SUS)	292.22	1336.96	Passed
Pm+Pl+Q (TOTAL)	2339.76	2673.92	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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ASME TS Calc: DESIGN CASE

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Input Echo, Tubesheet Number 1, Description: Fixed tubesheet

Shell Data:

Main Shell Description: Shell

Shell Maximum Design Pressure	Psd,max	10.20 kgf/cm ²
Shell Maximum Operating Pressure	Psox,max	10.20 kgf/cm ²
Shell Minimum Operating Pressure	Psox,min	0.00 kgf/cm ²
Shell Thickness	ts	10.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	700.000 mm
Shell Temperature for Internal Pressure	Ts	155.00 °C
Shell Material		SA-240 304L

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		S30403
Shell Allowable Stress at Temperature	Ss	1167.53 kgf/cm ²
Shell Allowable Stress at Ambient		1172.61 kgf/cm ²

Channel Data:

Channel Description: Chnl. rear

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	46.91 kgf/cm ²
Channel Maximum Operating Pressure	Ptox,max	46.91 kgf/cm ²
Channel Minimum Operating Pressure	Ptox,min	0.00 kgf/cm ²
Channel Thickness	tc	18.0000 mm
Channel Internal Corrosion Allowance	cac	1.5000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	700.000 mm
Channel Design Temperature	TEMPC	155.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	1407.13 kgf/cm ²
Channel Allowable Stress at Ambient		1407.13 kgf/cm ²

Tube Data:

Number of Tube Holes	Nt	545
Tube Wall Thickness	et	2.0000 mm
Tube Outside Diameter	D	19.0000 mm
Total Straight Tube Length	Lt	3300.00 mm
Straight Tube Length (bet. inner tubsht faces) L		3195.00 mm
Design Temperature of the Tubes		155.00 °C
Tube Material		SA-213 TP304L
Tube Material UNS Number		S30403
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		891.31 kgf/cm ²
Tube Allowable Stress At Ambient		1172.61 kgf/cm ²
Tube Yield Stress At design Temperature	Syt	1333.75 kgf/cm ²
Tube Pitch (Center to Center Spacing)	P	25.0000 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.0000 mm
Groove Weld Leg	ag	2.0000 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	330.948 mm
Largest Center-to-Center Tube Distance	Ul	0.0000 mm



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ASME TS Calc: DESIGN CASE

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Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	3121.6 cm ²
Perimeter of Tube Layout	Cp	2058.4 mm

Tubesheet Data:

Tubesheet TYPE: U-tube, Integral with Channel, Conf. e

Tubesheet Design Metal Temperature	T	155.00 °C
Tubesheet Material	SA-965 F304L	
Tubesheet Material UNS Number	S30403	
Tubesheet Allowable Stress at Temperature	S	891.31 kgf/cm ²
Tubesheet Allowable Stress at Ambient	Tt	1172.61 kgf/cm ²
Thickness of Tubesheet	h	105.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	0.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	1.5000 mm
Tubesheet Outside Diameter	A	860.000 mm

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped?	Yes
Is the Tubesheet Flat on Tubeside?	Yes
Is the Tubesheet Flat on Shellside?	No
Step 1 Diameter on the Shellside	ds1 763.00 mm
Step 1 Depth on the Shellside	hs1 8.00 mm
Step 2 Diameter on the Shellside	ds2 697.00 mm
Step 2 Depth on the Shellside	hs2 8.00 mm
Calculated Tubesheet Diameter per Div 2, 4.18.4	860.00 mm

Note: Tubesheet diameter is now: 860.000 mm Div 2, 4.18.4.

Area of the Untubed Lanes	AL	0.0 cm²
Junction Stress Reduction option	Perform Plastic Calculation	
Perform Differential Pressure Design	No	
Run Multiple Load Cases	Yes	

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side	Shell
Flange Outside Diameter	A 860.000 mm
Flange Inside Diameter	B 700.000 mm
Flange Face Outside Diameter	Fod 766.000 mm
Flange Face Inside Diameter	Fid 700.000 mm
Gasket Outside Diameter	Go 755.000 mm
Gasket Inside Diameter	Gi 729.000 mm
Small end Hub thk.	g0 6.0000 mm
Large end Hub thk.	g1 8.0000 mm
Gasket Factor,	m 3.00
Gasket Design Seating Stress	y 703.07 kgf/cm ²
Flange Facing Sketch	Code Sketch 1a
Gasket Thickness	tg 4.5000 mm
Full face Gasket Flange Option	Program Selects

Bolting Information:

Diameter of Bolt Circle	C	800.000	mm
Nominal Bolt Diameter	dB	25.4000	mm
Thread Series	TEMA Thread Series		
Number of Bolts	n	20	
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	1753.82	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1753.82	kgf/cm ²
Weld between Flange and Shell/Channel		0.0000	mm
Tubesheet Integral with		Channel	
Tubesheet Extended as Flange		Yes	



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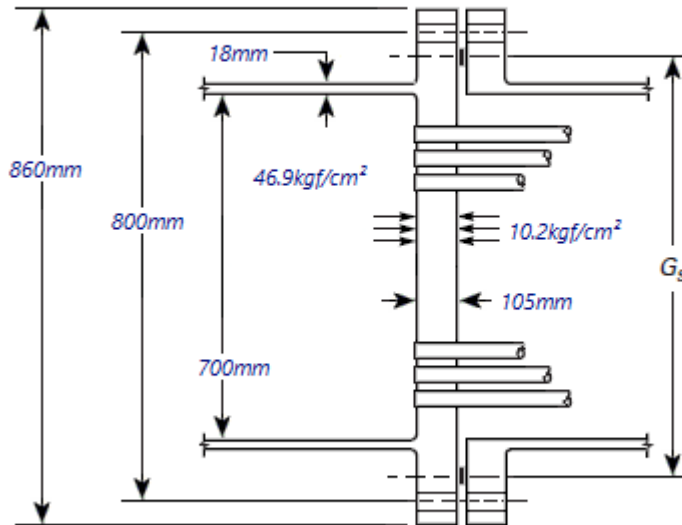
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ASME TS Calc: DESIGN CASE

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Thickness of Extended Portion of Tubesheet Tf 89.0000 mm
Is Bolt Load Transferred to the Tubesheet No

Tubesheet Gasketed w/Shell and Integral w/Channel, Extended as a Flange



Configuration e:

Messages regarding the Tubesheet & Exchanger Input Data:

Warning - Bolt load is not transferred to tubesheet which is extended for bolting. Req. Tubesheet extension thk. can't be computed. Please check input.

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Material Group G

Elastic Mod. at Design Temperature	155.0 °C	0.18937E+07 kgf/cm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.19897E+07 kgf/cm²

Channel - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	155.0 °C	0.19866E+07 kgf/cm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm²

Tubes - TM-1 Material Group G

Elastic Mod. at Tubsht. Design Temp.	155.0 °C	0.18937E+07 kgf/cm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.19897E+07 kgf/cm²

TubeSheet - TM-1 Material Group G

Elastic Mod. at Design Temperature	155.0 °C	0.18937E+07 kgf/cm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.19897E+07 kgf/cm²

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:



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ASME TS Calc: DESIGN CASE

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$$= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (46.91 \cdot (19.0/2 - 0)) / (891.31 \cdot 1.0 + 0.4 \cdot 46.91)$$

$$= 0.4897 + 0.0000 = 0.4897 \text{ mm}$$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart HA-3 at 155.00 °C
Elastic Modulus for Material 1864764.88 kgf/cm²

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.0000	19.00	3525.95	9.50	50.0000	0.0121884	699.79

EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 101.3333 \text{ kgf/cm}^2$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.3812	19.00	3525.95	49.84	50.0000	0.0004428	381.25

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 381.2455) / (3 \cdot 49.8432) = 10.1985 \text{ kgf/cm}^2$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.4897 mm
Allowable Internal Pressure at Corroded thickness	204.90 kgf/cm²
Required Internal Design Pressure	46.91 kgf/cm²
Allowable External Pressure at Corroded thickness	101.33 kgf/cm²
Required External Design Pressure	10.20 kgf/cm²
Required Thickness due to Shell Side pressure	0.3812 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Gasketed Tubesheets:

ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	$N = (G_{oc} - G_{ic}) / 2$	13.000 mm
Basic Gasket Width,	$b_0 = N / 2$	6.500 mm
Effective Gasket Width,	$b = \text{sqrt}(b_0) \cdot 2.5$	6.422 mm
Gasket Reaction Diameter,	$G = G_o - 2.0 \cdot b$	742.156 mm

Results for ASME U-tube Tubesheet Calculations for Configuration e:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot r_o + dt$$

$$= 2 \cdot 330.948 + 19.0$$

$$= 680.896 \text{ mm}$$

Basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (25.0 - 19.0) / 25.0$$

$$= 0.240$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$$= G_s / D_o \text{ (Configurations d, e, f)}$$

$$= 742.1556 / 680.896$$

$$= 1.0900$$

Ratio [pc]:



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ASME TS Calc: DESIGN CASE

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= Dc / Do (Configurations a, e, f)
= 700.0/680.896
= 1.0281

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [ρ]:

= ltx / h
= 0/105.0
= 0.000 (must be $0 \leq \rho \leq 1$)

Effective tube hole Diameter [d*]:

= max(dt - 2tt(EtT/E) • (StT/S) • (ρ); dt - 2tt)
= max(19.0 - 2 • 2.0 (1893659/1893659) •
(891/891) • (0); 19.0 - 2 • 2.0)
= 19.0000 mm

Effective tube pitch [p*]:

= p / $\sqrt{(1 - 4 \cdot \min(AI; 4 \cdot Do \cdot p) / (\pi \cdot Do^2))}$
= 25.0 / $\sqrt{(1 - 4 \cdot \min(0; 4 \cdot 680.896 \cdot 25.0) / (\pi \cdot 680.896^2))}$
= 25.0000 mm

Effective ligament efficiency for bending [μ*]:

= (p* - d*) / p*
= (25.0 - 19.0) / 25.0
= 0.2400

h/p = 4.200000 ; μ* = 0.240000 ; E* = 386361 kgf/cm²

E*/E = 0.204029 ; ν* = 0.407183

Note: As h/p (4.200) is > 2, data values for h/p = 2 were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Intermediate parameters for Tubesheet Gasketed on the Shell Side:

βs, ks, λs, δs and ws are all = 0

Channel coefficient [βc]:

= $((12(1 - \nu_c^2))^{1/4}) / \sqrt{(Dc + tc)tc}$
= $((12(1 - 0.3^2))^{1/4}) / \sqrt{(700.0 + 18.0)18.0}$
= 0.0160 1/mm

Channel coefficient [kc]:

= βc • Ec • tc³ / (6(1 - νs²))
= 0.016 • 1986601 • 18.0³ / (6(1 - 0.3²))
= 33930696.0000 kgf/cm²•mm²

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

CqDelta = CqTheta = CmTheta = CmDelta = 1

Channel coefficient [λc]:

= 6(Dc + tc/cos(theta)) • kc/h³ • (CmTheta +
h • βc • (CmΔ + CqTheta)/2 + (h • βC)²/2 • CqΔ)
= 6(700.0 + 18.0/cos(0)) • 33930696/105.0³ • (1.0 +
105.0 • 0.016 • (1.0 + 1.0)/2 + (105.0 • 0.016)²/2 • 1.0)
= 516252.4 kgf/cm²

Channel coefficient [δc]:

= Dc² / (4 • Ec • tc) • (1 - Dc • νc / (2(Dc + tc)))
= 700.0² / (4 • 1986601 • 18.0) • (1 - 700.0 • 0.3 / (2(700.0 + 18.0)))
= 0.29248E-02 mm/kgf/cm²

Channel coefficient [zetac]:

zetac = 0



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ASME TS Calc: DESIGN CASE

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Parameter [wc]:

$$\begin{aligned}
 &= pc \cdot kc [\delta C (CmTheta \cdot \beta C + Cq\Delta \cdot h \cdot \beta C^2) - \\
 &\quad \zeta c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta C / 2)) \\
 &= 1.028 \cdot 33930696 [0.002925 (1.0 + 0.016 \cdot 1.0 \cdot 105.0) - \\
 &\quad 0 (1.0/105.0 + 1.0 \cdot 0.016/2)) \\
 &= 4370.4482 \text{ mm}^2
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$\begin{aligned}
 &= A / Do \\
 &= 860.0/680.896 \\
 &= 1.2630
 \end{aligned}$$

Coefficient [F]:

$$\begin{aligned}
 &= (1 - v*) / E* \cdot (\lambda c + E \cdot \ln(K)) \\
 &= (1 - 0.41) / 386361 \cdot (516252 + 1893659 \cdot \ln(1.26)) \\
 &= 1.4706
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$\begin{aligned}
 &= Do^2/16 \cdot [(ps-1) \cdot (ps^2+1) \cdot Ps - (pc-1) \cdot (pc^2+1) \cdot Pt] \\
 &= 680.896^2/16 \cdot [(1.09 - 1) \cdot (1.09^2 + 1) \cdot 10.197 - \\
 &\quad (1.028 - 1) \cdot (1.028^2 + 1) \cdot 46.908] \\
 &= -20274.1992 \text{ kgf/cm}^2 \cdot \text{mm}^2
 \end{aligned}$$

Moment acting on unperforated tubesheet rim [M*]:

$$\begin{aligned}
 &= Mts + (wc \cdot Pt) + W* \cdot (C - Gs) / (2 \cdot \pi \cdot Do) \\
 &= -20274 + (205009) + 53274 \cdot (800.0 - 742.156) / (2 \cdot \pi \cdot 680.896) \\
 &= 256766.5938 \text{ kgf/cm}^2 \cdot \text{mm}^2
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$\begin{aligned}
 &= ((M*) - Do^2/32 \cdot F (Ps - Pt)) / (1 + F) \\
 &= ((256766.59) - 680.896^2/32 \cdot 1.471 (10.2 - 46.91)) / (1 + 1.47) \\
 &= 420519.0938 \text{ kgf/cm}^2 \cdot \text{mm}^2
 \end{aligned}$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$\begin{aligned}
 &= Mp + Do^2/64 \cdot (3 + v*) \cdot (Ps - Pt) \\
 &= 420519.09 + 680.896^2/64 \cdot (3 + 0.407) \cdot (10.2 - 46.91) \\
 &= -485566.2188 \text{ kgf/cm}^2 \cdot \text{mm}^2
 \end{aligned}$$

Maximum bending moment acting on tubesheet [M]:

$$\begin{aligned}
 &= \max(| Mp |; | Mo |) \\
 &= \max(| 420519.094 |; | -485566.21 |) \\
 &= 485566.2188 \text{ kgf/cm}^2 \cdot \text{mm}^2
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$\begin{aligned}
 &= 6 \cdot M / (\mu* (h - hg')^2) \\
 &= 6 \cdot 485566.219 / (0.24 (105.0 - 0)^2) \\
 &= 1101.0573 \text{ kgf/cm}^2
 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$\begin{aligned}
 &= 2 \cdot S \\
 &= 2 \cdot 891.31 \\
 &= 1782.61 \text{ kgf/cm}^2
 \end{aligned}$$

Tubesheet bending stress at final thickness [σ]:

$$\begin{aligned}
 &= 6 \cdot M / (\mu* (h - hg')^2) \\
 &= 6 \cdot 476215.562 / (0.24 (81.7674 - 0)^2) \\
 &= 1780.6675 \text{ kgf/cm}^2
 \end{aligned}$$



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ASME TS Calc: DESIGN CASE

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Required tubesheet thickness for Bending Stress [HreqB]:

$$\begin{aligned} &= h + \text{Cats} + \text{Catc} \\ &= 81.7674 + 0 + 0 \\ &= 81.7674 \text{ mm} \end{aligned}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$\begin{aligned} &= \max(\text{HreqB}; \text{HreqS}) \\ &= \max(81.7674; 32.6316) \\ &= 81.7674 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [| Ps - Pt |]:

$$\begin{aligned} &= | \text{Ps} - \text{Pt} | \\ &= | 10.2 - 46.91 | \\ &= 36.711 \text{ kgf/cm}^2 \end{aligned}$$

Shear Stress check [τlimit]:

$$\begin{aligned} &= 4 \cdot \mu \cdot h / \text{Do} \cdot \min(0.8 \cdot S; 0.533 \cdot S_y) \\ &= 4 \cdot 0.24 \cdot 105.0 / 680.9 \cdot \min(0.8 \cdot 891.306; 0.533 \cdot 1333.752) \\ &= 105.240 \text{ kgf/cm}^2 \end{aligned}$$

| Ps - Pt | is ≤ the Shear Stress limit (τlimit). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$\begin{aligned} &= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot (4 \cdot \text{Ap} / \text{Cp}) \cdot | \text{Ps} - \text{Pt} | + \text{Cats} + \text{Catc} \\ &= 1 / (4 \cdot 0.24 \cdot 710.89) \cdot (4 \cdot 3121.6 / 2058.35) \cdot | 10.2 - 46.91 | + 0 \\ &= 32.6316 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Channel thickness of: 18.000 mm

Min. Channel len. adjacent to the tubesheet is: 202.049mm

Axial Channel Membrane Stress [σc,m]:

$$\begin{aligned} &= \text{Dc}^2 / (4 \cdot \text{tc}(\text{Dc} + \text{tc}) \cdot \cos(\theta)) \cdot \text{Pt} \\ &= 700.0^2 / (4 \cdot 18.0(700.0 + 18.0) \cdot \cos(0)) \cdot 46.908 \\ &= 444.617 \text{ kgf/cm}^2 \end{aligned}$$

Axial Channel Bending Stress [σc,b]:

$$\begin{aligned} &= 6 \cdot \text{kc} / \text{tc}^2 \cdot [(\delta_c \cdot \text{Cm}\Delta \cdot \beta_c - \zeta_c \cdot \text{Cm}\Theta) \cdot \text{Pt} - 6((1 - \nu^*) / E^*) \cdot \text{Do} / h^3 \cdot (\text{Cm}\Theta + h \cdot \beta_c / 2 \cdot \text{Cm}\Delta) \cdot (\text{Mp} + \text{Do}^2 / 32 \cdot (\text{Ps} - \text{Pt}))] \\ &= 6 \cdot 748043 / 18.0^2 \cdot [(0.002925 \cdot 1.0 \cdot 0.016 - 0 \cdot 1.0) \cdot 46.908 - 6((1 - 0.407) / 386361) \cdot 680.896 / 105.0^3 \cdot (1.0 + 0.016 \cdot 105.0 / 2 \cdot 1.0) \cdot (420519.094 + 680.896^2 / 32 \cdot (10.197 - 46.908))] \\ &= 2075.358 \text{ kgf/cm}^2 \end{aligned}$$

Channel Membrane + Bending Stress [σc]:

$$\begin{aligned} &= | \sigma_{c,m} | + | \sigma_{c,b} | \text{ (Should be } \leq 1.5 \cdot S_c) \\ &= 2519.975 \text{ kgf/cm}^2 \text{ (Should be } \leq 2110.70) \end{aligned}$$

Step 11, The Cylinder-to-Tubesheet Junction is Overstressed.

Performing a simplified Elastic-Plastic calculation

(option 3) to reduce the overstress condition.

Modify Es and/or Ec and recompute from Step 4 onwards.

Modified Elasticity [Ec]:

$$\begin{aligned} &= E_c \cdot \sqrt{(1.5 \cdot S_c / \sigma_c)} \\ &= 1986601 \cdot \sqrt{(1.5 \cdot 1407.13 / 2519.98)} \\ &= 1818134.8 \text{ kgf/cm}^2 \end{aligned}$$



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Tubesheet Bending Stress at Original Thk., after the Elastic-Plastic iteration:

$$= 6 \cdot M / (\mu \cdot (h - h'g)^2)$$

$$= 6 \cdot 488678.312 / (0.24 \cdot (105.0 - 0)^2)$$

$$= 1108.1 \text{ kgf/cm}^2 \text{ (Should be } \leq 1782.6 \text{)}$$



Note: The tubesheet is Not overstressed after the Elastic-Plastic iteration.
The design is acceptable. Recomputing tubesheet required thickness.

Required Tubesheet Thk., for Bending Stress after the Elastic-Plastic iteration [HreqB]:

$$= h + Cats + Catc = 82.1151 + 0 + 0 = 82.1151 \text{ mm}$$

Required Tubesheet Thk. after the Elastic-Plastic iteration (includes CA) [Hreq]:

$$= \max(HreqB; HreqS) = \max(82.1151; 32.6316) = 82.1151 \text{ mm}$$

Tube Metal Area [At]:

$$= \pi (do - t) t$$

$$= \pi (19.0 - 2.0) 2.0$$

$$= 1.068 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (do - t) S$$

$$= \pi \cdot 2.0 (19.0 - 2.0) 891.306$$

$$= 952.040 \text{ kgf}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot ag (do + 0.67 \cdot ag) Sw \text{ (but not } > Ft)$$

$$= 0.85 \cdot \pi \cdot 2.0 (19.0 + 0.67 \cdot 2.0) 891.31$$

$$= 952.0400 \text{ kgf}$$

Weld Strength Factor [fw]:

$$= S / Sw \text{ where } Sw = \min(S \text{ and } St)$$

$$= 891.306 / 891.306$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot af (do + 0.67 \cdot af) Sw \text{ (but not } > Ft)$$

$$= 0.55 \cdot \pi \cdot 2.0 (19.0 + 0.67 \cdot 2.0) 891.31$$

$$= 626.4983 \text{ kgf}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{(0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) fw \cdot fd} - 0.75 \cdot do \right)$$

$$= 2 \left(\sqrt{(0.75 \cdot 19.0)^2 + 1.07 \cdot 2.0 \cdot (19.0 - 2.0) 1.0 \cdot 1.0} - 0.75 \cdot 19.0 \right)$$

$$= 2.448 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.2239 mm

Minimum Required Groove Weld Leg [agr]: 1.2239 mm

Tube-Tubesheet Jt allowable, 952.04 is $\geq$ tube strength 952.04 kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
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ASME TS Calc: DESIGN CASE

Case: 1 9:43a Aug 12,2026

Tubesheet bend. stress	1108.1	<=	1782.6	kgf/cm ²	Ok
Tubesheet shear stress	220.9	<=	710.9	kgf/cm ²	Ok
Stress in Channel at Tubesheet	2520.0	<=	4715.2	kgf/cm ²	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	82.115	105.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.224	2.000	Ok
Tube-Tubesheet Groove Weld Leg :	1.224	2.000	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 6 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend Allwd	Stresses Shear Allwd	Case Type	Pass/ Fail		
D1uc	100.249	...	1625	1783	282	711	Ps+Pt D1	Ok
D2uc	45.457	...	491	1783	61	711	Ps+Pt D2	Ok
D3uc	82.115	...	1108	1783	221	711	Ps+Pt D3	Ok
D1c	103.348	...	1726	1783	286	711	Ps+Pt-c D1	Ok
D2c	48.402	...	517	1783	62	711	Ps+Pt-c D2	Ok
D3c	85.216	...	1181	1783	224	711	Ps+Pt-c D3	Ok
Max:	103.3482	... mm	0.968	0.403	(Str. Ratio)			

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

Case#	-- Integral Cyl. Reqd. Thk. Shell Channel	Shell Stress	Stress Allwd	Channel Stress	Stress Allwd	Pass Fail
D1uc	...	...	...	3224	4715	Ok
D2uc	...	...	...	704	2111	Ok
D3uc	...	...	...	2520	4715	Ok
D1c	...	...	...	3547	4715	Ok
D2c	...	...	...	744	2111	Ok
D3c	...	...	...	2803	4715	Ok
Max	...	...	...	0.752		

Note:

Because there was no net pressure differential across the tubesheet due to the specified pressures psd,min and ptd,min; load case D4 was not processed as the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 6 Load Cases:



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ASME TS Calc: DESIGN CASE

Case: 1 9:43a Aug 12, 2026

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	103.348	105.000	Ok
Tubesheet thickness flanged extension :	0.000	89.000	Ok
Tube thickness :	0.490	2.000	Ok
Tube-tubesheet fillet weld leg :	1.224	2.000	Ok
Tube-tubesheet groove weld leg :	1.224	2.000	Ok

Given Shell Thickness: 10.0000 mm

Given Channel Thickness: 18.0000 mm

Min Channel length of thk, (18.000) adj. to tubesheet: 202.049 mm

Note: This is a full strength Tube to Tubesheet Joint.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	48.327	0.999	34.697	1.000
Tubesheet Shear Stress	116.437	1.000	116.437	1.000
Tubesheet-Channel Junction	48.327	0.999	34.697	1.000
Tube Pressure Stress	204.897	1.000	101.332	1.000
Minimum MAWP	48.327		34.697	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	66.579	0.999	47.056	1.000
Tubesheet Shear Stress	153.522	1.000	153.522	1.000
Tubesheet-Channel Junction	66.579	0.999	47.056	1.000
Tube Pressure Stress	269.565	1.000	137.024	1.000
Minimum MAPnc	66.579		47.056	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

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Problems/Failures Summary:

Step: 32 9:43am Aug 12, 2026

Noted issues for the current model:

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

**** Warning: The net force on one of the saddles putting it in tension!**

Please review all reports carefully!

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PV Elite® Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: